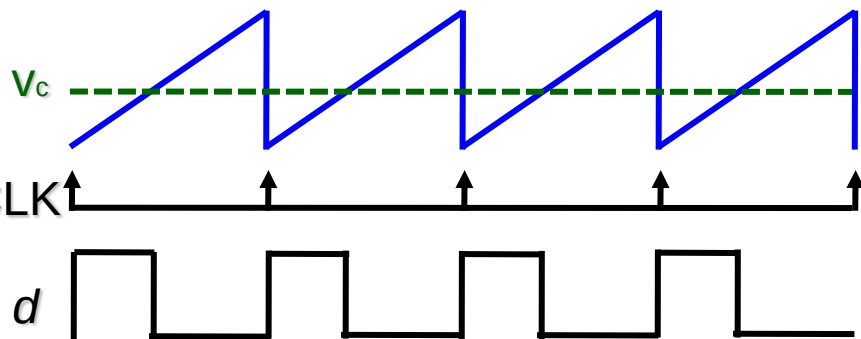
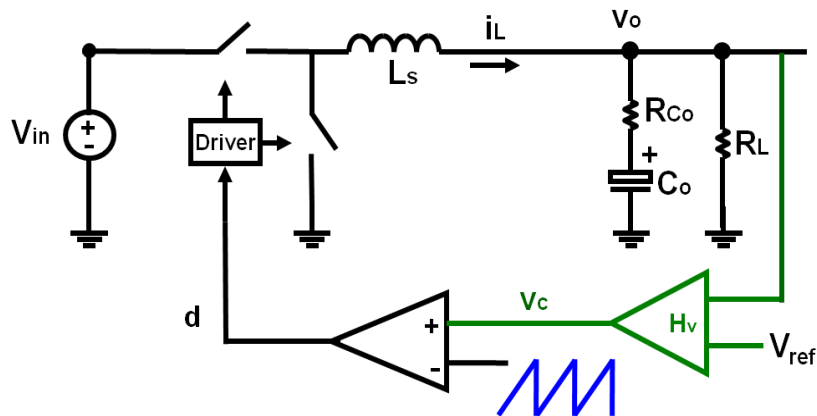


Chapter 5

New Current-Mode Control Modeling based on Describing Function Method

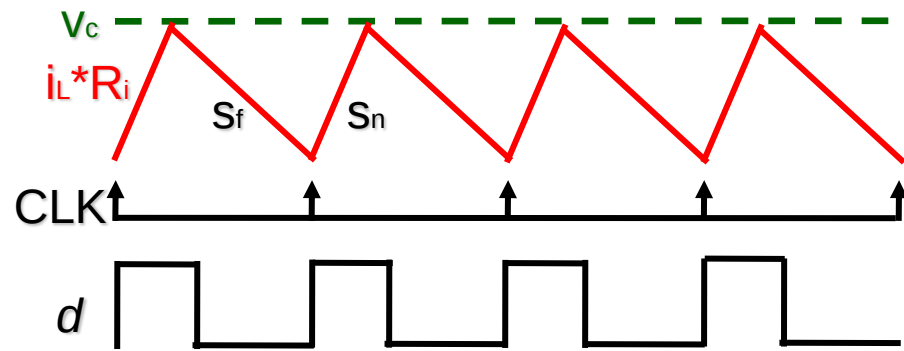
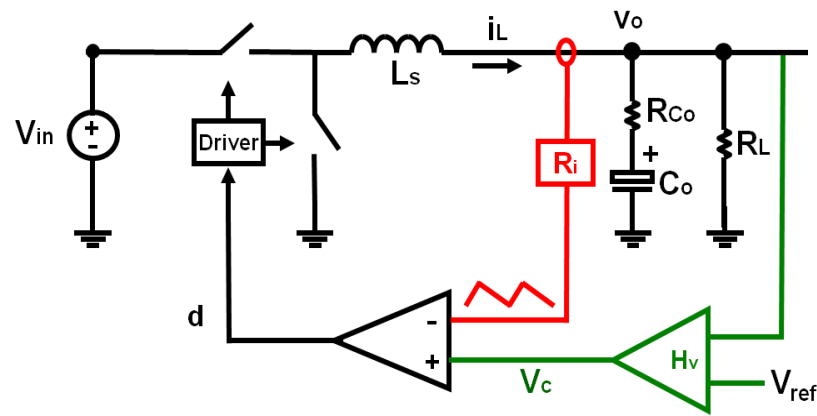
Voltage-Mode vs. Current-Mode

➤ Voltage-Mode Control



External ramp is used

➤ Peak Current-Mode Control

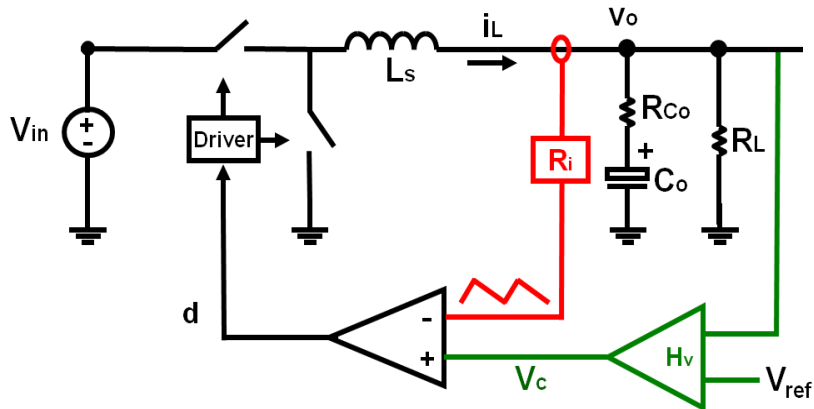


Inductor current ramp is used

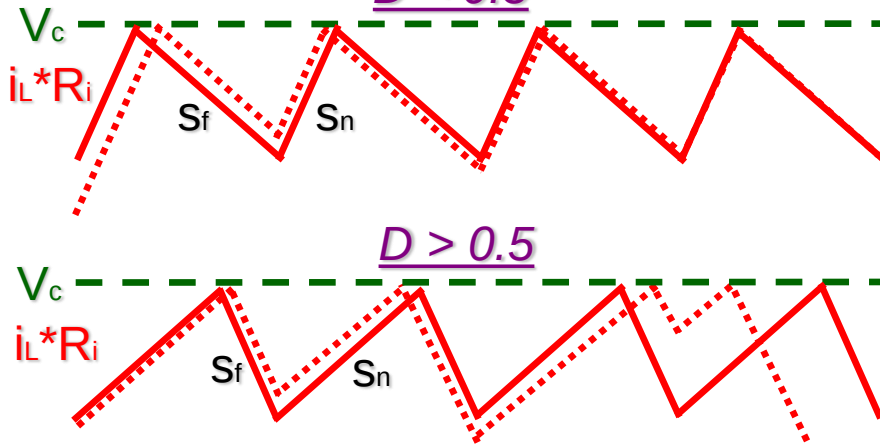
**In current-mode control,
state variable i_L is used in the pulse-width modulator**

Subharmonic Oscillations in Current-Mode Control

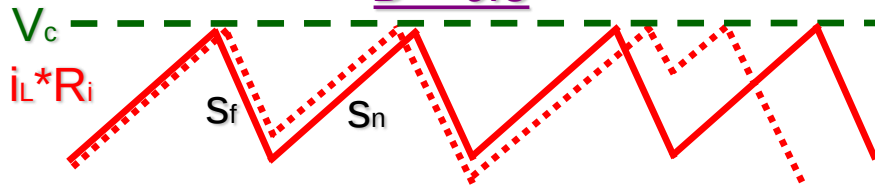
➤ Peak Current Mode Control



$D < 0.5$

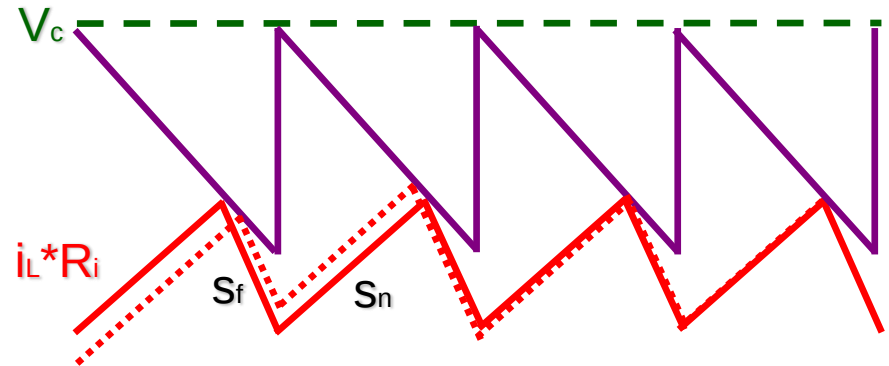
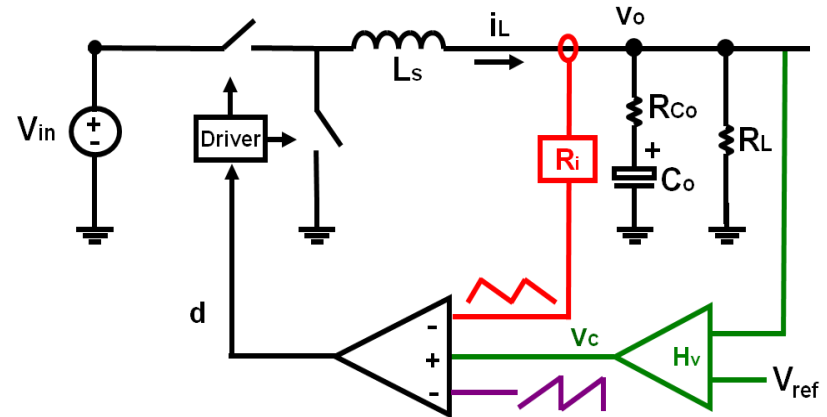


$D > 0.5$



Subharmonic Oscillations

➤ Adding an External Ramp



External ramp is used to eliminate subharmonic oscillations

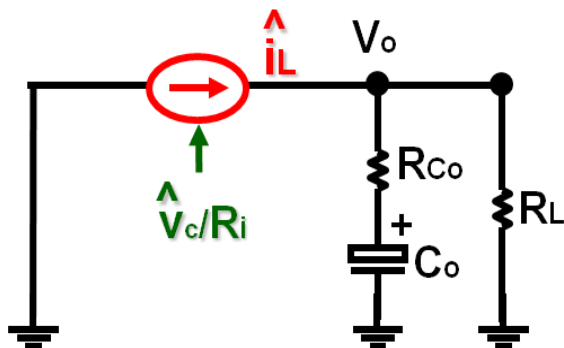
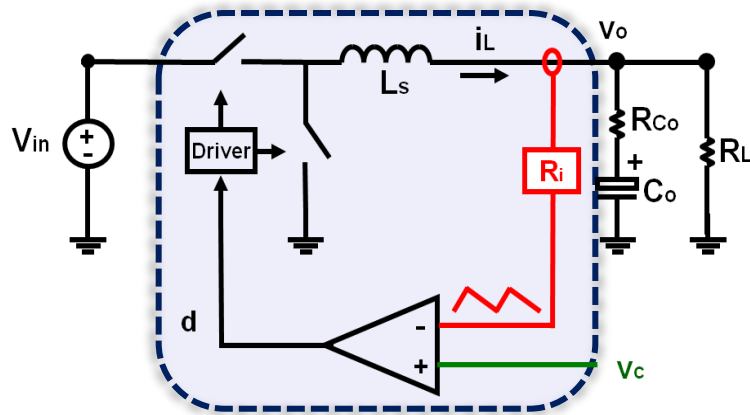
An accurate model is indispensable to system design

Review of Previous Models for Peak Current-Mode Control

- “Current Source” Model
- Average Model
- Discrete-Time Model
- Sample-Data Model
- Modified Average Model
- Other Models

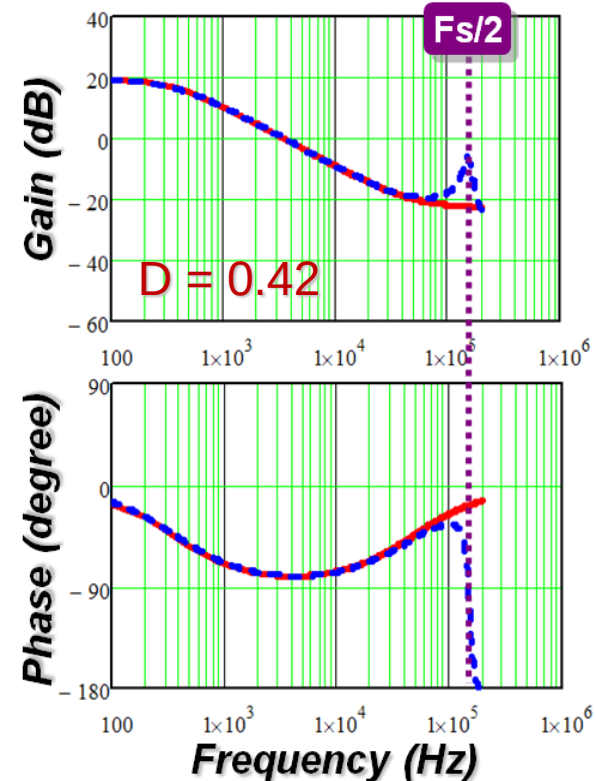
The Concept of “Current Source”*

- The current loop is treated as one entity to model



A 1st order system

Control-to-output Transfer Function

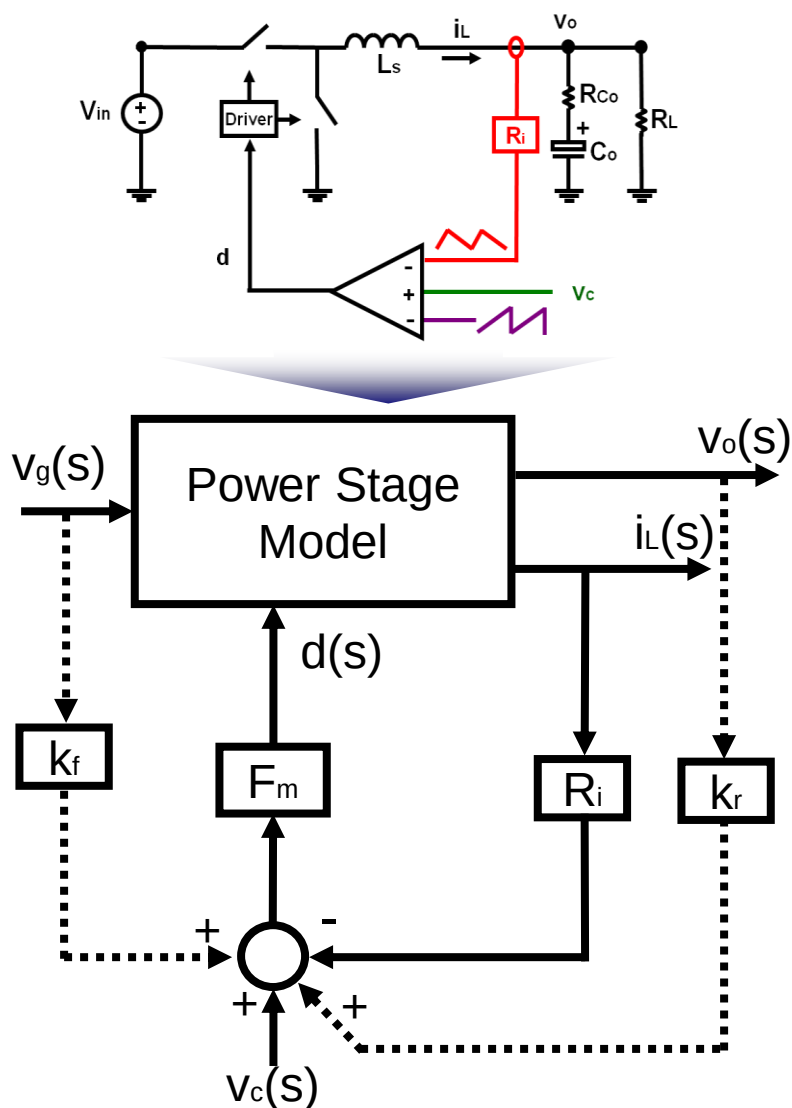


- ◆ “Current Source” Model
- ◆ Simplis Simulation $\approx$ Measurement

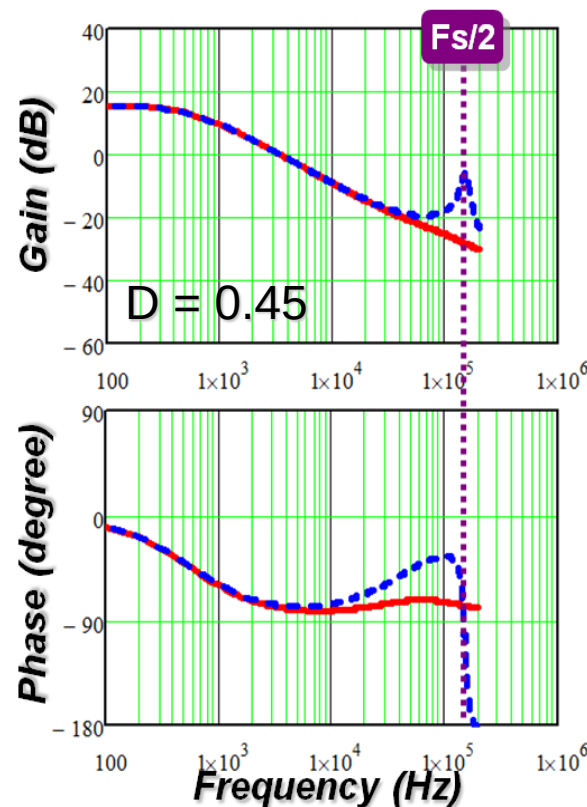
This concept is too simple to predict subharmonic oscillations

* S. P. Hsu, R. D. Middlebrook, etc. “Modeling and analysis of switching dc-to-dc converters...,” PESC’79

Average Model



Control-to-output Transfer Function



◆ *Average Model*
◆ *Simplis Simulation $\approx$ Measurement*

The average model can't predict the response @ high frequency

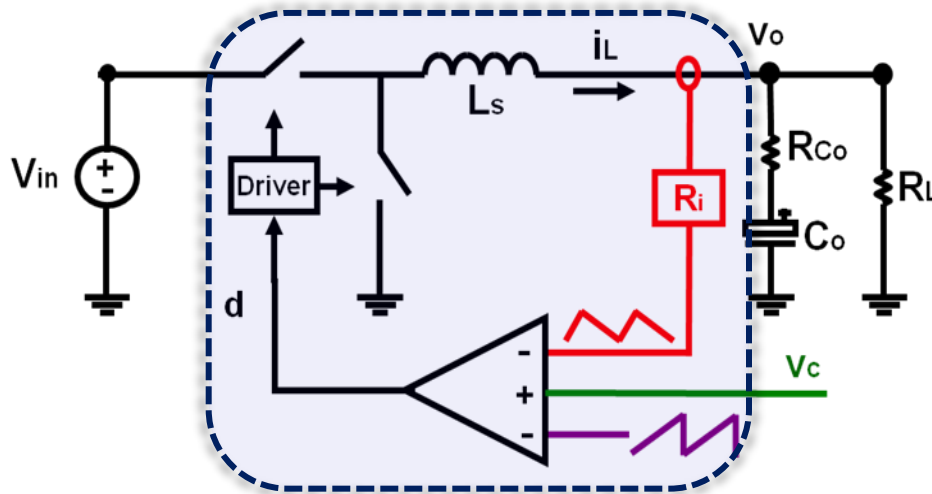
R. D. Middlebrook, etc. "Topics in Multiple-Loop Regulators and Current Mode Programming," PESC'85

Sample-Data Model

for Peak Current-Mode Control

by Arthur Brown and Middlebrook

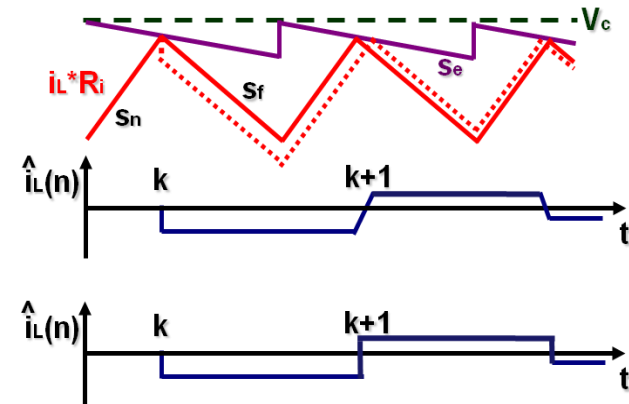
➤ The current loop is treated as one entity to model



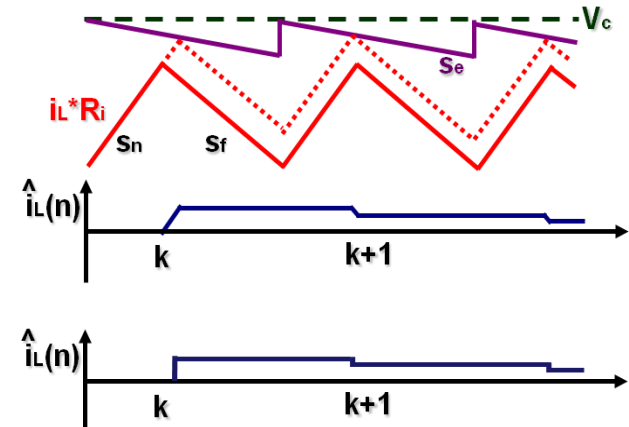
$$H(z) = \frac{\hat{i}_L(z)}{\hat{v}_c(z)} = \frac{1+\alpha}{R_i} \frac{z}{z+\alpha}$$

where, $\alpha = (s_f - s_e)/(s_n + s_e)$

1. Natural Response



2. Forced Response



Assumption: inductor current slopes are constant

Current loop instability can be shown in this model

Modified Average Model

R. Ridley's Frequency-Domain Analysis

➤ Discrete time

$$H(z) = \frac{\hat{i}_L(z)}{\hat{v}_c(z)} = \frac{1+\alpha}{R_i} \frac{z}{z+\alpha}$$

to

➤ Continuous

$$\frac{i_L(s)}{v_c(s)} = \frac{1}{R_i} \frac{1+\alpha}{sT_{sw}} \frac{e^{sT_{sw}} - 1}{e^{sT_{sw}} + \alpha}$$

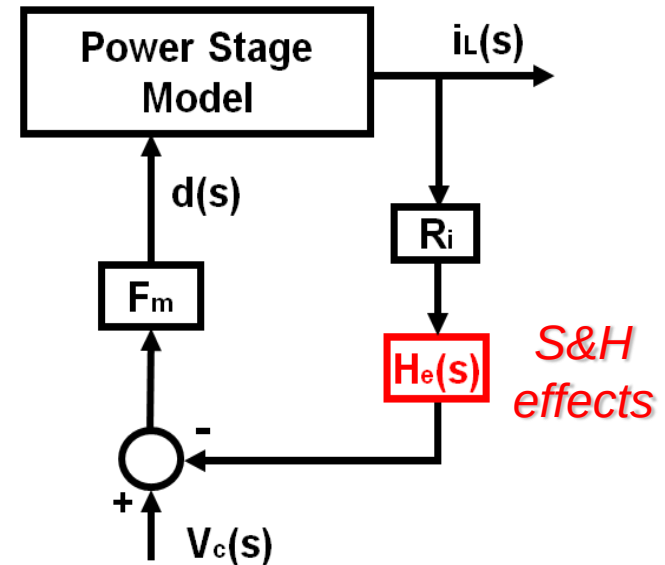
=

$$\frac{i_L(s)}{v_c(s)} = \frac{F_m F_i(s)}{1 + F_m F_i(s) R_i H_e(s)}$$



$$H_e(s) = \frac{sT_{sw}}{e^{sT_{sw}} - 1}$$

➤ Modified Average model

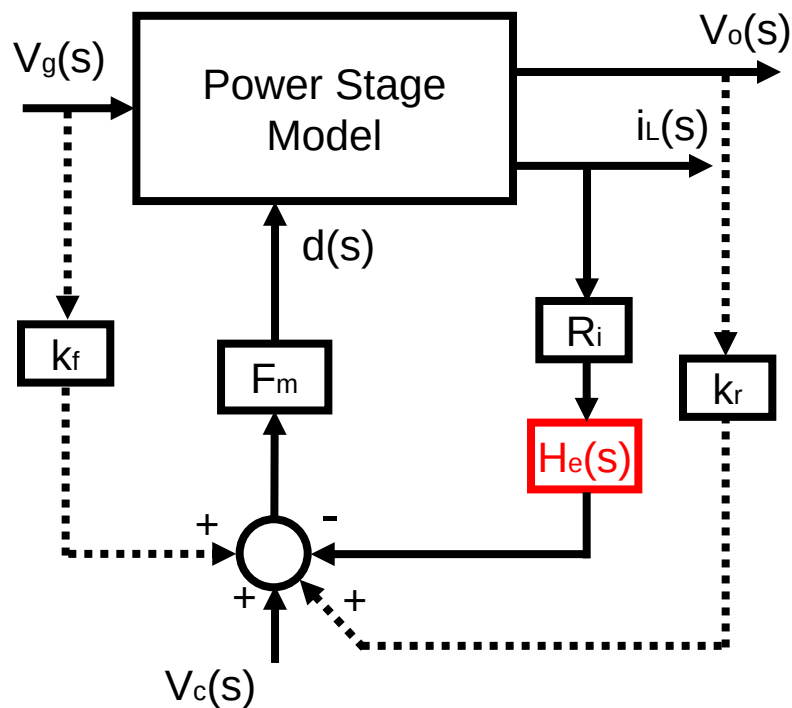


Hypothesis method is used in R. Ridley's model

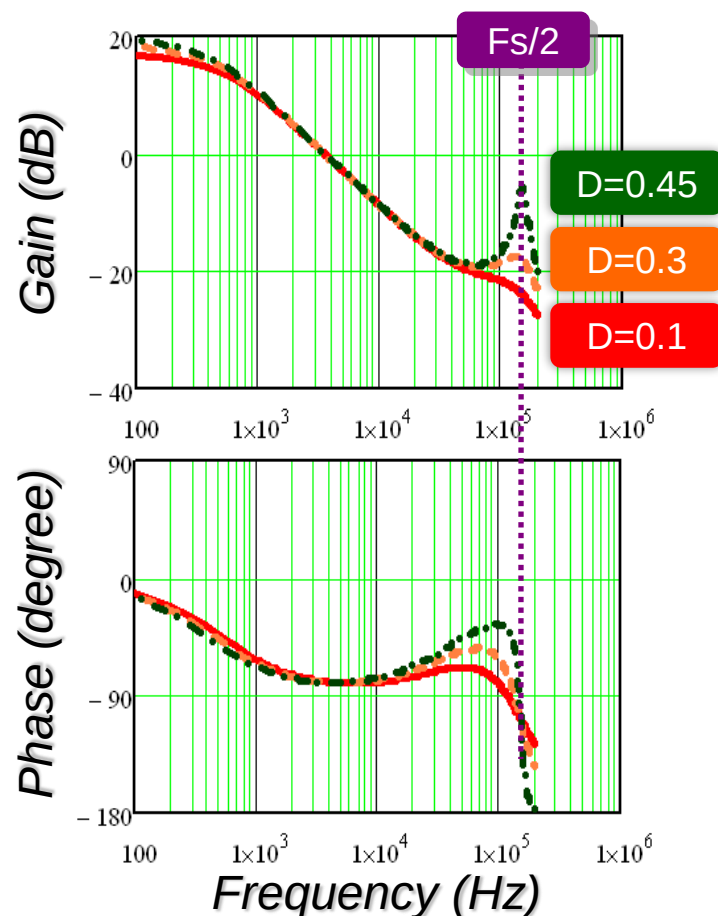
R. B. Ridley, "A new, continuous-time model for current-mode control," IEEE Trans. on Power Electronics, 1991.

R. Ridley's Model

for Peak Current-Mode Control

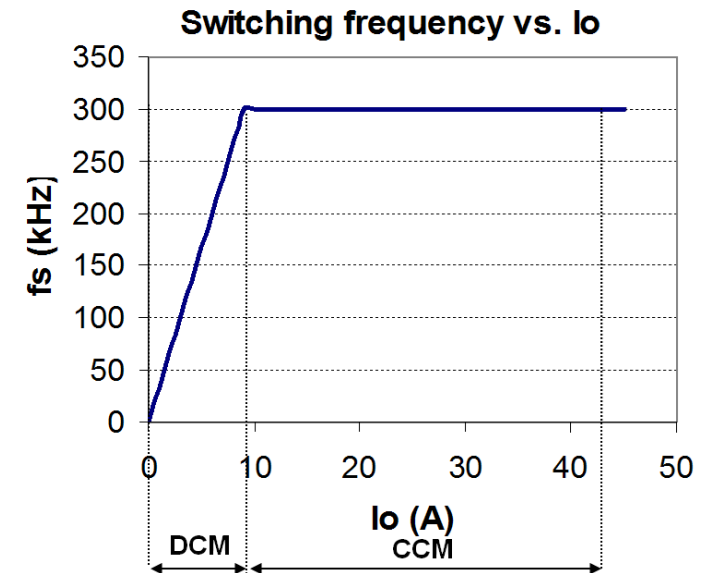
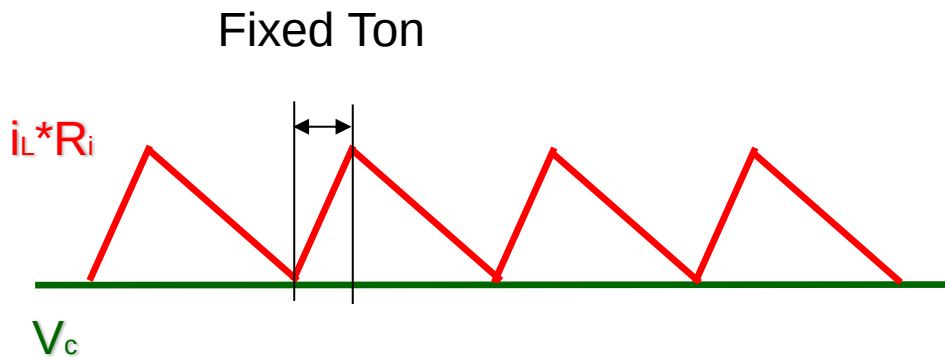
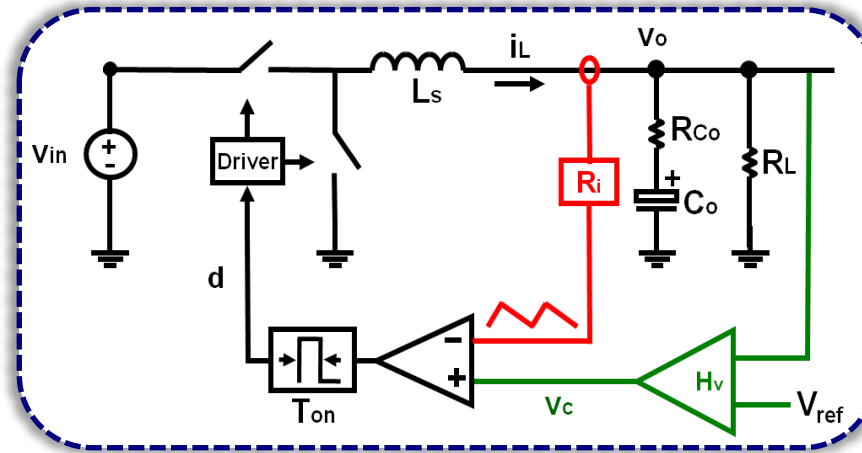


Control to v_o Transfer Function



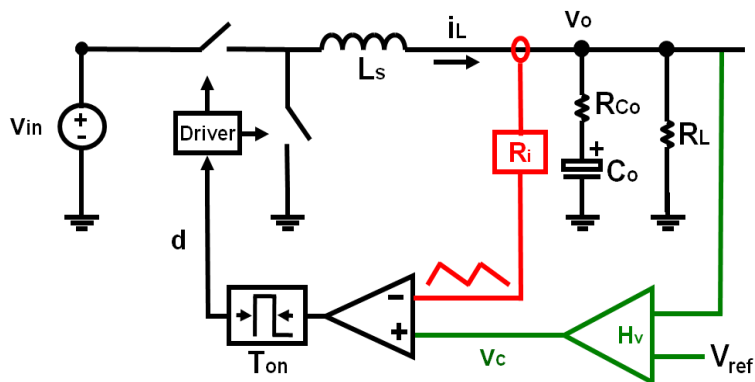
- ◆ R. Ridley's model is good for constant frequency case:
 1. Peak current model control
 2. Valley current mode control

Constant On-time Control

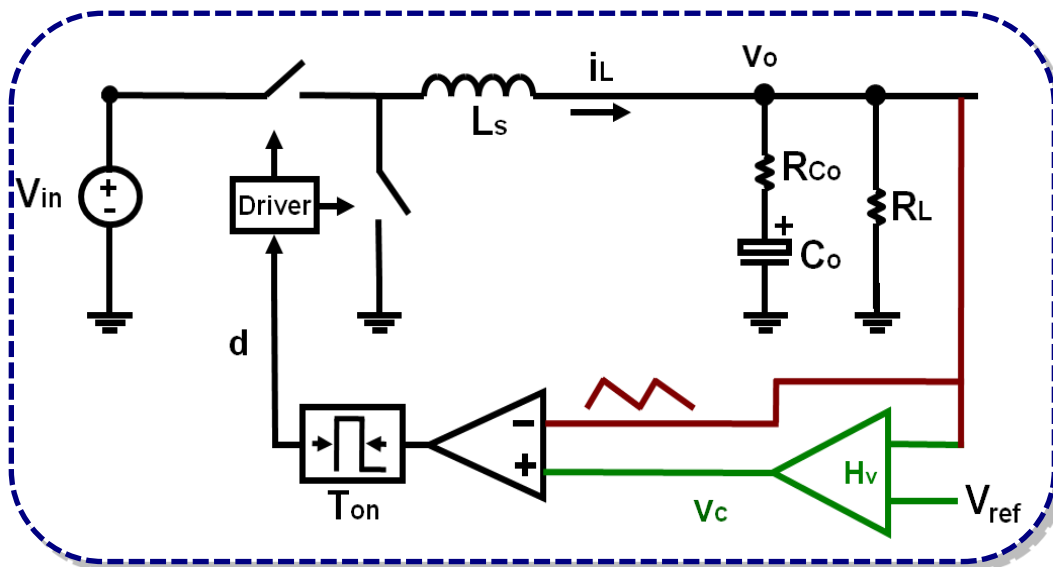


Constant on-time control is widely used to improve light-load efficiency

V² Type Implementation



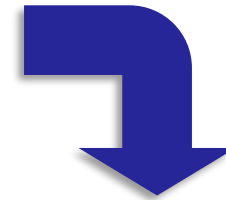
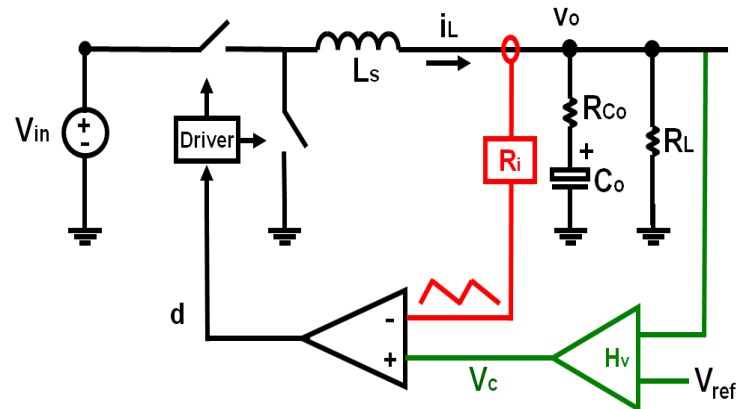
Using ESR as the sensing resistor



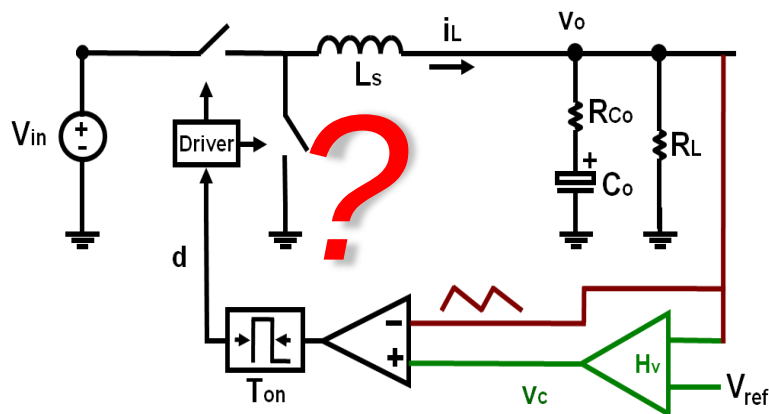
Benefits: 1. Simple implementation; 2. Good efficiency @ light load
3. Fast transient response

Can R. Ridley's Model be Extended to Variable Freq.?

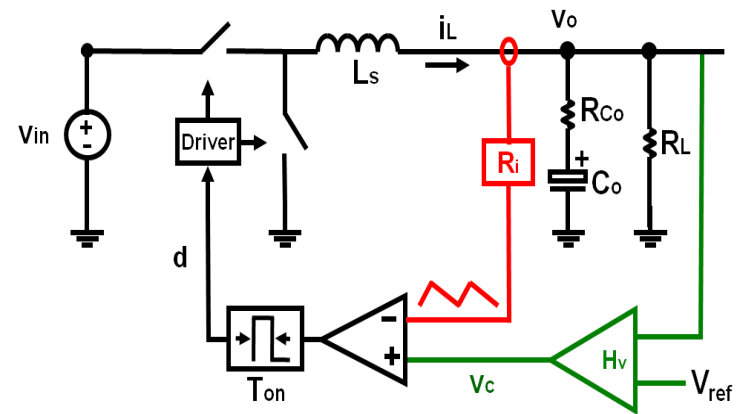
➤ Peak Current-Mode Control



➤ V^2 Constant On-time Control



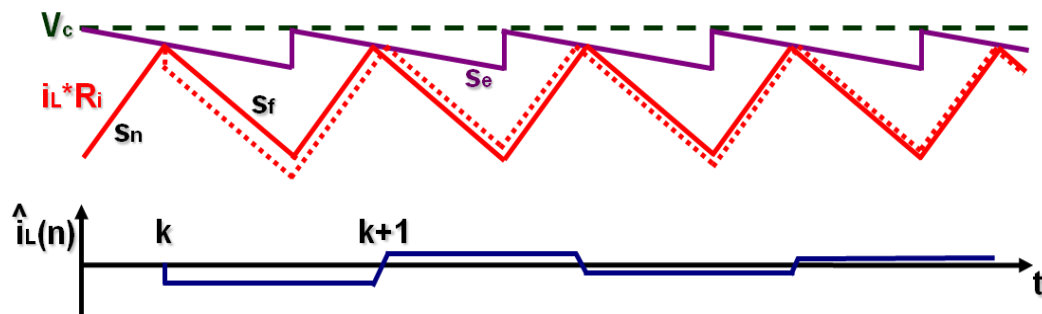
➤ Constant On-time Control



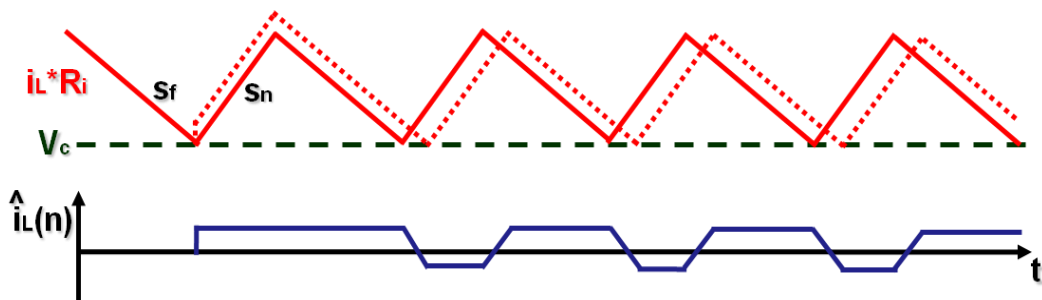
Extension of Ridley's Model

to Constant On-time Control*

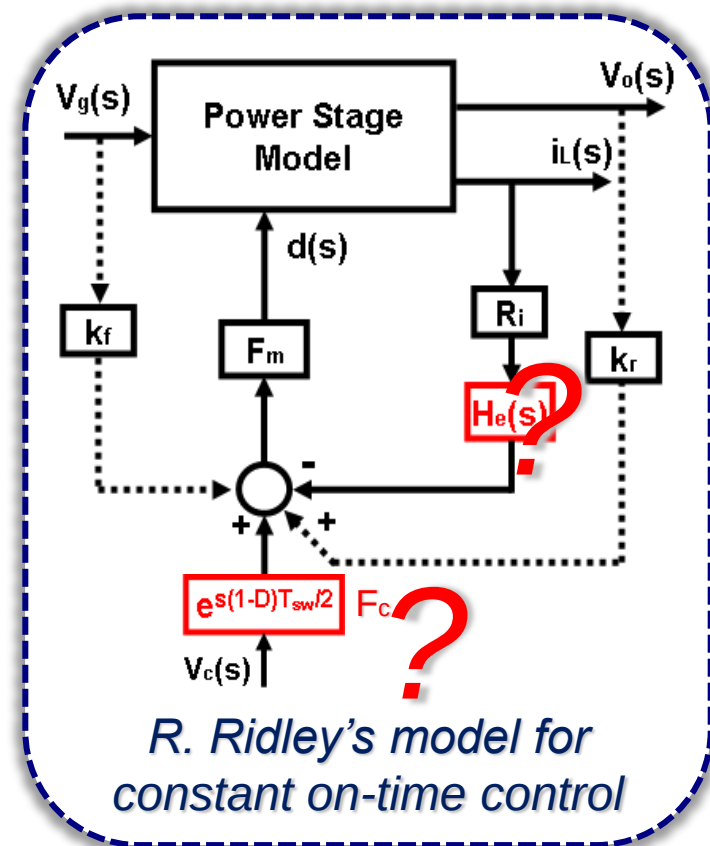
◆ Peak Current-Mode control



◆ Constant On-time Control



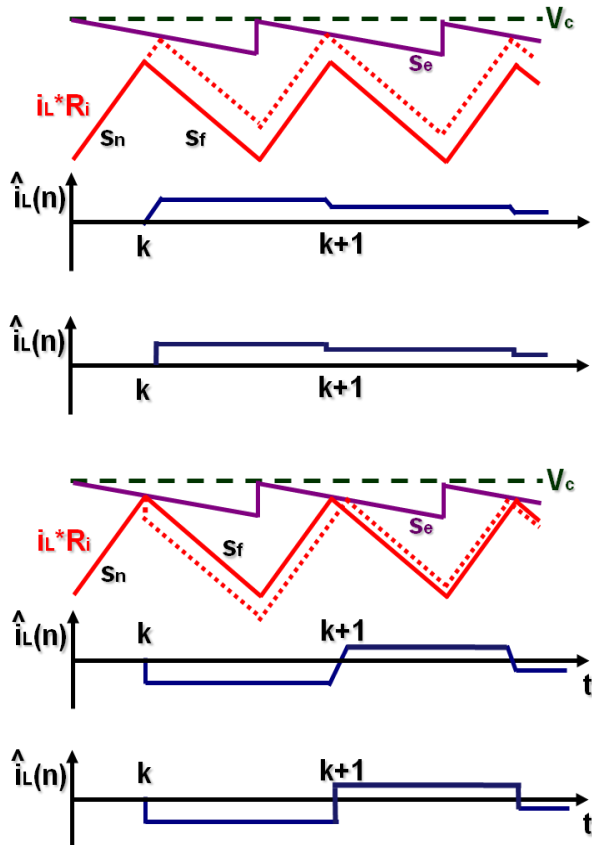
**Variable frequency,
No dynamic exhibited,
No sample and hold effect.**



*R. Ridley's model for
constant on-time control*

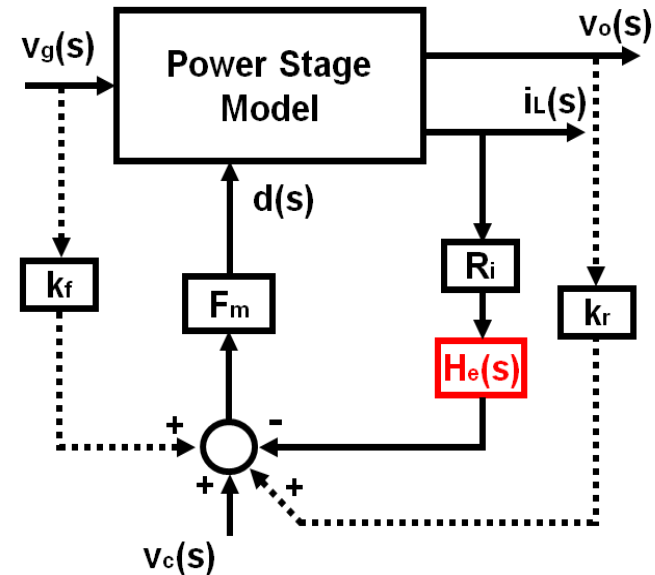
Fundamental Limitation of R. Ridley's Model

➤ Discrete-time analysis (Sample-Data Model)



$$H(z) = \frac{\hat{i}_L(z)}{\hat{v}_c(z)} = \frac{1 + \alpha}{R_i} \frac{z}{z + \alpha}$$

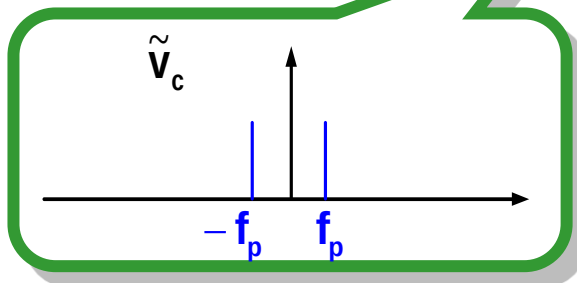
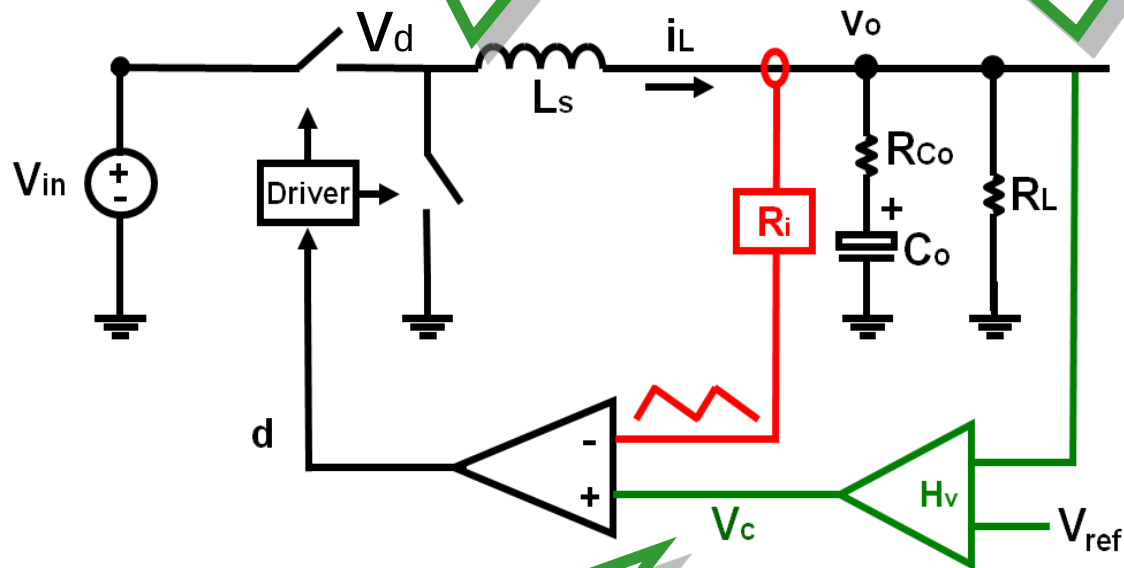
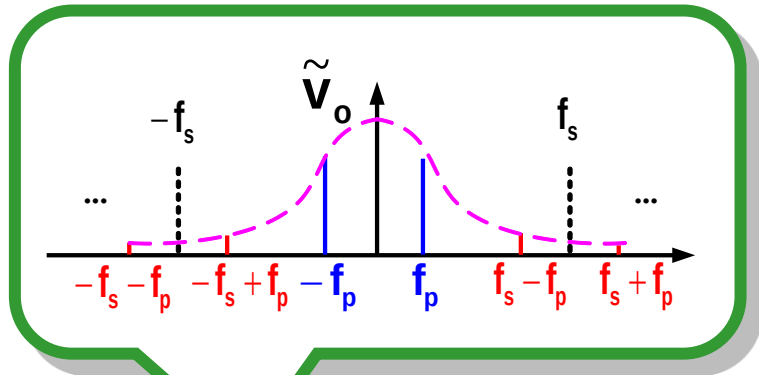
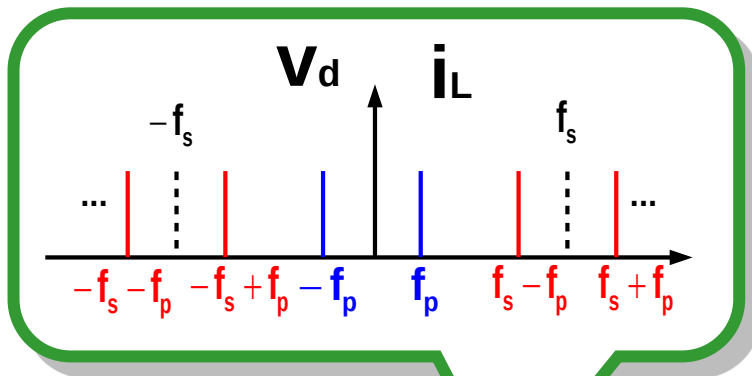
➤ R. Ridley's model



1. R. Ridley's model is based on discrete-time analysis (Sample-Data Model);
2. Discrete-time analysis is not applicable to Variable-frequency modulation;

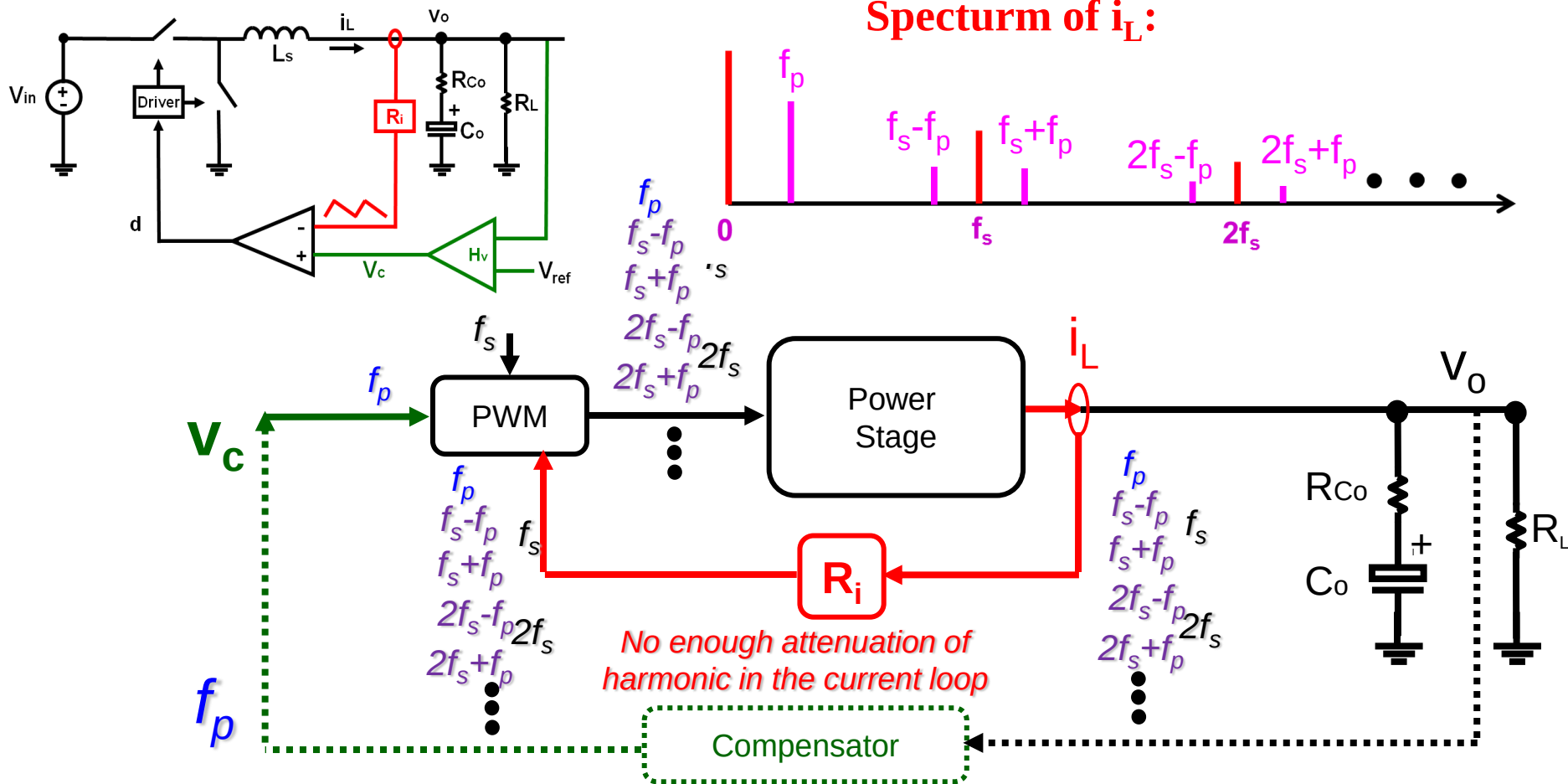
New Modeling Approach for Current-Mode Control

Frequency Spectrum in Voltage feedback



- In the voltage feedback loop, low pass filters exist both in the power stage as well as in the feedback loop, thus, higher order harmonics can be ignored.

Complexities Introduced by Current feedback

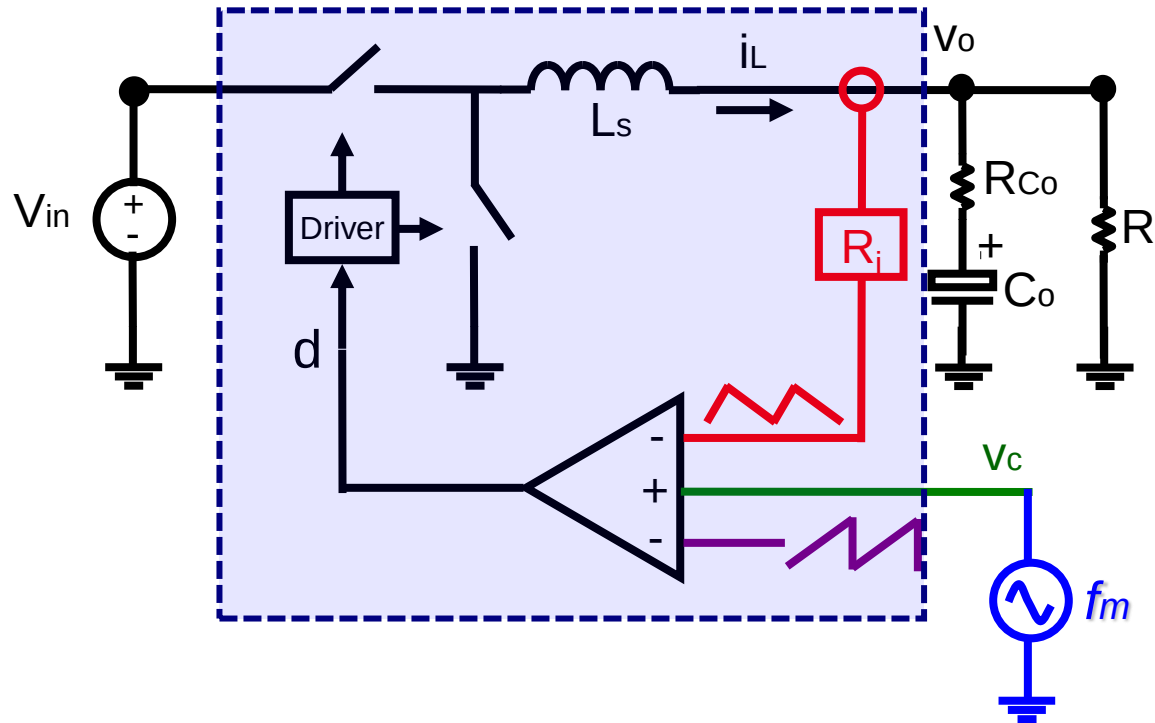


- No low-pass filter exists in the current feedback loop, thus higher order harmonics should be included in the modeling process
- **Current-loop has to be considered as one single entity**

New Modeling Strategy*

for Peak Current-Mode Control

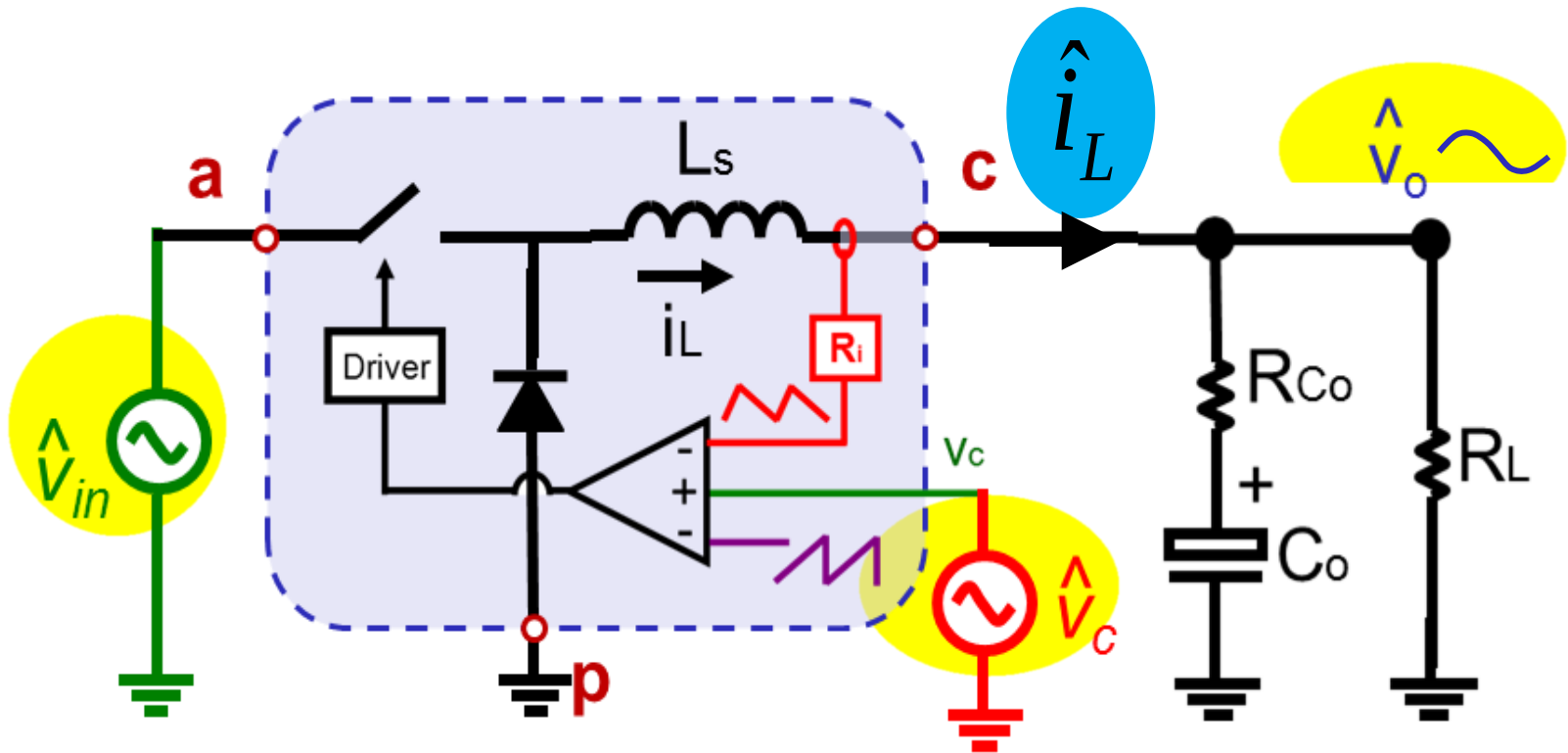
by Jian Li



- ◆ *Treat the inductor, modulator and switches as a single entity*
- ◆ *Describing function method is used*

* Jian Li; Lee, F.C.; , "New Modeling Approach and Equivalent Circuit Representation for Current-Mode Control," *Power Electronics, IEEE Transactions on* , vol.25, no.5, pp.1218-1230, May 2010

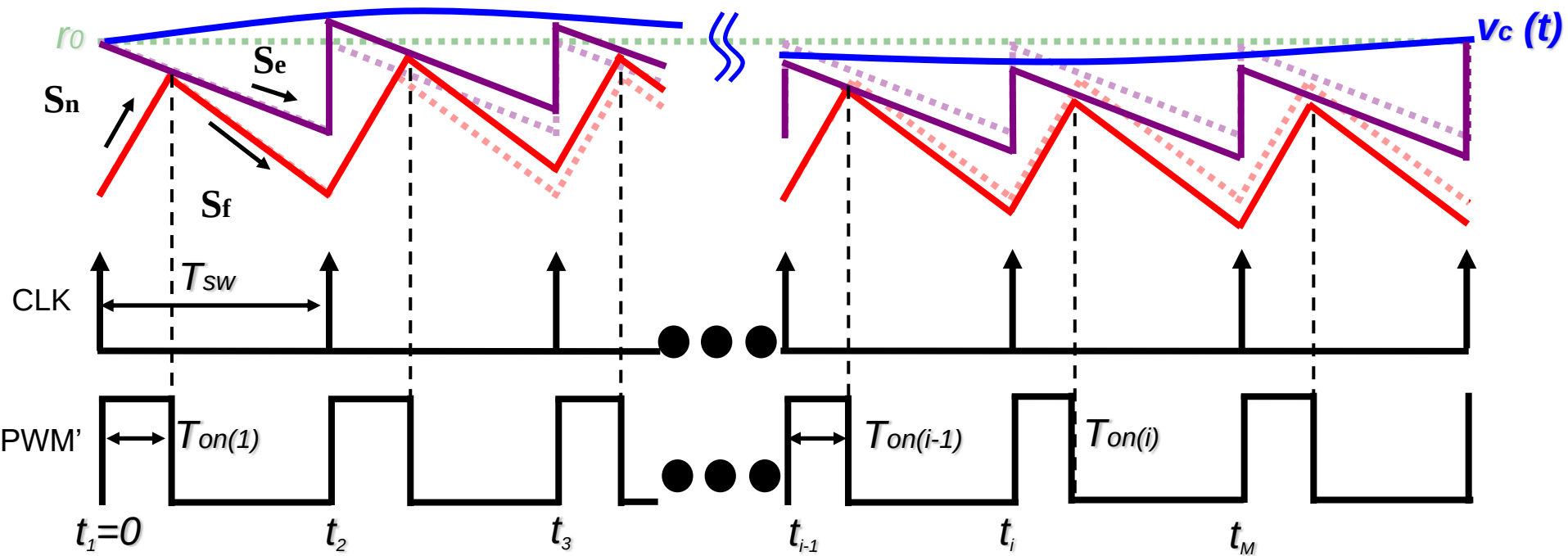
Modeling in Time Domain



$$\hat{i}_L(s) \approx K_{vc}(s)\hat{v}_c(s) + K_{vo}(s)\hat{v}_o(s) + K_{vin}(s)\hat{v}_{in}(s)$$

K_{vc} , K_{vo} and K_{vin} are derived separately

Small Signal Perturbation on v_c



➤ *Sinusoidal perturbation (small signal):*

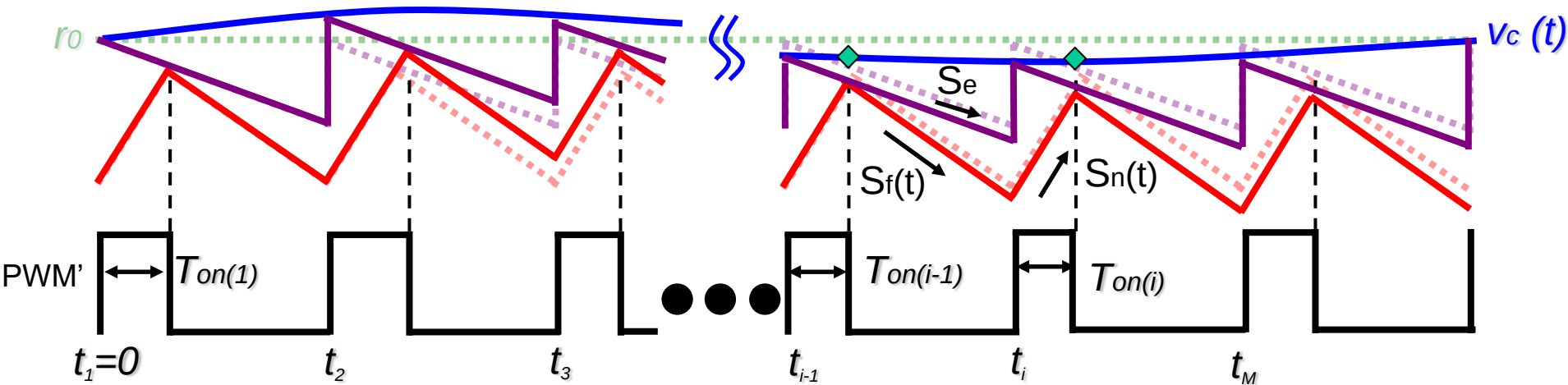
$$v_c(t) = r_0 + \hat{r} \sin(2\pi f_m \cdot t + \theta)$$

➤ On-time calculation: $T_{on(i)} = T_{on} + \Delta T_{on(i)}$

where

$$(s_e + s_n)\Delta T_{on(i)} = -\hat{v}_c(t_{i-1} + T_{on(i-1)}) + \hat{v}_c(t_i + T_{on(i)}) + (s_e - s_f)\Delta T_{on(i-1)}$$

Calculation of Inductor Current



➤ Calculation of Duty cycle waveform

$$d(t) \Big|_{0 \leq t \leq NT_m} = \sum_{i=1}^M [u(t - t_i) - u(t - t_i - T_{on(i)})]$$

Where, $u(t)=1$, when $t>0$.

➤ Calculation of inductor current:

$$i_L(t) = \int_0^t \left\{ \frac{V_{in} - V_o}{L_s} d(t) - \frac{V_o}{L_s} [1 - d(t)] \right\} \cdot dt + I_{L0}$$

$$= \int_0^t \left\{ \frac{V_{in}}{L_s} d(t) - \frac{1}{L_s} V_o \right\} \cdot dt + I_{L0}$$

Where, I_{L0} is the initial value of the inductor current.

Fourier Analysis of Inductor Current

➤ Calculation of the Fourier coefficient at f_m for inductor current $i_L(t)$ and $v_c(t)$:

For $\hat{i}_{L(f_m)} \rightarrow c_{m(i_L)} = j \frac{2f_m}{N} \int_0^{NT_m} i_L(t) \cdot e^{-j2\pi f_m t} dt$

↓

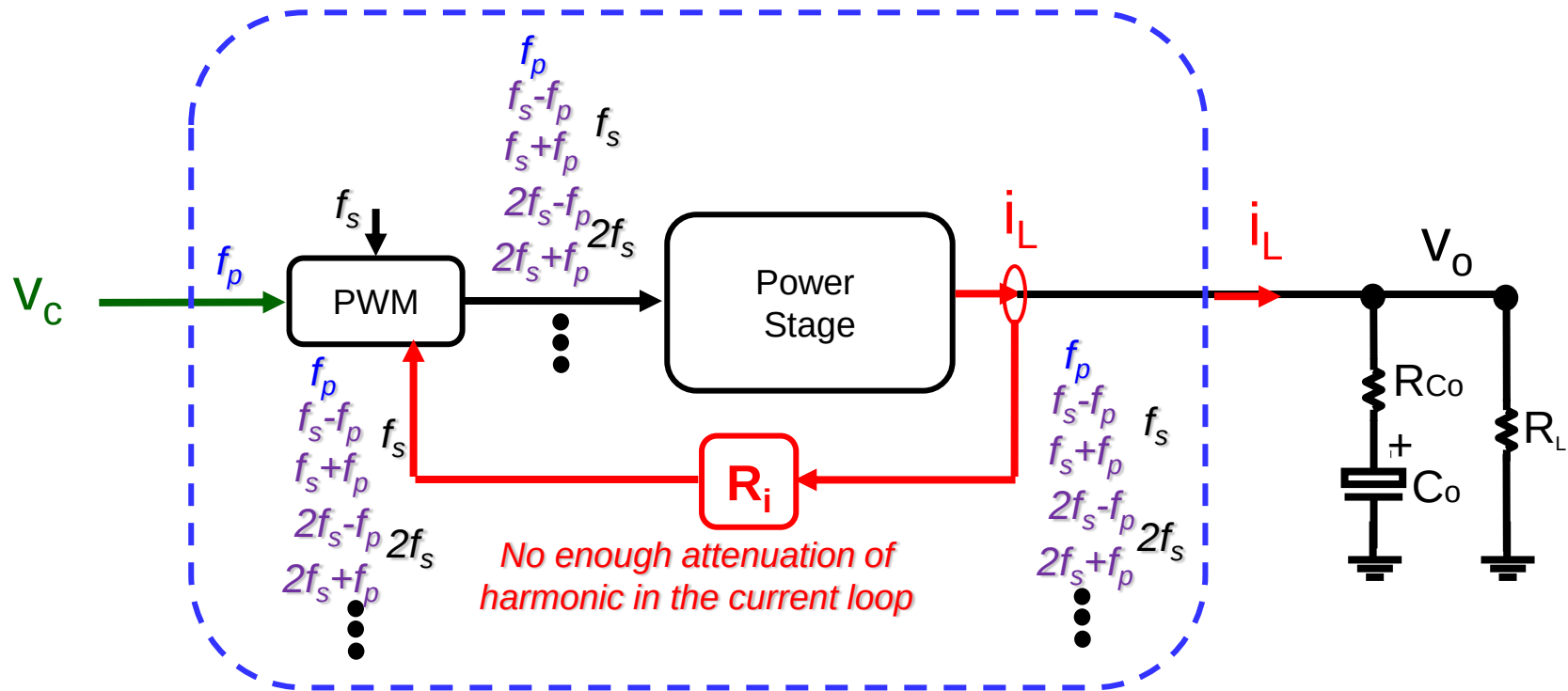
$$c_{m(i_L)} = \hat{r} \cdot e^{-j\theta} \cdot \frac{f_s(1 - e^{-j2\pi f_m T_{sw}})}{(s_n + s_e) + (s_f - s_e)e^{-j2\pi f_m T_{sw}}} \frac{V_{in}}{L_s \cdot j2\pi f_m}$$

For $\hat{v}_{c(f_m)} \rightarrow c_{m(v_c)} = \hat{r}e^{-j\theta}$

➤ Calculation of describing function:

$$\frac{\hat{i}_{L(f_m)}}{\hat{v}_{c(f_m)}} = \frac{c_{m(i_L)}}{\hat{r}e^{-j\theta}} = \frac{f_s(1 - e^{-j2\pi f_m T_{sw}})}{(s_n + s_e) + (s_f - s_e)e^{-j2\pi f_m T_{sw}}} \frac{V_{in}}{L_s \cdot j2\pi f_m}$$

Control-to- i_L Transfer Function

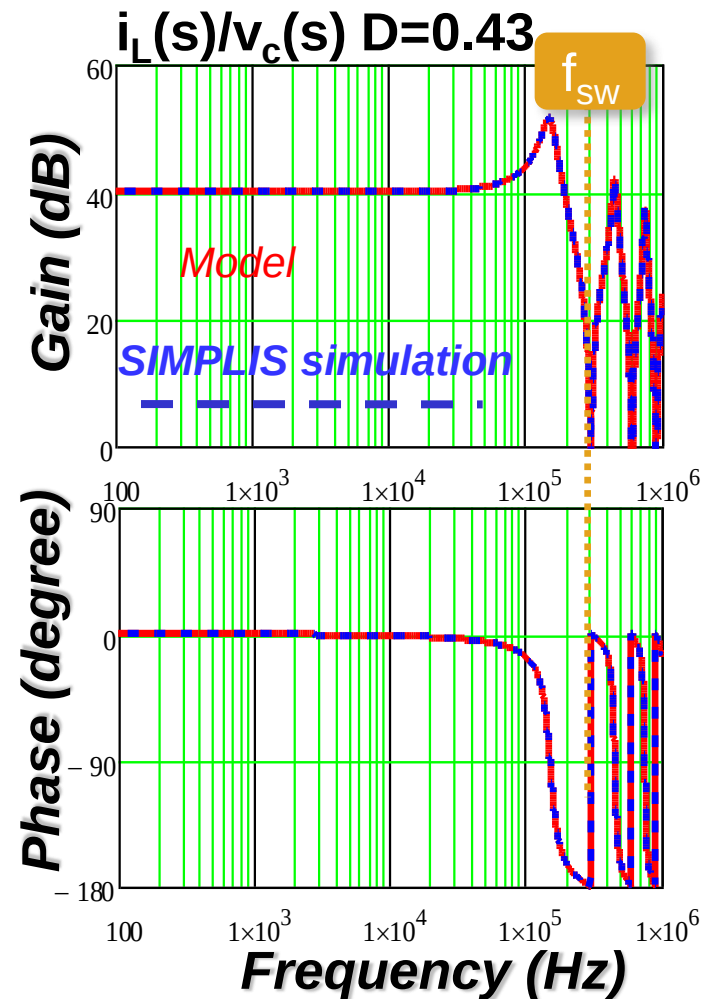
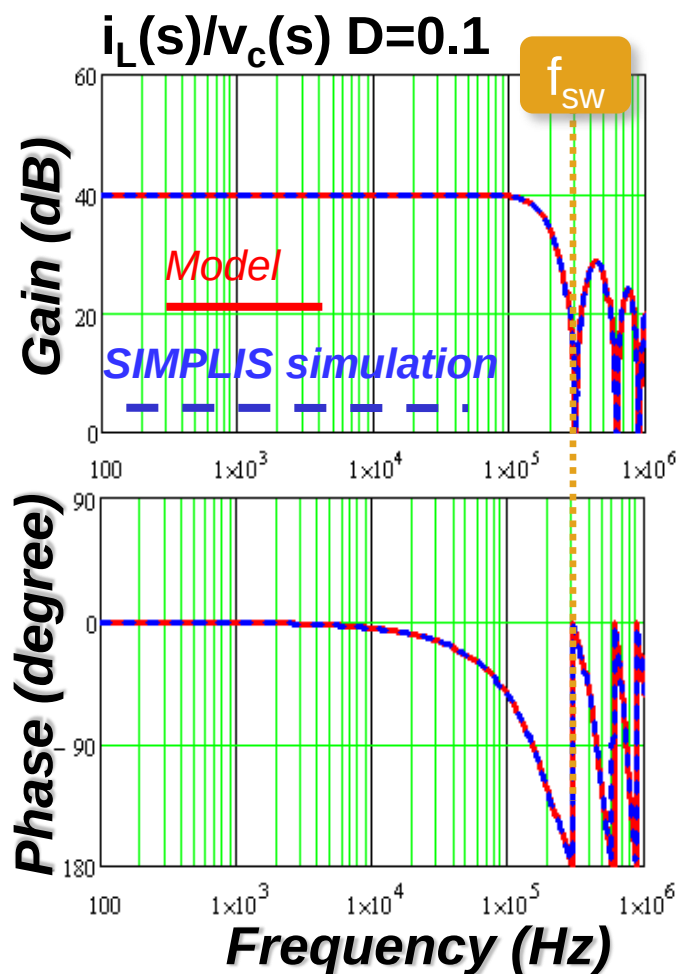


$$\frac{\hat{i}_L(s)}{\hat{v}_c(s)} = \frac{f_s(1 - e^{-sT_{sw}})}{(s_n + s_e) + (s_f - s_e)e^{-sT_{sw}}} \frac{V_{in}}{L_s s}$$

$$s = j2\pi f_m$$

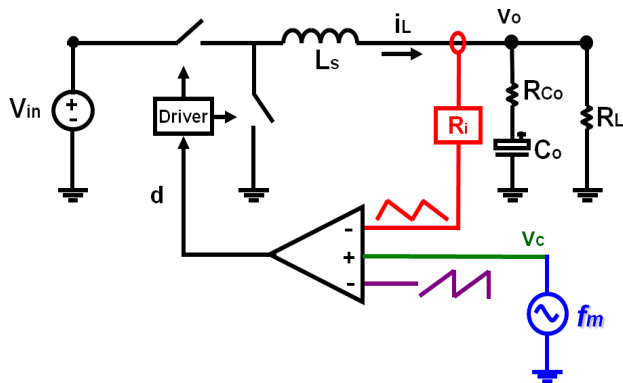
Control-to- i_L Transfer Function

$$\frac{\hat{i}_L(s)}{\hat{v}_c(s)} = \frac{f_s}{(s_n + s_e) + (s_f - s_e)e^{-sT_{sw}}} \frac{V_{in}}{L_s s} (1 - e^{-sT_{sw}})$$



Control-to- i_L Transfer Function

➤ New modeling result



$$\frac{\hat{i}_L(s)}{\hat{v}_c(s)} = \frac{f_s(1 - e^{-sT_{sw}})}{(s_n + s_e) + (s_f - s_e)e^{-sT_{sw}}} \frac{V_{in}}{L_s s}$$

$$\frac{\hat{i}_L(s)}{\hat{v}_c(s)} = \frac{1}{R_i} \frac{1 + \alpha}{sT_{sw}} \frac{e^{sT_{sw}} - 1}{e^{sT_{sw}} + \alpha}$$

where, $\alpha = (s_f - s_e)/(s_n + s_e)$

➤ Sample-Data Model result

$$H(z) = \frac{\hat{i}_L(z)}{\hat{v}_c(z)} = \frac{1 + \alpha}{R_i} \frac{z}{z + \alpha}$$

to

➤ Continuous

$$\frac{\hat{i}_L(s)}{\hat{v}_c(s)} = \frac{1}{R_i} \frac{1 + \alpha}{sT_{sw}} \frac{e^{sT_{sw}} - 1}{e^{sT_{sw}} + \alpha}$$

The result is the same as previous Sample-Data analysis

Approximation

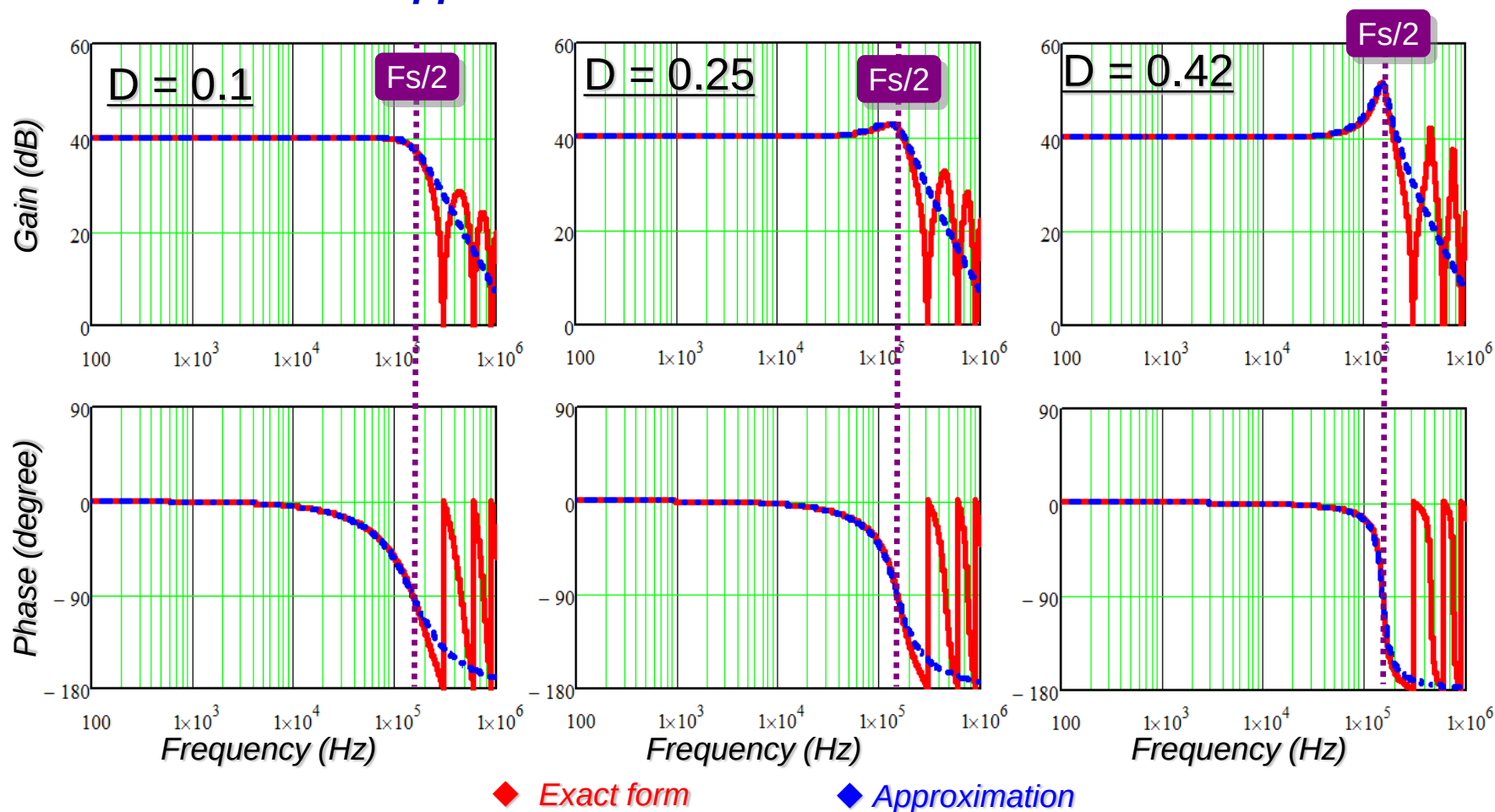
$$\frac{\hat{i}_L(s)}{\hat{v}_c(s)} = \frac{1}{R_i} \frac{1+\alpha}{sT_{sw}} \frac{e^{sT_{sw}} - 1}{e^{sT_{sw}} + \alpha}$$

$$\frac{\hat{i}_L(s)}{\hat{v}_c(s)} \approx \frac{1}{R_i} \frac{1}{1 + \frac{s}{Q_2\omega_2} + \frac{s^2}{\omega_2^2}}$$

where,

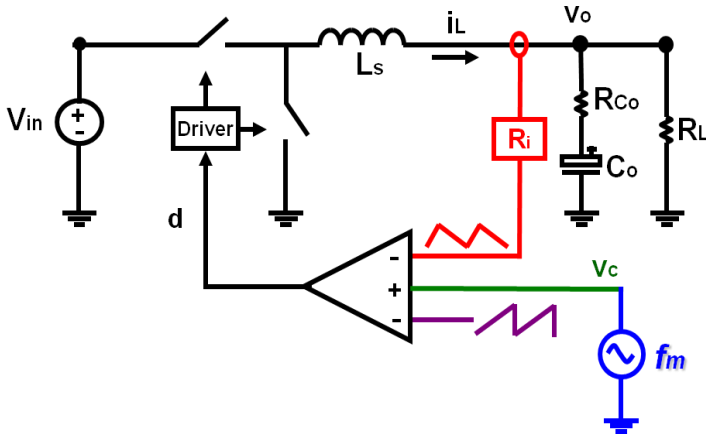
$$\omega_2 = \frac{\pi}{T_{sw}} \quad Q_2 = \frac{1}{\pi \left(\frac{s_n + s_e}{s_n + s_f} - 0.5 \right)}$$

Pade approximation



Fourier Analysis of Output Voltage

- Calculation of the Fourier coefficient at f_m for output voltage $v_o(t)$:

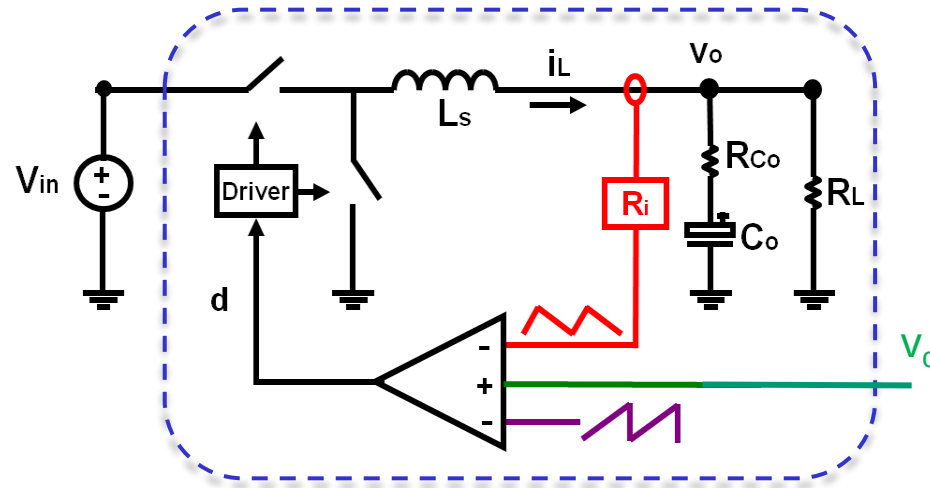


$$C_{m(v_o)} = C_{m(i_L)} \frac{R_L (R_{Co} C_o \cdot j2\pi f_m + 1)}{(R_L + R_{Co}) C_o \cdot j2\pi f_m + 1}$$

- Calculation of describing function:

$$\frac{\hat{V}_{o(f_m)}}{\hat{V}_{c(f_m)}} = \frac{f_{sw} \cdot [1 - e^{-j(2\pi f_m \cdot T_{sw})}]}{(s_n + s_e) + (s_f - s_e) e^{-j \cdot 2\pi f_m \cdot T_{sw}}} \cdot \frac{V_{in}}{j2\pi f_m \cdot L_s} \cdot \frac{R_L (R_{Co} C_o \cdot j2\pi f_m + 1)}{(R_L + R_{Co}) C_o \cdot j2\pi f_m + 1}$$

Simplified Control-to- v_o Transfer Function



$$\frac{\hat{v}_o(s)}{\hat{v}_c(s)} \approx K'_p \frac{R_{Co}C_o s + 1}{s/\omega'_p + 1} \frac{1}{1 + \frac{s}{Q_2\omega_2} + \frac{s^2}{\omega_2^2}}$$

where, $K'_p = \frac{R_L}{R_i}$ $\omega'_p = \frac{1}{(R_L + R_{Co})C_o}$ $\omega_2 = \frac{\pi}{T_{sw}}$ $Q_2 = \frac{1}{\pi(M_c D' - 0.5)}$

Peak Current Mode Control

w/o External Ramp

➤ Peak Current Mode Control

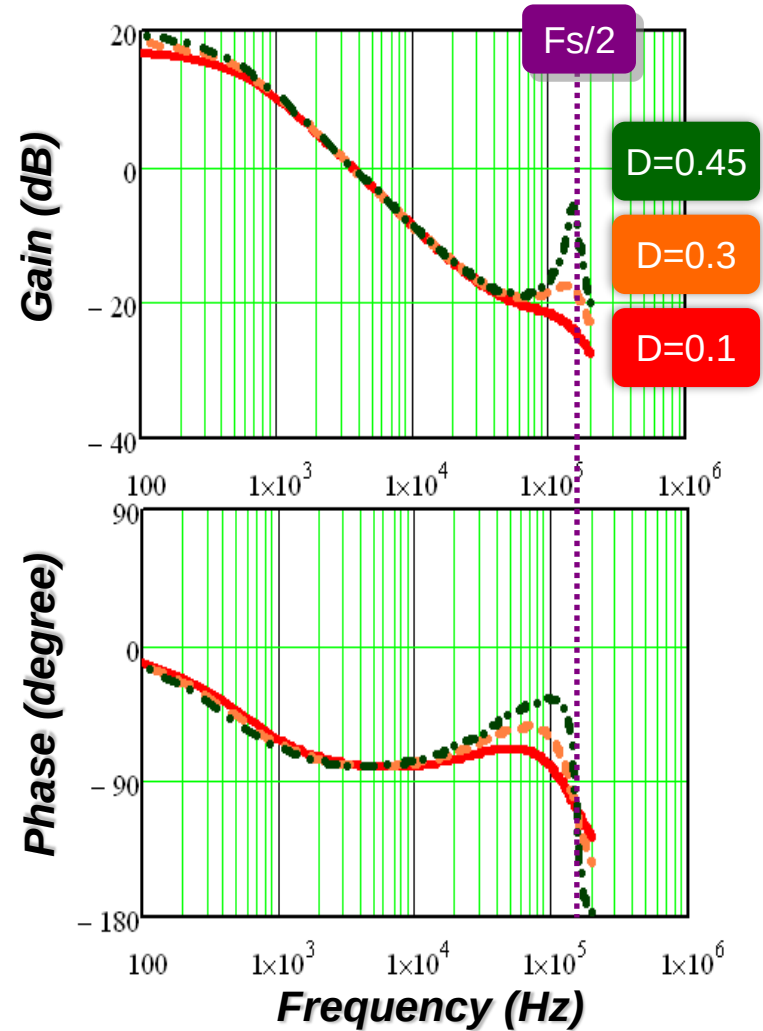
$$\frac{v_o(s)}{v_c(s)} \approx K'_p \frac{R_{Co}C_o s + 1}{s/\omega'_p + 1} \frac{1}{1 + \frac{s}{Q_2\omega_2} + \frac{s^2}{\omega_2^2}}$$

where

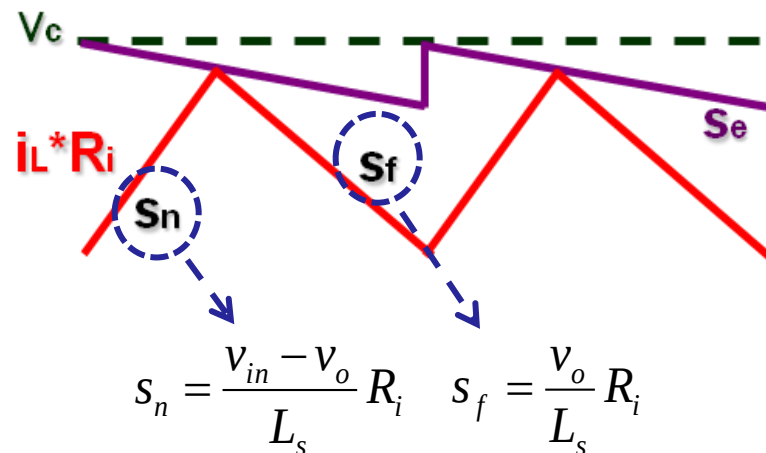
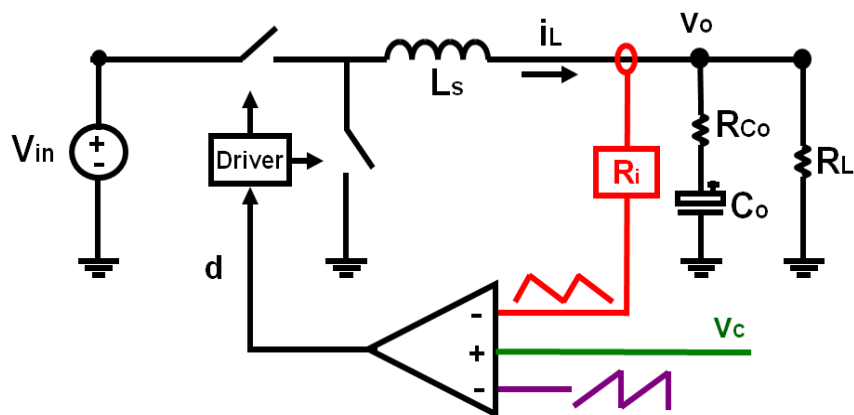
$$K'_p = \frac{R_L}{R_i} \quad \omega'_p = \frac{1}{(R_L + R_{Co})C_o}$$

$$\omega_2 = \frac{\pi}{T_{sw}} \quad Q_2 = \frac{1}{\pi(M_c D' - 0.5)}$$

Control-to-output Transfer Function



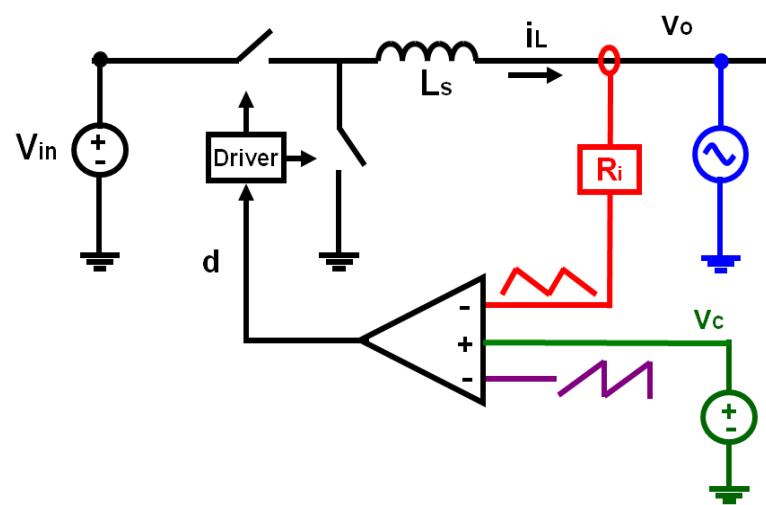
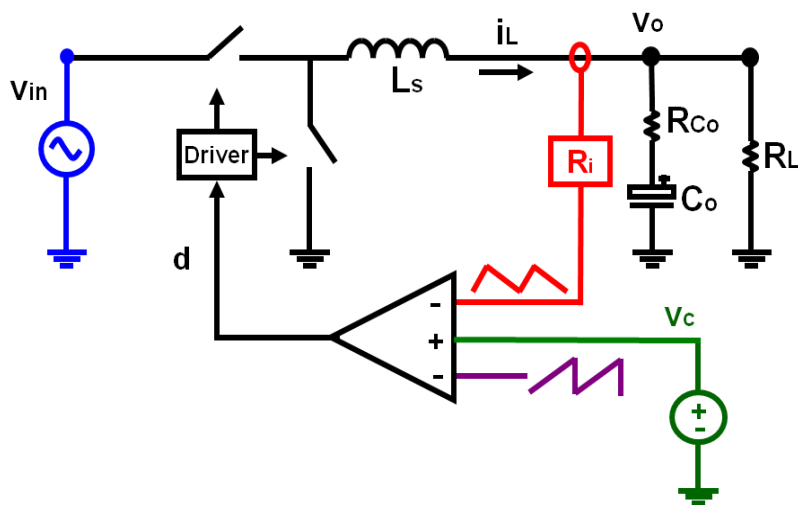
Inductor Current Slope Variation



➤ Additional feed-forward and feedback gain should be considered

◆ The influence of the input voltage

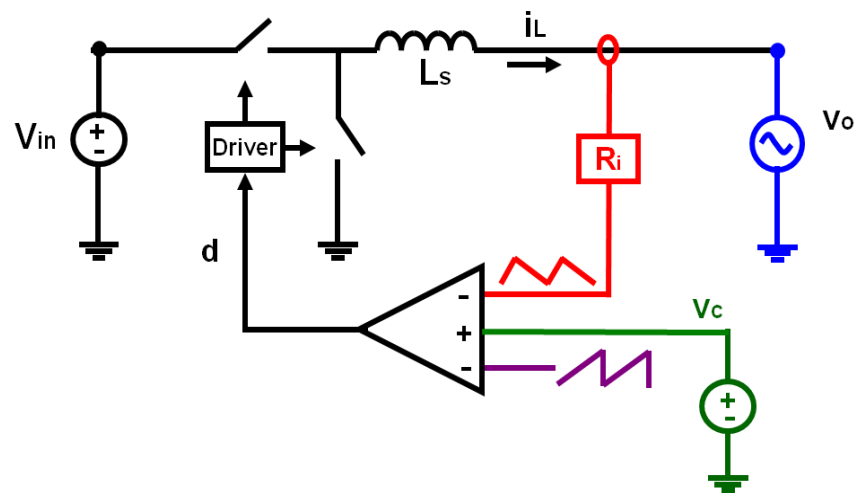
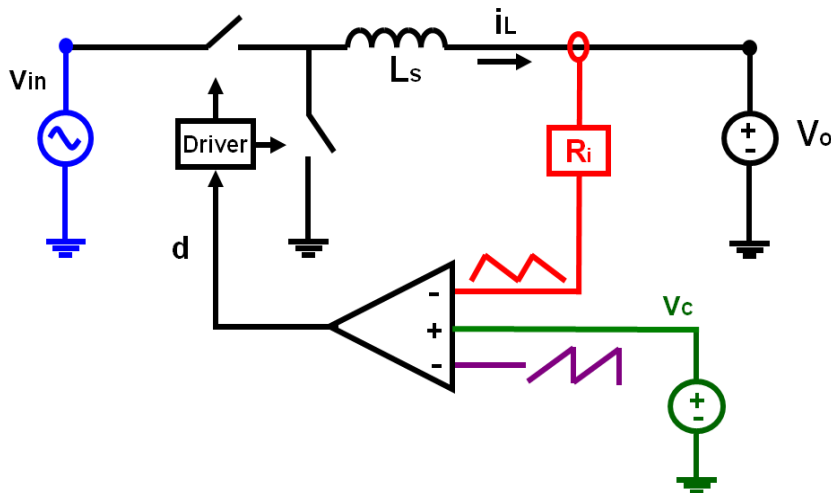
◆ The influence of the output voltage



Similar Derivation Process

◆ The influence of the input voltage

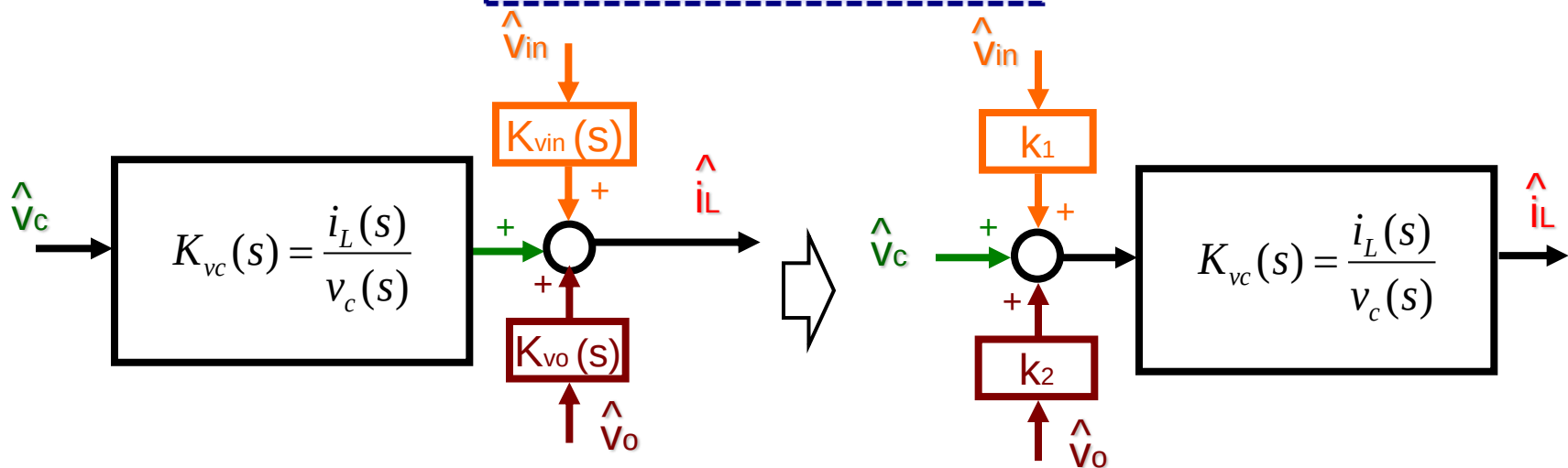
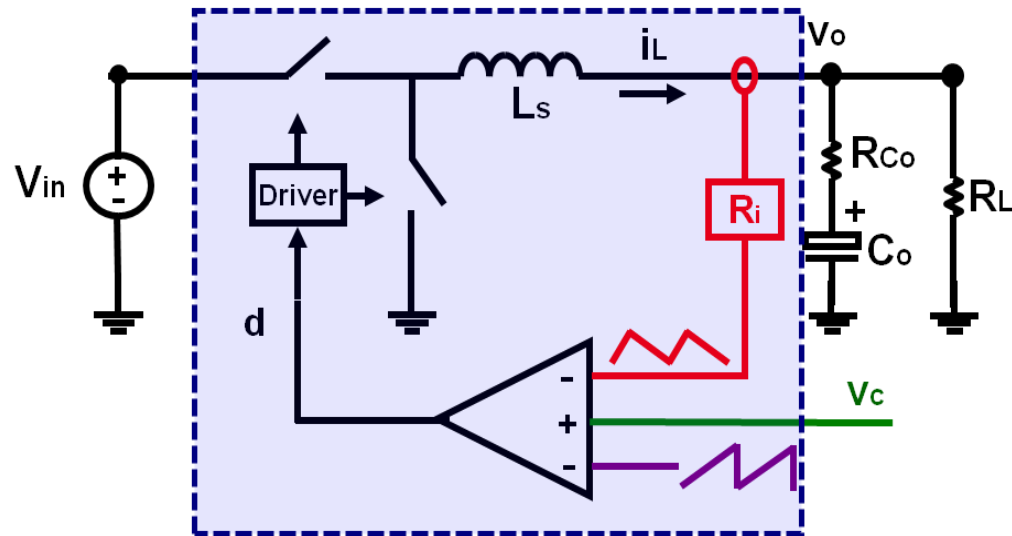
◆ The influence of the output voltage



$$K_{vin}(s) = \frac{i_L(s)}{v_{in}(s)} = \frac{1}{L_s s} \left[-\frac{f_s}{(s_n + s_e) + (s_f - s_e)e^{-sT_{sw}}} \frac{(1 - e^{-sT_{on}})}{s \cdot L_s / R_i} \cdot V_{in} + D \right]$$

$$K_{vo}(s) = \frac{i_L(s)}{v_o(s)} = \frac{1}{L_s s} \cdot \left[\frac{f_s (1 - e^{-sT_{sw}})}{(s_n + s_e) + (s_f - s_e)e^{-sT_{sw}}} \cdot \frac{1}{s \cdot L_s / R_i} \cdot V_{in} - 1 \right]$$

Complete Model

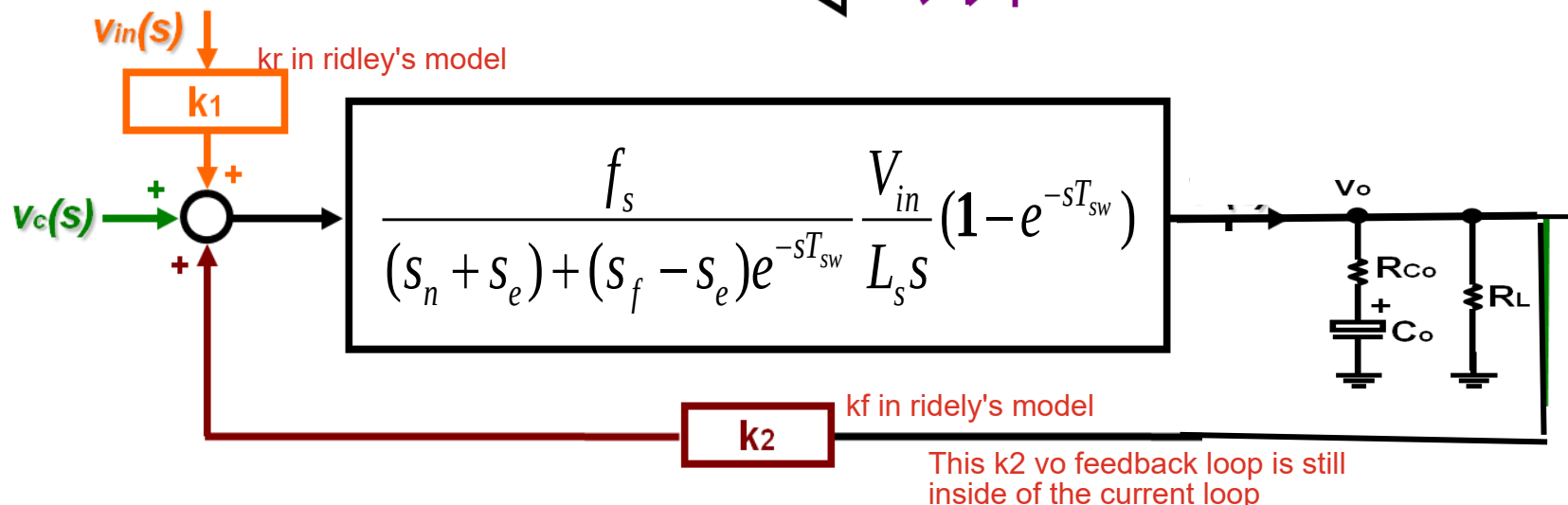
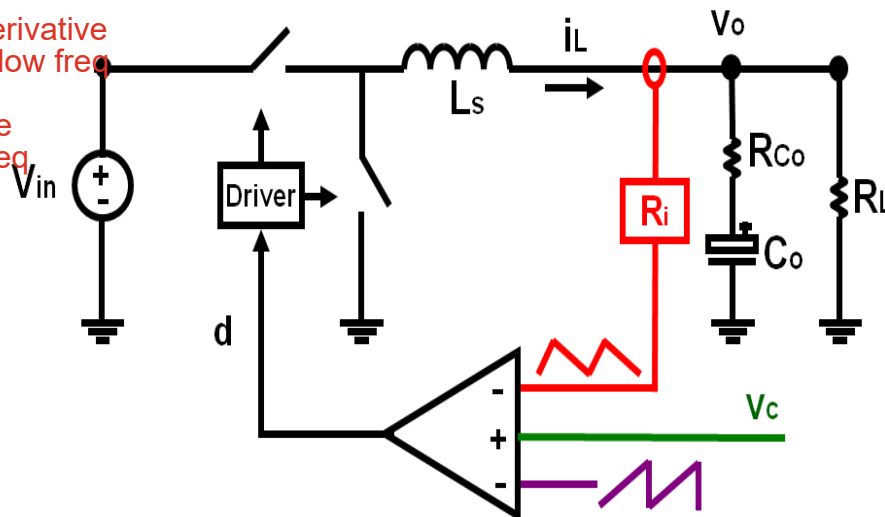


$$k_1(s) = \frac{K_{vin}(s)}{K_{vc}(s)} \approx \frac{DR_i}{L_s} \left[\frac{1}{Q_2 \omega_2} - \frac{T_{off}}{2} \right] \doteq k_1$$

$$k_2(s) = \frac{K_{vo}(s)}{K_{vc}(s)} \approx -\frac{R_i}{L_s Q_2 \omega_2} \doteq k_2$$

Complete Model

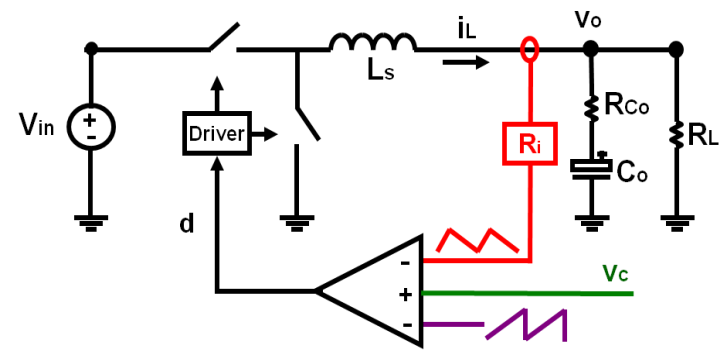
k₁ and k₂ are derived by describing function as partial derivative the relationship of high freq to low freq components are maintained; but Ridley's model use average model in kr, kf process, high freq components are rule out



$$k_1(s) = \frac{K_{vin}(s)}{K_{vc}(s)} \approx \frac{DR_i}{L_s} \left[\frac{1}{Q_2 \omega_2} - \frac{T_{off}}{2} \right] \doteq k_1$$

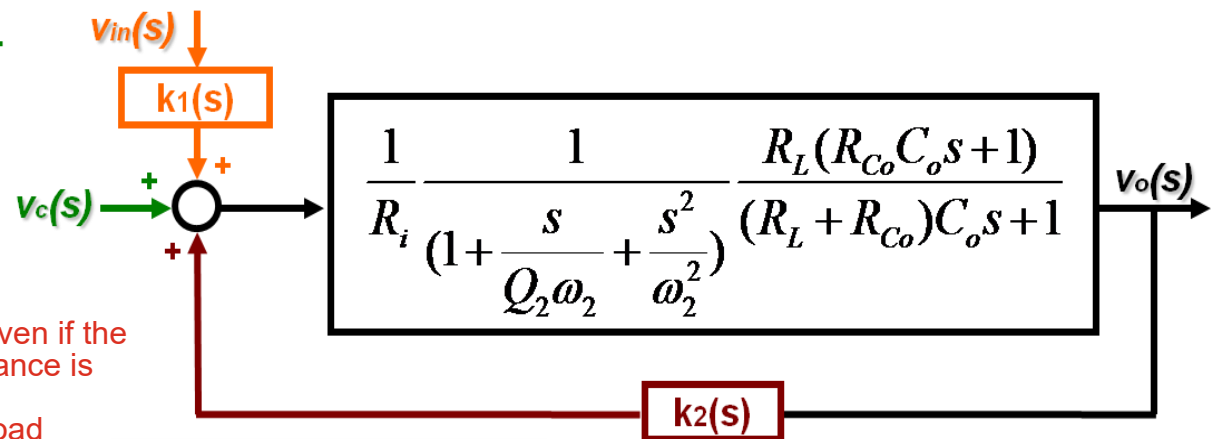
$$k_2(s) = \frac{K_{vo}(s)}{K_{vc}(s)} \approx -\frac{R_i}{L_s Q_2 \omega_2} \doteq k_2$$

Complete Model with Approximation



$$k_1(s) \approx \frac{DR_i}{L_s} \left[\frac{1}{Q_2 \omega_2} - \frac{T_{off}}{2} \right]$$

$$k_2(s) \approx -\frac{R_i}{L_s Q_2 \omega_2}$$



when ask for $Z_o(s)$ equation, $v_{in}(s)$ is 0, even if the $k_2(s)$ is different o $k_f(s)$, the output impedance is almost same.

$Z_o(s) = I_L(\text{current source}) // (1/sC + esr) // R_{load}$
in low freq, the I_L is infinite large, no different; in high freq, the esr dominant the impedance, basically same

with COT control:

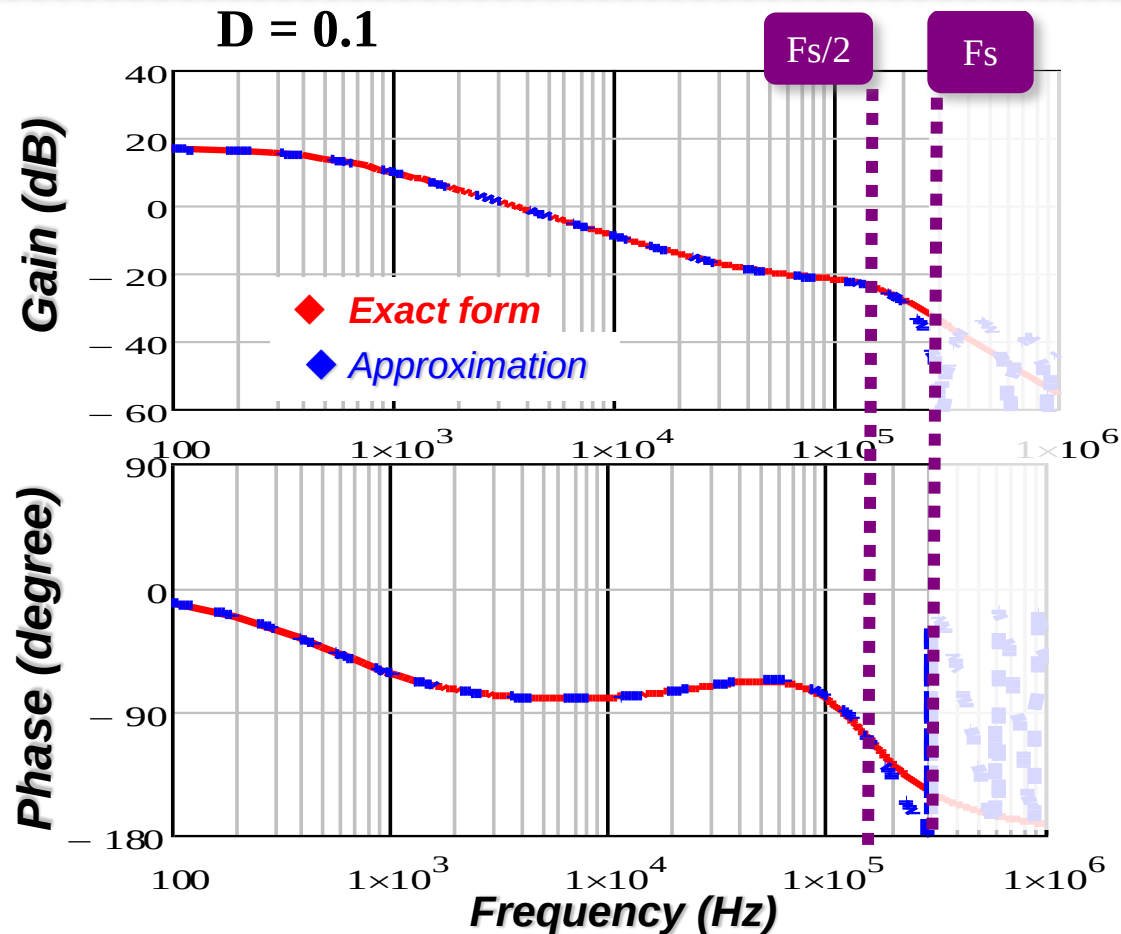
1. double pole gets push to high freq, good for design
2. current loop is always stable
3. the wp is no longer moving

$$\frac{v_o(s)}{v_c(s)} \approx K_p \frac{R_{Co} C_o s + 1}{s / \omega_p + 1} \frac{1}{1 + \frac{s}{Q_2 \omega_2} + \frac{s^2}{\omega_2^2}}$$

where, $K_p = \frac{R_L}{R_i(1 - k_2 / R_i \cdot R_L)}$ $\omega_p = \frac{1 - R_L k_2 / R_i}{(R_L + R_{Co})C_o - R_L R_{Co} C_o k_2 / R_i}$ $\omega_2 = \frac{\pi}{T_{sw}}$ $Q_2 = \frac{1}{\pi \left(\frac{s_n + s_e}{s_n + s_f} - 0.5 \right)}$

Simplified Model

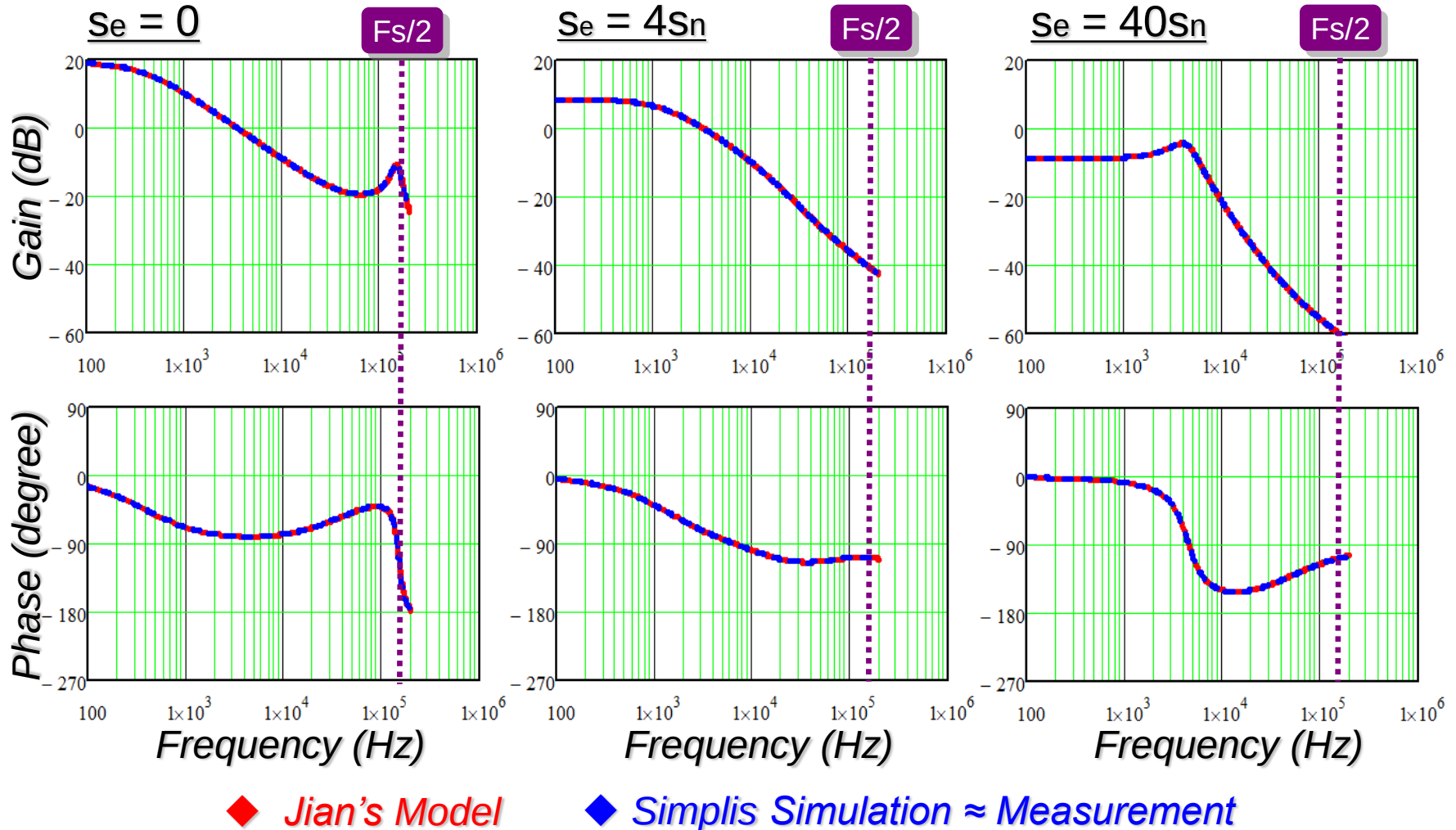
$$\frac{v_o(s)}{v_c(s)} \approx K_p \frac{R_{Co} C_o s + 1}{s / \omega_p + 1} \frac{1}{1 + \frac{s}{Q_2 \omega_2} + \frac{s^2}{\omega_2^2}} \quad \text{where,} \quad \omega_2 = \frac{\pi}{T_{sw}} \quad Q_2 = \frac{1}{\pi \left(\frac{s_n + s_e}{s_n + s_f} - 0.5 \right)}$$



Model accurate at $1/2 f_s$, and useful up to f_s

Simulation Verification for Peak Current-Mode Control

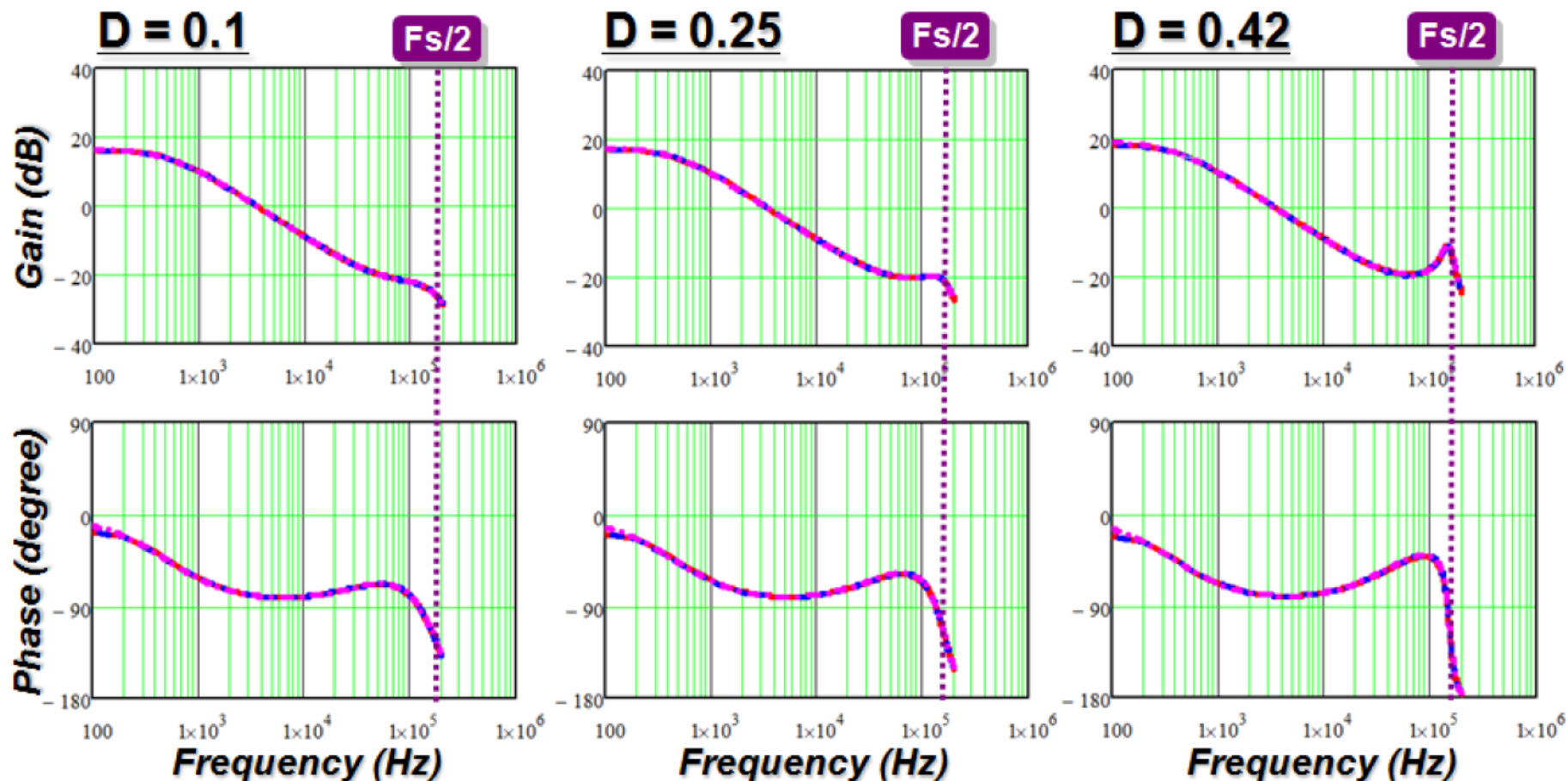
Control-to-output Transfer Function ($D = 0.42$)



Model Comparison

Control-to-output transfer function of Peak Current Mode Control

$S_e = 0$

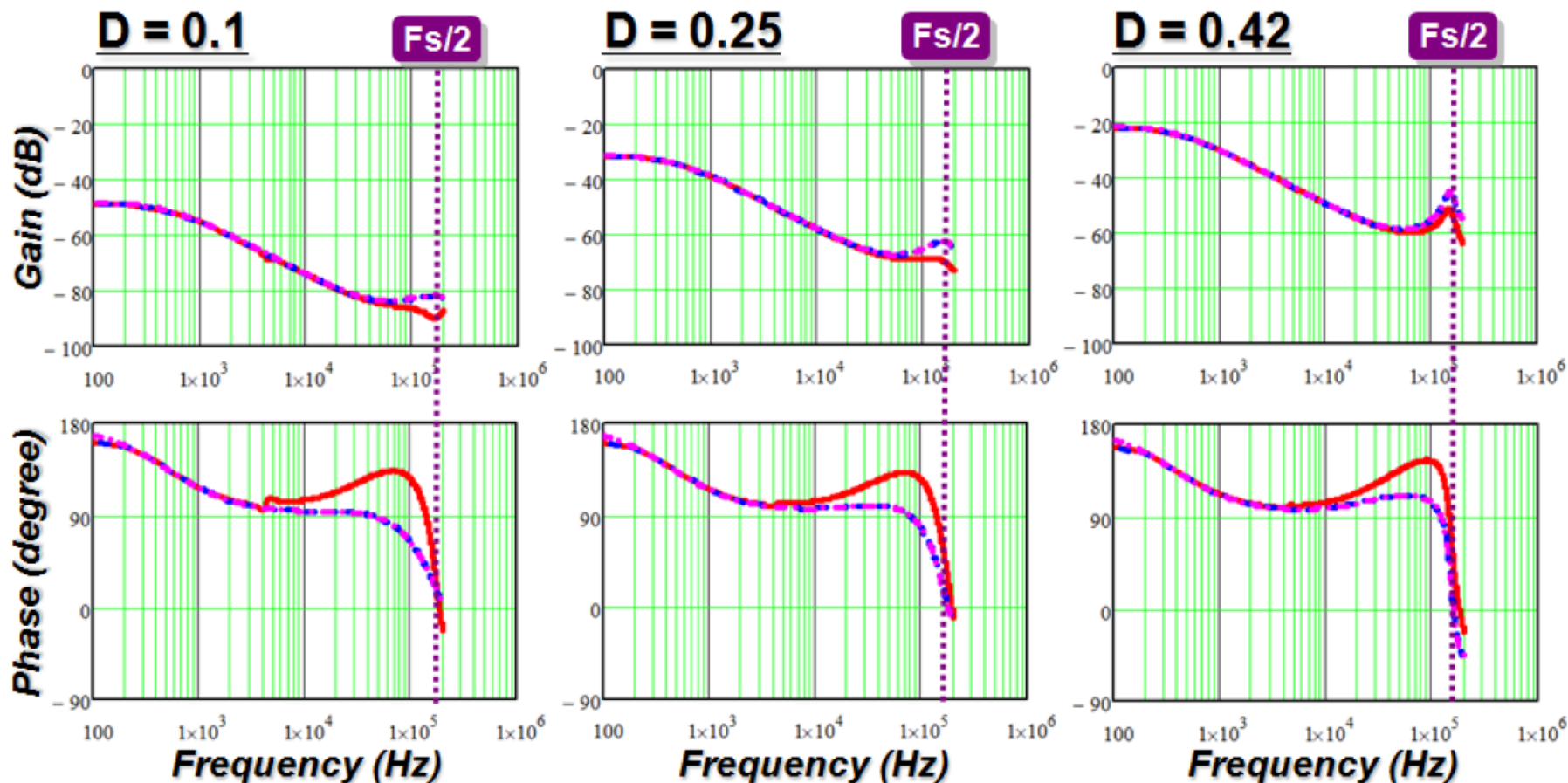


◆ *R. Ridley's Model* ◆ *Jian's Model* ◆ *Simplis Simulation $\approx$ Measurement*

Model Comparison

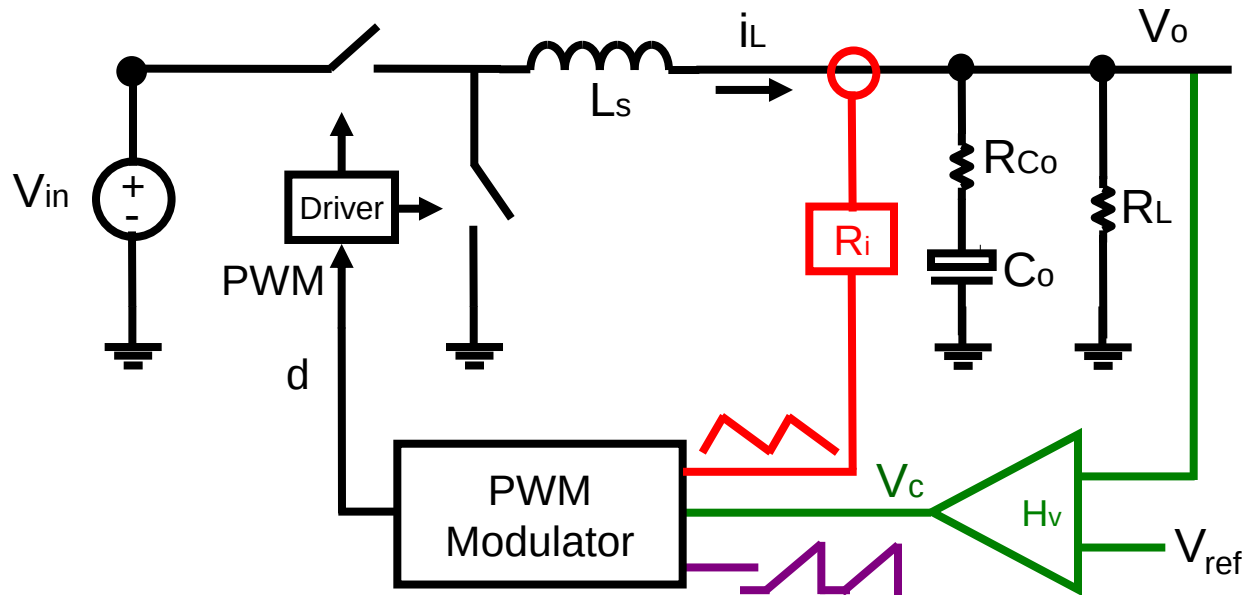
V_{in} -to- V_o transfer function of Peak Current Mode Control

Gvd has no significant different but Gvv has variation, reason (see p33): Ridley model is average model, high freq components in kf,kr are ignored Se = 0



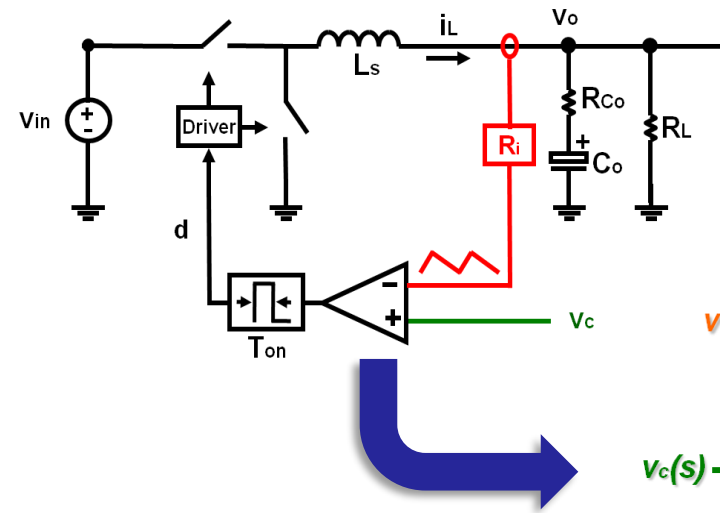
◆ R. Ridley's Model ◆ Jian's Model ◆ Simplis Simulation $\approx$ Measurement

Extension to Other Current Mode Controls



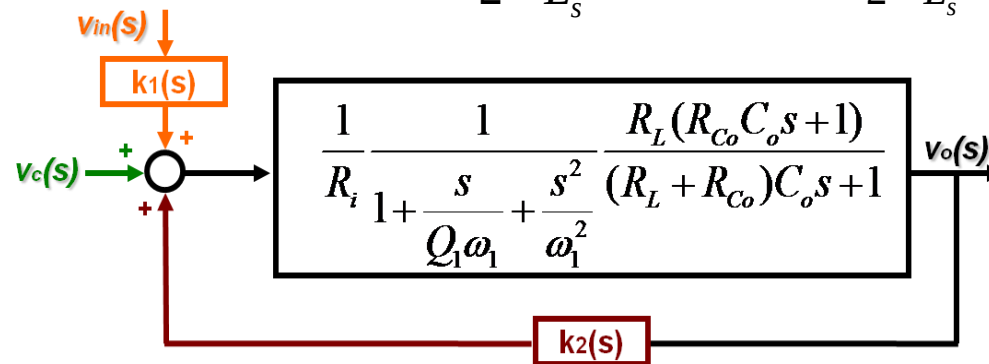
- ◆ *Valley Current-Mode Control*
- ◆ *Constant On-time Control*
- ◆ *Constant Off-time Control*
- ◆ *Charge Control*

Model Extension to Constant on-time control



$$k_1(s) = \frac{1}{2} \frac{T_{on} R_i}{L_s}$$

$$k_2(s) = -\frac{1}{2} \frac{T_{on} R_i}{L_s} = -\frac{R_i}{L_s Q_1 \omega_1}$$



$$\frac{v_o(s)}{v_c(s)} \approx K_c \frac{R_{Co} C_o s + 1}{s / \omega_a + 1} \frac{1}{1 + \frac{s}{Q_1 \omega_1} + \frac{s^2}{\omega_1^2}}$$

$$\text{where, } K_c = \frac{R_L}{R_i (1 - k_2 / R_i \cdot R_L)} \quad \omega_a = \frac{1 - R_L k_2 / R_i}{(R_L + R_{Co}) C_o - R_L R_{Co} C_o k_2 / R_i} \quad \omega_1 = \frac{\pi}{T_{on}} \quad Q_1 = \frac{2}{\pi}$$

Simulation Verification for Constant On-time Control

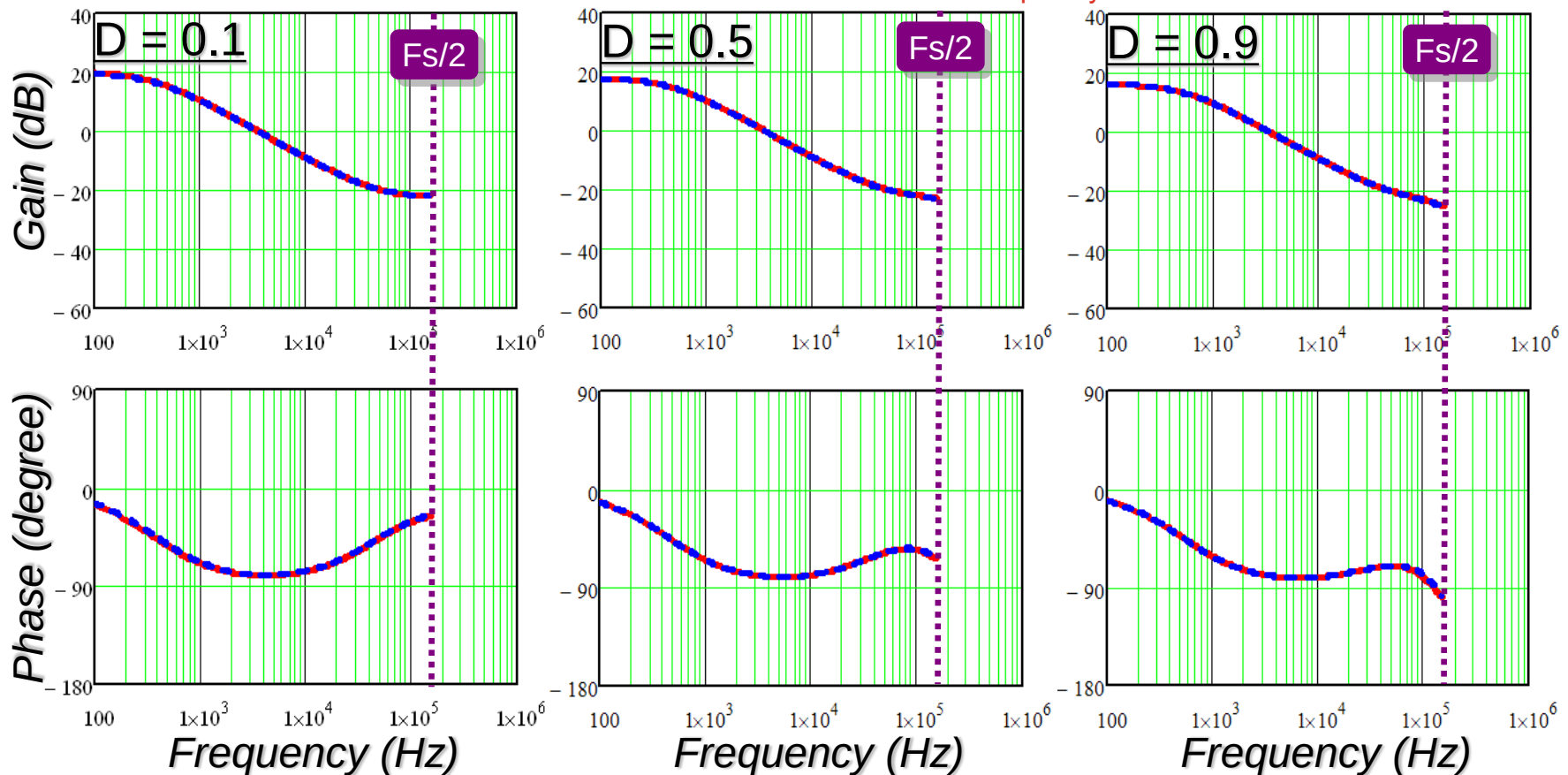
Control-to-output Transfer Function

$$\frac{v_o(s)}{v_c(s)} \approx K_c \frac{R_{Co}C_o s + 1}{s/\omega_a + 1} \frac{1}{1 + \frac{s}{Q_1\omega_1} + \frac{s^2}{\omega_1^2}}$$

$$\omega_1 = \frac{\pi}{T_{on}}$$

$$Q_1 = \frac{2}{\pi}$$

keep T_{on} , larger D bring ω_1 to lower frequency



◆ *Proposed Model* ◆ *Simplis Simulation $\approx$ Measurement*

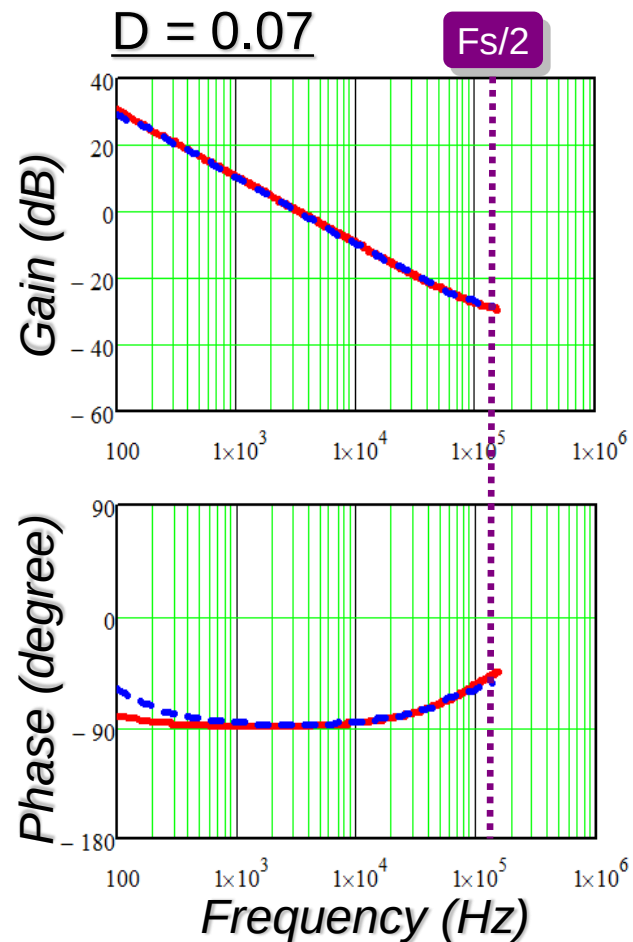
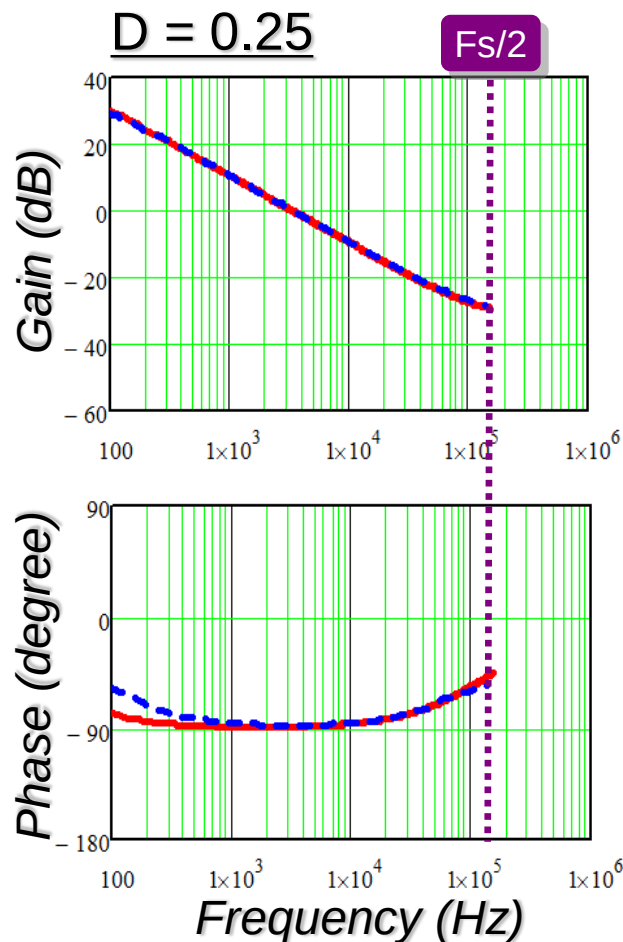
Experiment Verification

for Constant On-time Control

➤ Based on the demo-board from 

*LTC3812 Const. On-time Controller $V_o = 3.3V$, $T_{on} = 0.26\mu s$, $L_s = 3.1\mu F$, $C_o = 2 \times 150\mu F$, and $R_{Co} = 9/2m\Omega$.

Control-to-output Transfer Function



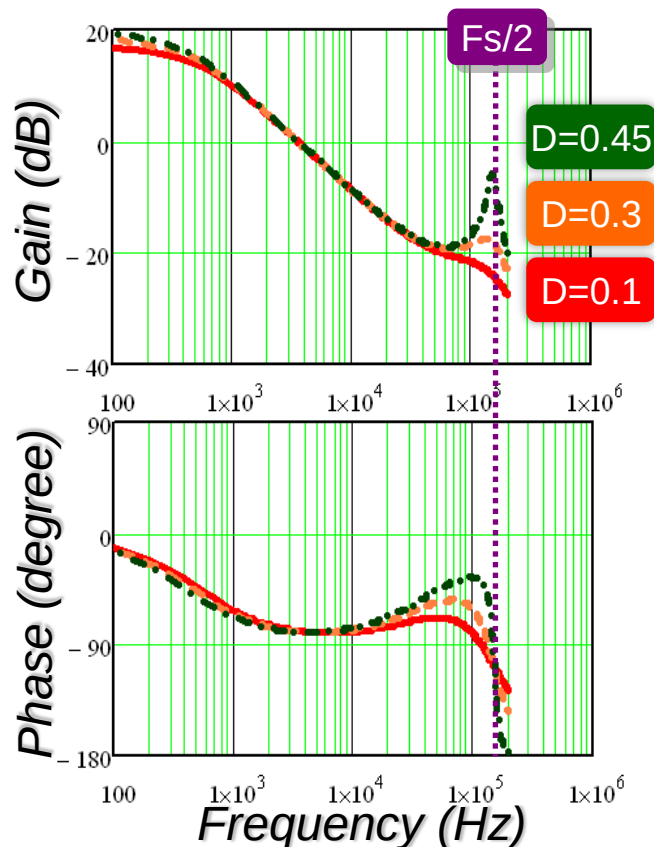
◆ Proposed Model ◆ Measurement

Transfer Function Comparison

➤ Peak Current Mode-Control

$$\frac{v_o(s)}{v_c(s)} \approx K_p \frac{R_{Co}C_o s + 1}{s/\omega_p + 1} \frac{1}{1 + \frac{s}{Q_2\omega_2} + \frac{s^2}{\omega_2^2}}$$

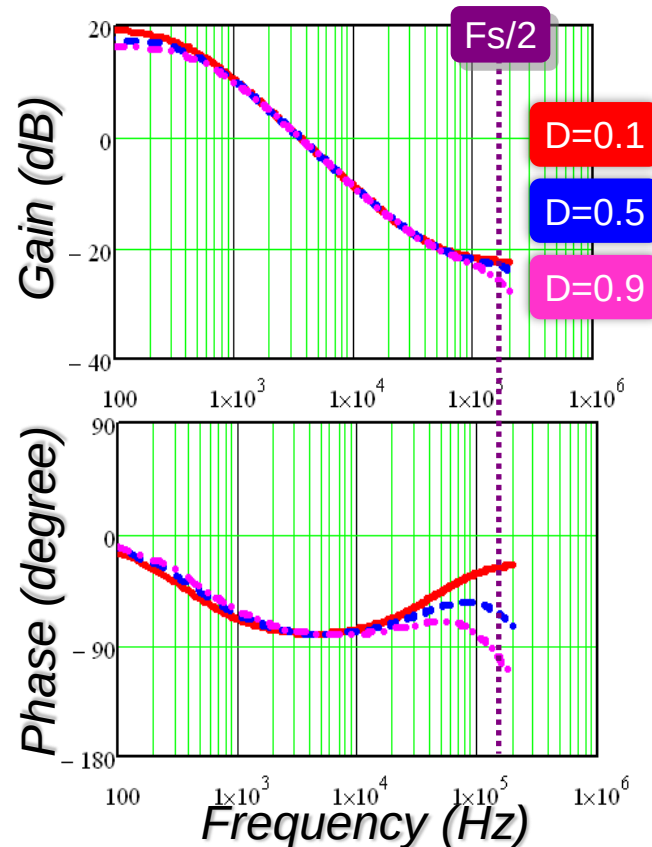
$$\omega_2 = \frac{\pi}{T_{sw}} \quad Q_p = \frac{1}{\pi[(s_n + s_e)/(s_n + s_f) - 0.5]}$$



➤ Constant On-time Control

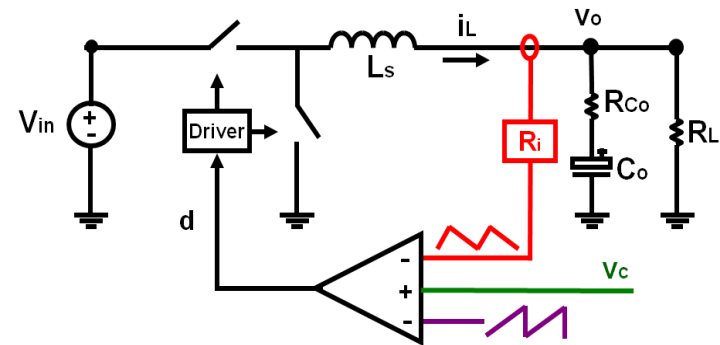
$$\frac{v_o(s)}{v_c(s)} \approx K_c \frac{R_{Co}C_o s + 1}{s/\omega_a + 1} \frac{1}{1 + \frac{s}{Q_1\omega_1} + \frac{s^2}{\omega_1^2}}$$

$$\omega_1 = \frac{\pi}{T_{on}} \quad Q_1 = \frac{2}{\pi}$$

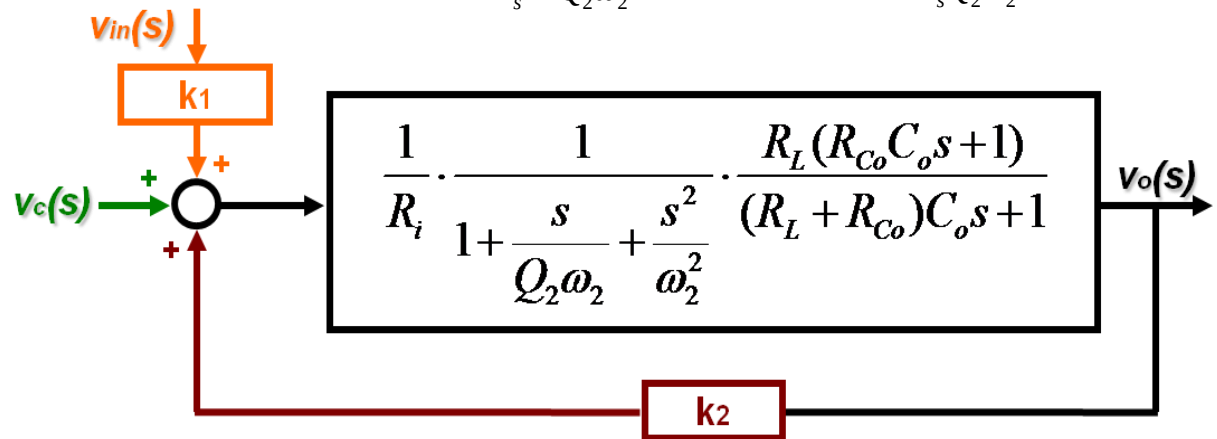


Equivalent Circuit Representation of Current-Mode Control

Complete Model for Peak Current-Mode control



$$k_1(s) \approx \frac{DR_i}{L_s} \left[\frac{1}{Q_2 \omega_2} - \frac{T_{off}}{2} \right] \quad k_2(s) \approx -\frac{R_i}{L_s Q_2 \omega_2}$$

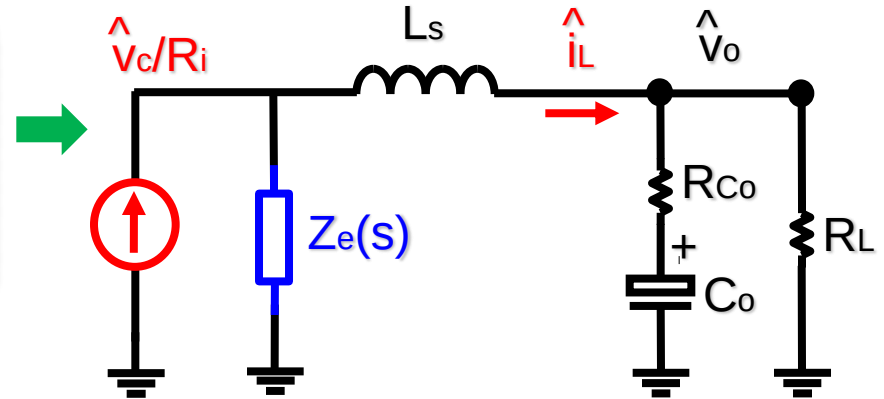


$$\frac{v_o(s)}{v_c(s)} = \frac{\frac{1}{R_i} \cdot \frac{1}{1 + \frac{s}{Q_2 \omega_2} + \frac{s^2}{\omega_2^2}} \cdot \frac{R_L (R_{Co} C_o s + 1)}{(R_L + R_{Co}) C_o s + 1}}{1 - k_2 \frac{1}{R_i} \cdot \frac{1}{1 + \frac{s}{Q_2 \omega_2} + \frac{s^2}{\omega_2^2}} \cdot \frac{R_L (R_{Co} C_o s + 1)}{(R_L + R_{Co}) C_o s + 1}}$$

Equivalent Circuit Model

for Peak Current-Mode Control

$$v_o(s) = \frac{v_c(s)}{R_i} \cdot \frac{Z_e(s) \cdot \frac{R_L(R_{Co}C_o s + 1)}{(R_L + R_{Co})C_o s + 1}}{Z_e(s) + L_s s + \frac{R_L(R_{Co}C_o s + 1)}{(R_L + R_{Co})C_o s + 1}}$$



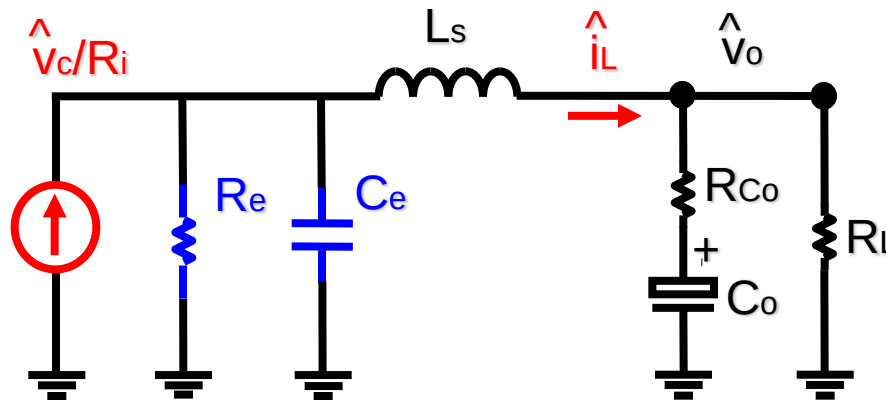
if no simplification in $Z_e(s)$ (Z_e inequal to $R_e//C_e$)
Everything is the same only impedance is different

$$Z_e(s) = \frac{1}{\frac{1}{L_s Q_2 \omega_2} + \frac{s}{L_s \omega_2^2}}$$

where

$$Q_2 = \frac{1}{\pi \left(\frac{s_n + s_e}{s_n + s_f} - 0.5 \right)}$$

$$\omega_2 = \frac{\pi}{T_{sw}}$$

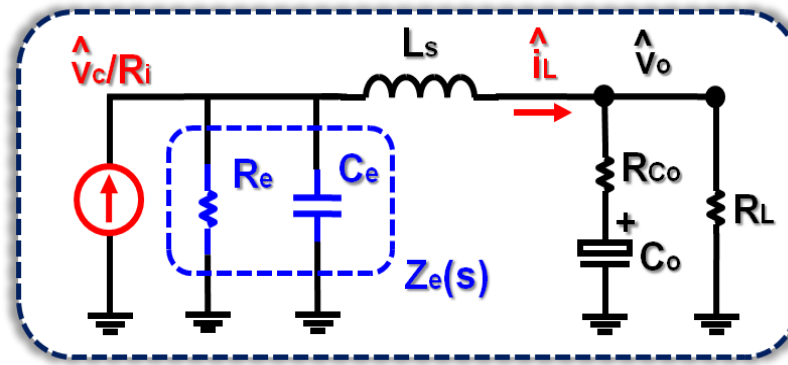


$$C_e = \frac{1}{L_s \omega_2^2} = \frac{T_{sw}^2}{L_s \pi^2}$$

$$R_e = L_s Q_2 \omega_2 = L_s / \left[T_{sw} \left(\frac{s_n + s_e}{s_n + s_f} - 0.5 \right) \right]$$

Equivalent Circuit Model

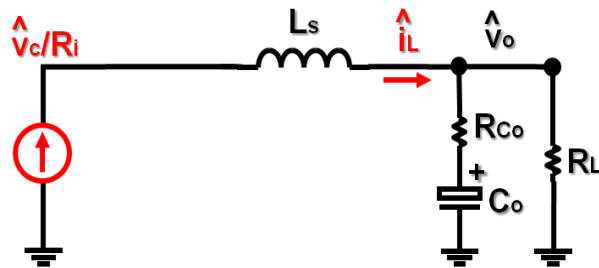
for Peak Current-Mode Control



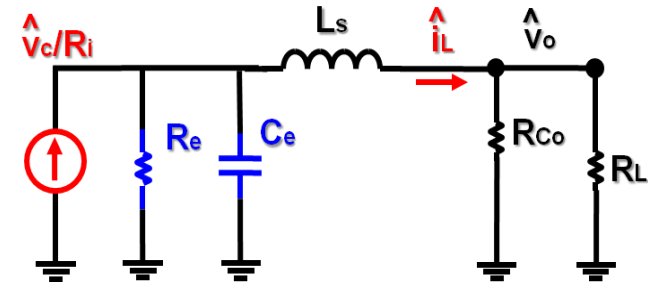
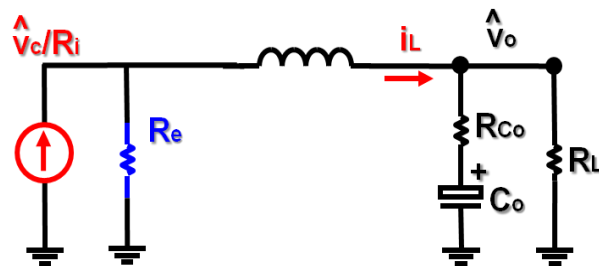
➤ R_e : the current loop effect

➤ C_e : the high frequency response

1. R_e is large: current-loop effect is strong



2. R_e is small: current-loop effect is weak

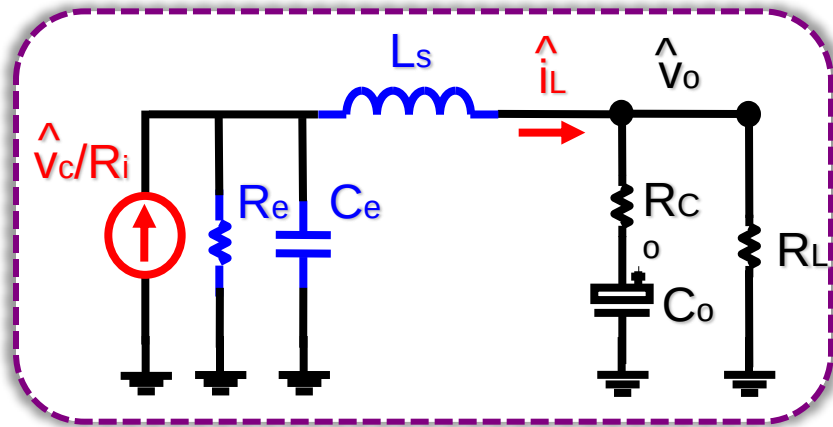


1. The resonant frequency between L_s and C_e is $f_s/2$

2. R_e may be negative which results in instability issue

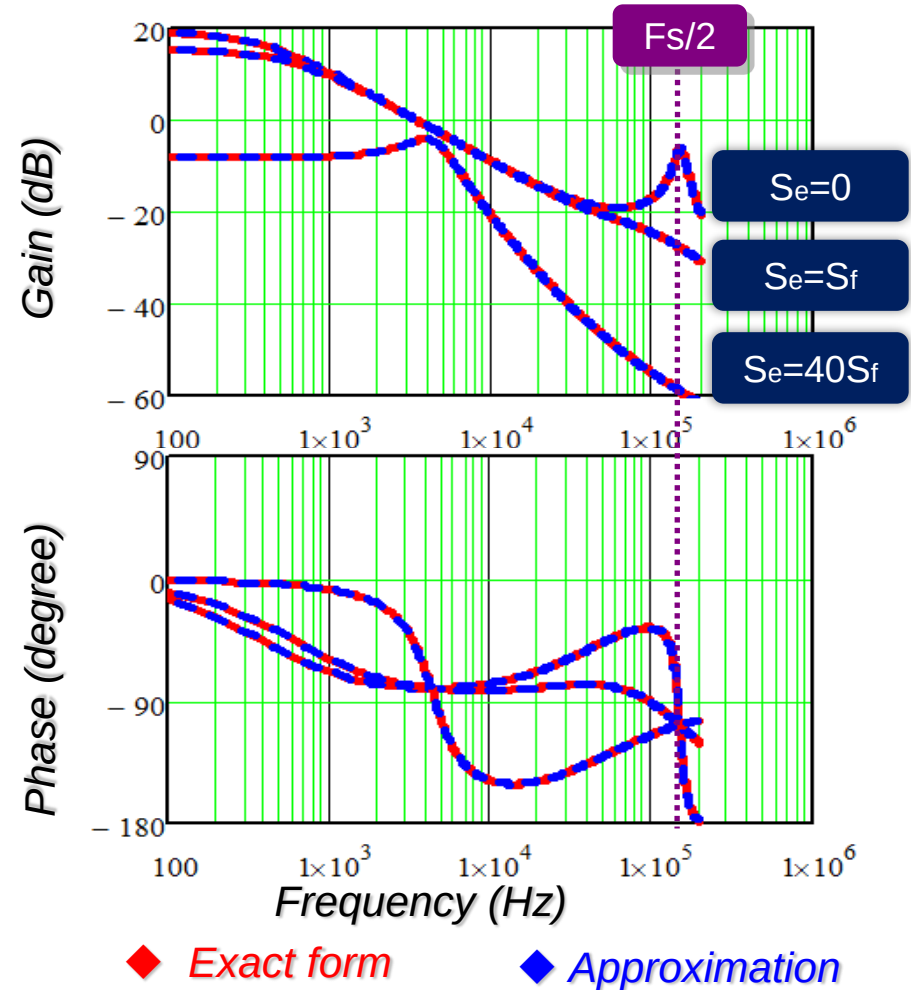
Equivalent Circuit Model

for Peak Current-Mode Control



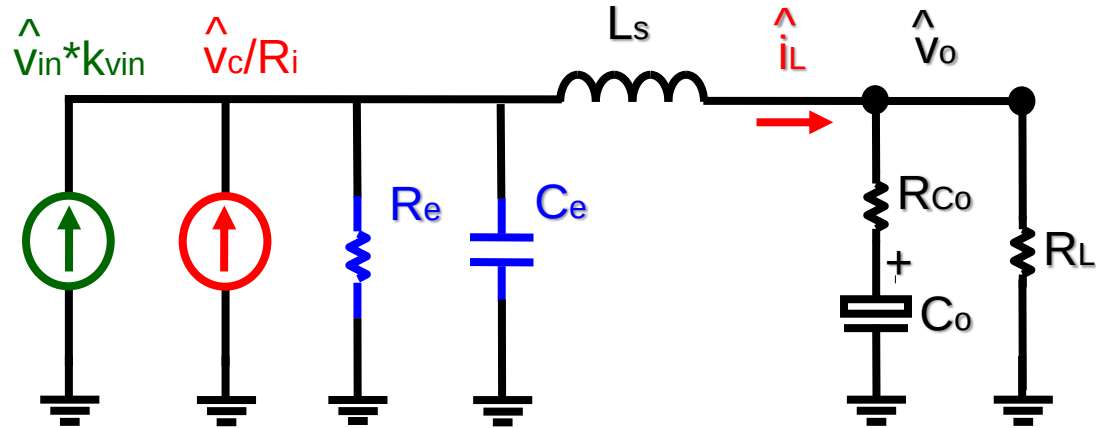
$$R_e = L_s / [T_{sw} (\frac{s_n + s_e}{s_n + s_f} - 0.5)]$$

$$C_e = T_{sw}^2 / (L_s \pi^2)$$



Complete Equivalent Circuit Model

for Peak Current-Mode Control



$$C_e = \frac{1}{L_s \omega_2^2} = \frac{T_{sw}^2}{L_s \pi^2}$$

$$R_e = L_s Q_2 \omega_2 = L_s / [T_{sw} (\frac{s_n + s_e}{s_n + s_f} - 0.5)]$$

$$k_{vin} = \frac{D}{L_s} \left[\frac{1}{Q_2 \omega_2} - \frac{T_{off}}{2} \right]$$

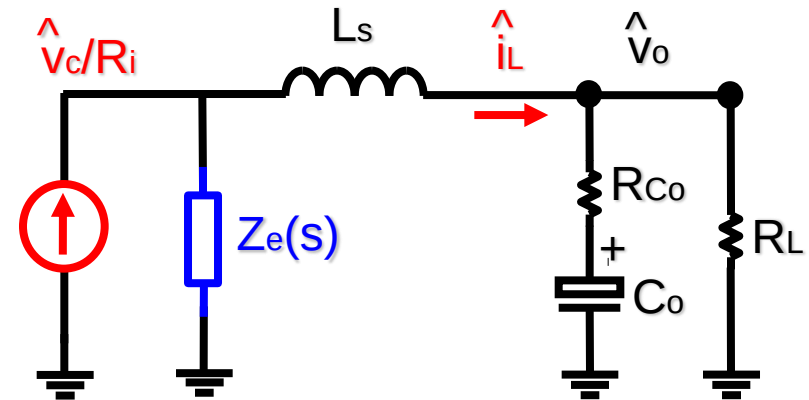
where $Q_2 = \frac{1}{\pi (\frac{s_n + s_e}{s_n + s_f} - 0.5)}$ $\omega_2 = \frac{\pi}{T_{sw}}$

Equivalent Circuit Model

for Constant On-time Control

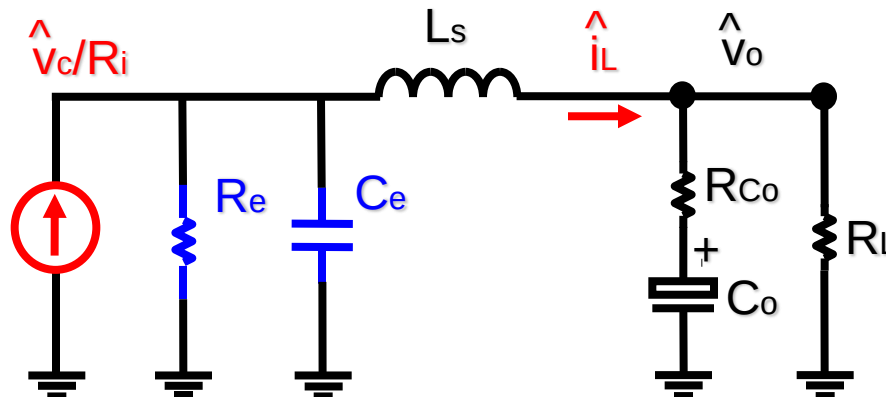
$$v_o(s) = \frac{v_c(s)}{R_i} \cdot \frac{Z_e(s) \cdot \frac{R_L(R_{Co}C_o s + 1)}{(R_L + R_{Co})C_o s + 1}}{Z_e(s) + L_s s + \frac{R_L(R_{Co}C_o s + 1)}{(R_L + R_{Co})C_o s + 1}}$$

$$Z_e(s) = 1 / \left(\frac{1}{\omega_1 Q_1 L_s} + \frac{s}{\omega_1^2 L_s} \right)$$



where

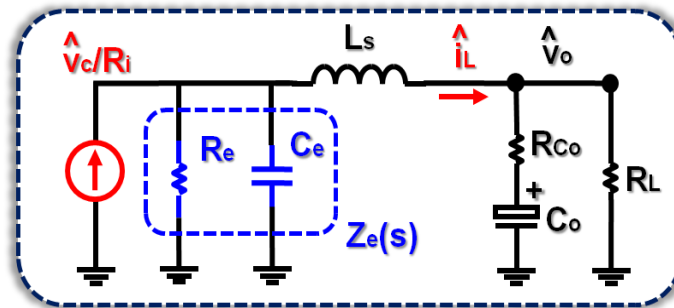
$$\omega_1 = \frac{\pi}{T_{on}} \quad Q_1 = \frac{2}{\pi}$$



$$C_e = T_{on}^2 / (L_s \pi^2) \quad R_e = 2L_s / T_{on}$$

R_e is always positive in constant on-time control

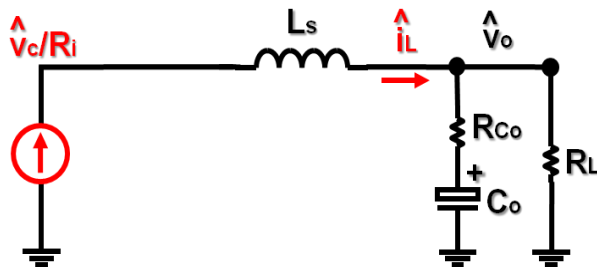
Equivalent Circuit Model for Constant On-time Control



➤ R_e : the current loop effect

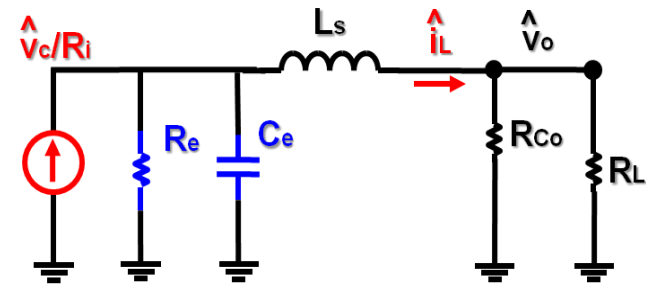
$$R_e = 2L_s / T_{on}$$

R_e is large (no external ramp)



➤ C_e : the high frequency response

$$C_e = T_{on}^2 / (L_s \pi^2)$$



1. The resonant frequency between L_s and C_e is $1/(2T_{on})$
2. R_e is always positive which results no instability issue

Extension to Other Current-Mode Controls

➤ Constant On-time Control

$$C_e = T_{on}^2 / (L_s \pi^2)$$

$$R_e = 2L_s / T_{on}$$

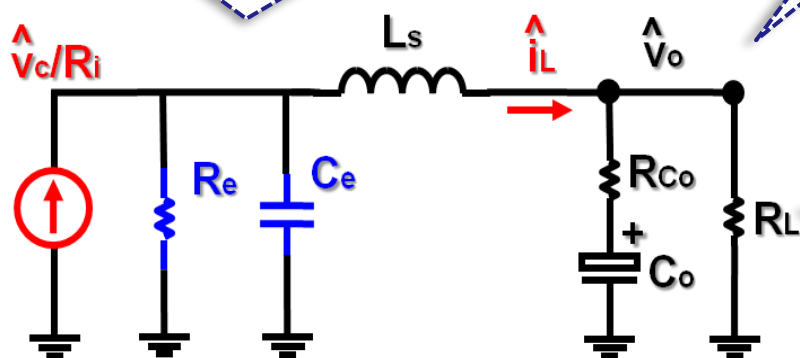
R_e is always positive

➤ Constant Off-time Control

$$C_e = T_{off}^2 / (L_s \pi^2)$$

$$R_e = 2L_s / T_{off}$$

R_e is always positive



➤ Charge Control

$$C_e = T_{sw}^2 / (L_s \pi^2)$$

$$R_e = L_s / [T_{sw} (\frac{L_s f_s}{R_L} - \frac{D}{2} + \frac{L_s C_T f_s}{V_o R_i} s_e)]$$

R_e can be positive or negative

➤ Peak Current-Mode Control

$$C_e = T_{sw}^2 / (L_s \pi^2) \quad \text{fixed freq} \rightarrow \text{a fixed double pole at half of switch freq}$$

$$R_e = L_s / [T_{sw} (\frac{s_n + s_e}{s_n + s_f} - 0.5)]$$

R_e can be positive or negative

negative R_e results unstable system

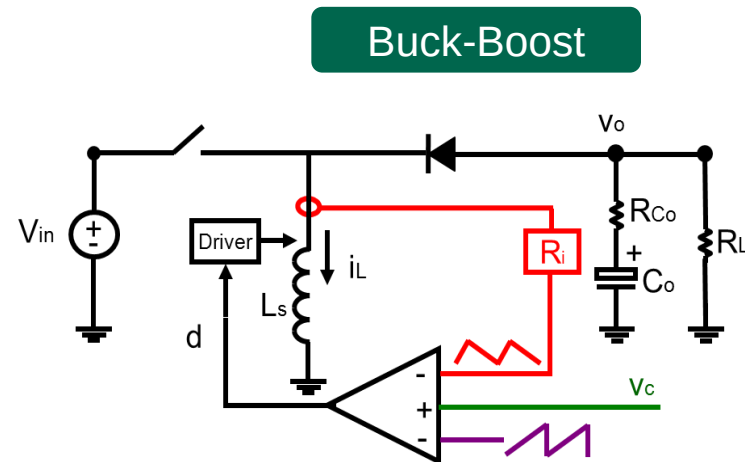
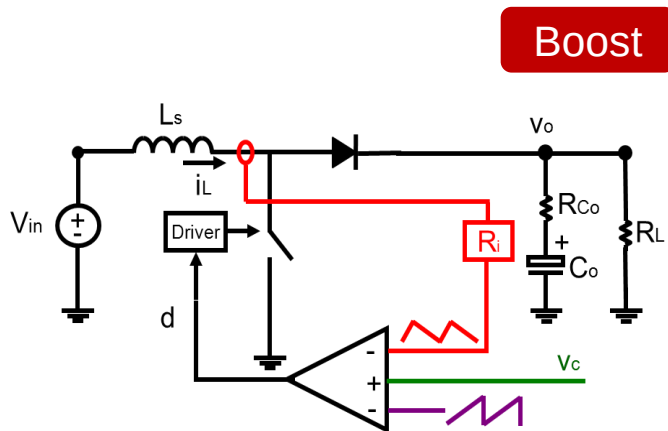
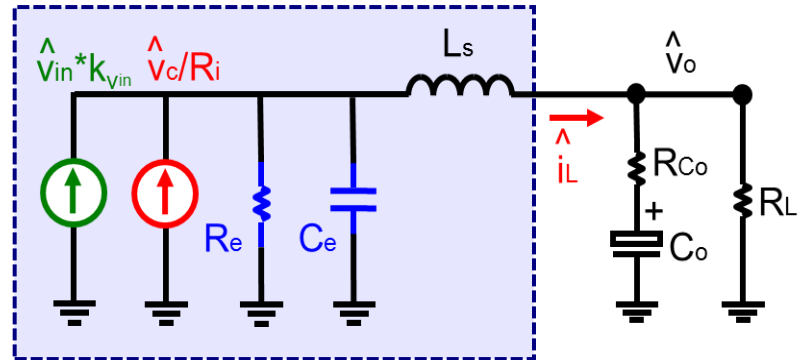
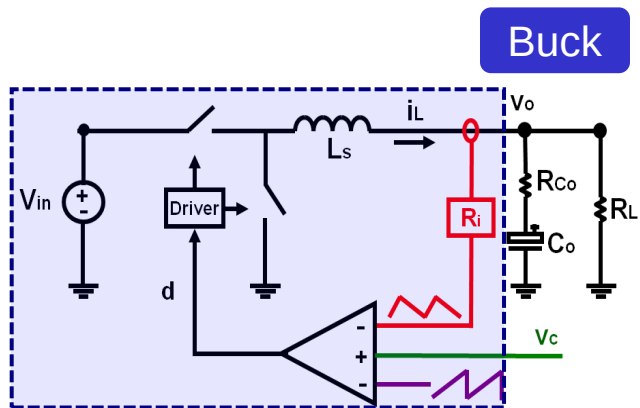
➤ Valley Current-Mode Control

$$C_e = T_{sw}^2 / (L_s \pi^2)$$

$$R_e = L_s / [T_{sw} (\frac{s_f + s_e}{s_n + s_f} - 0.5)]$$

R_e can be positive or negative

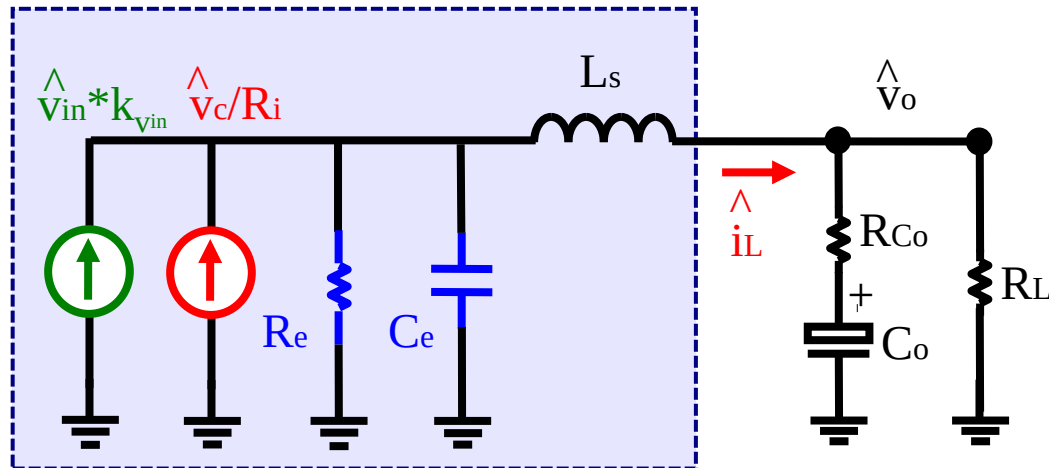
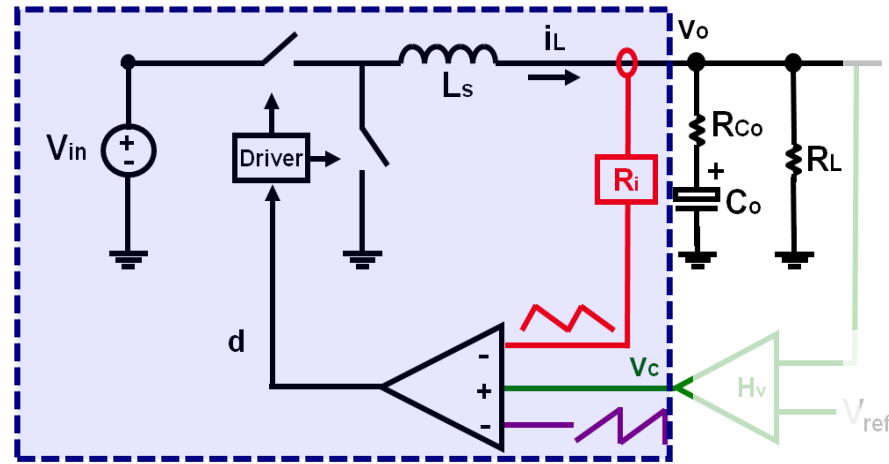
Extend to Other Converters ?



No equivalent circuit for other current mode control converters

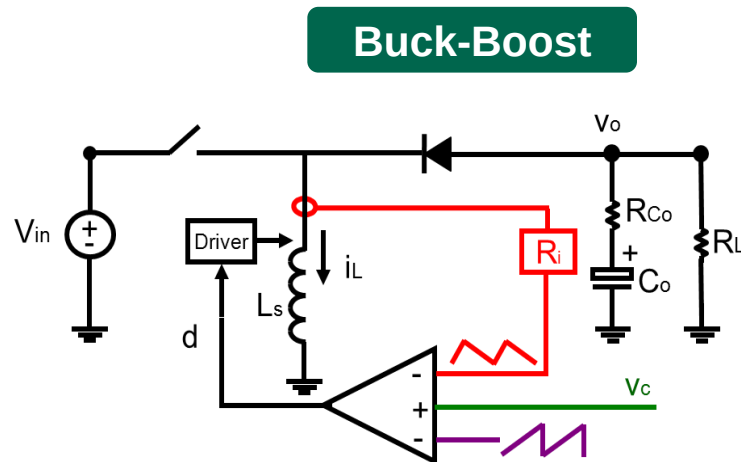
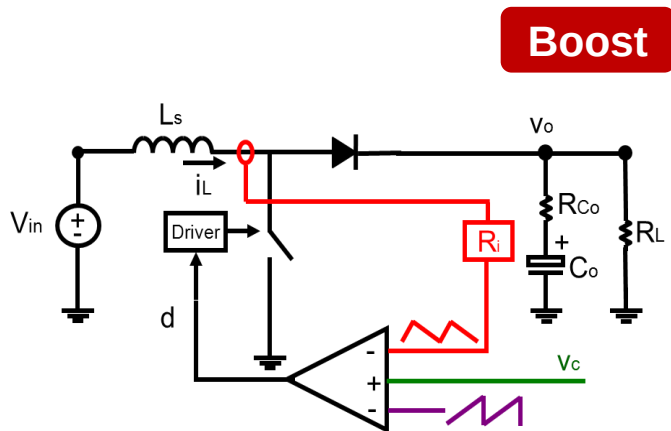
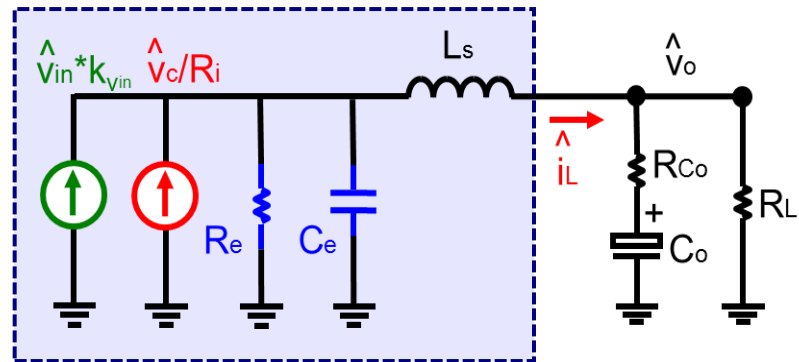
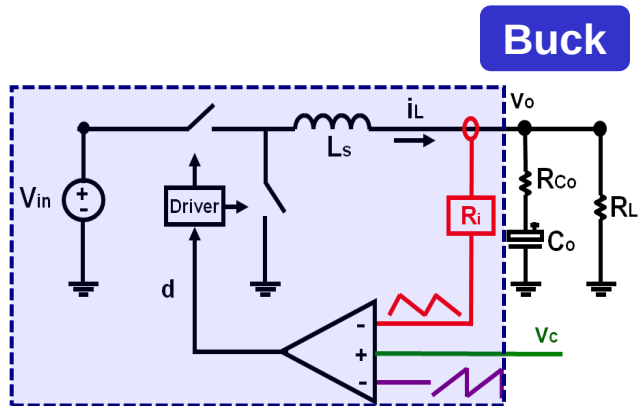
Unified Three-Terminal Switch Model for Current Mode Controls

Motivation 1: Complete the Equivalent Circuit



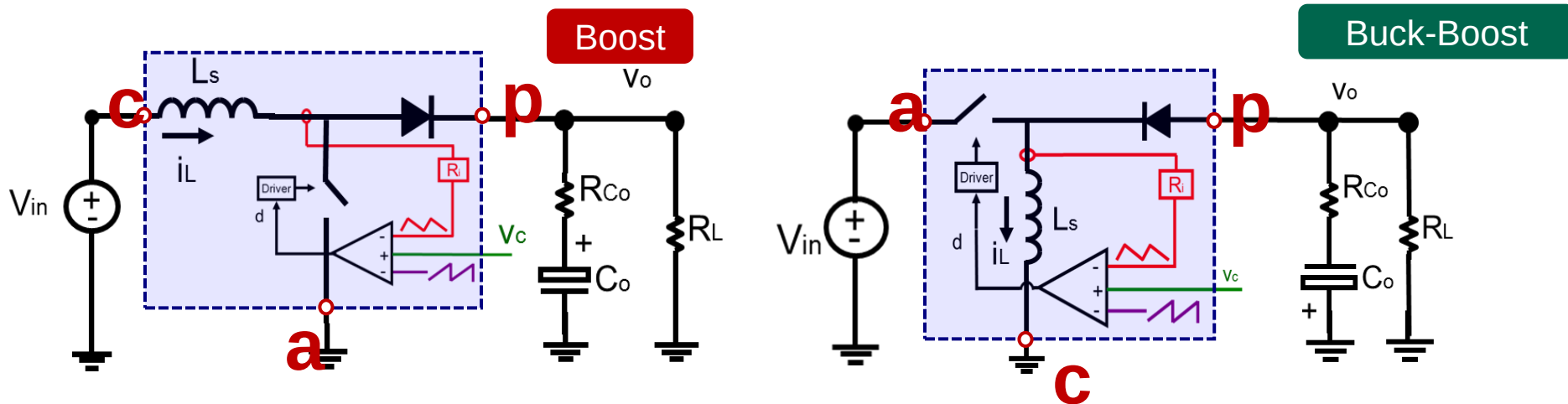
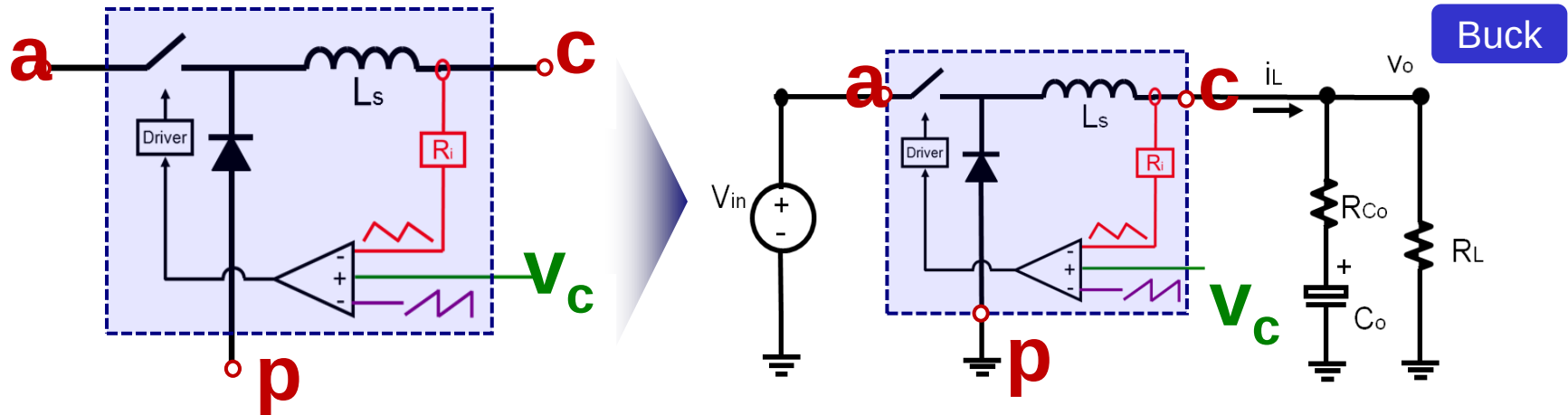
The model only shows relationship between v_{in} , v_c and v_o , but input current property is lost.

Motivation 2: Extend to Other Converters



No equivalent circuit for other current mode control converters

Objective: Model the Common Structure

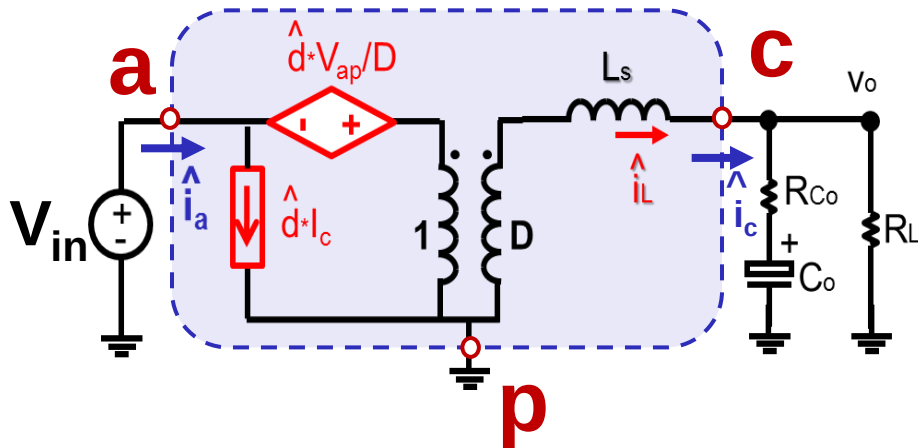
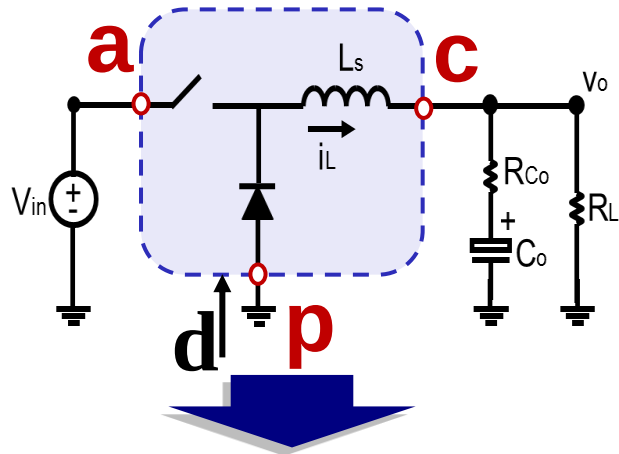


Objectives:

1. To correctly represent input and output characteristic of Buck converter
2. To develop a unified equivalent circuit applicable to other topologies

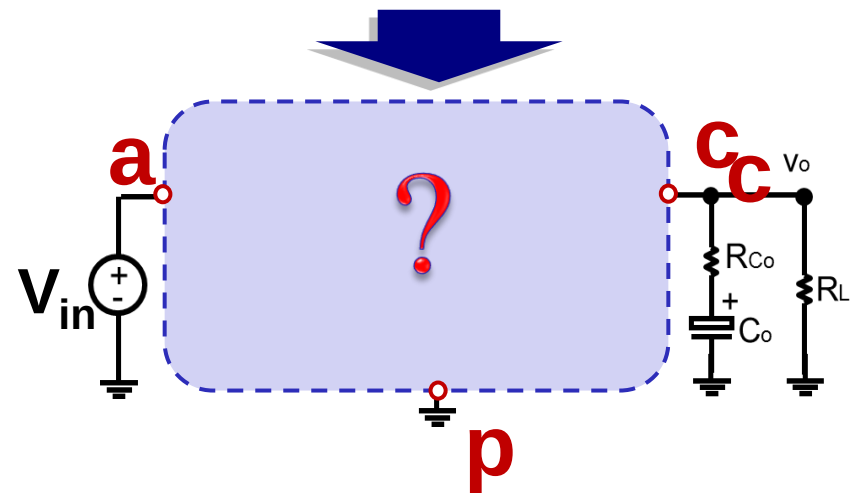
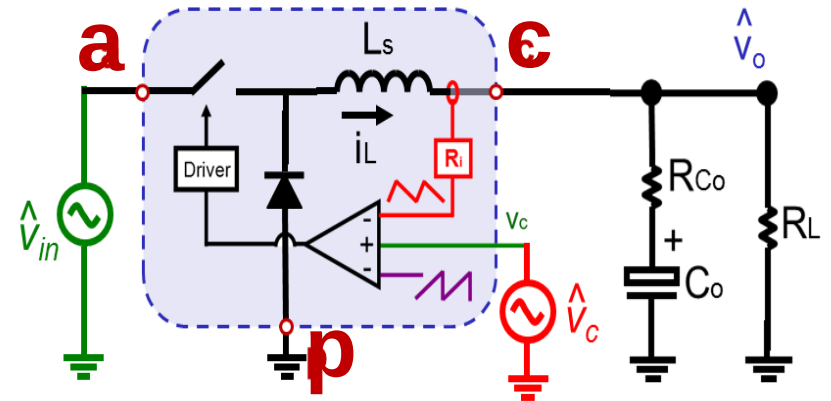
Three-terminal Switch Model Concept

➤ 3-terminal switch model



➤ 3-terminal switch model

with current feedback

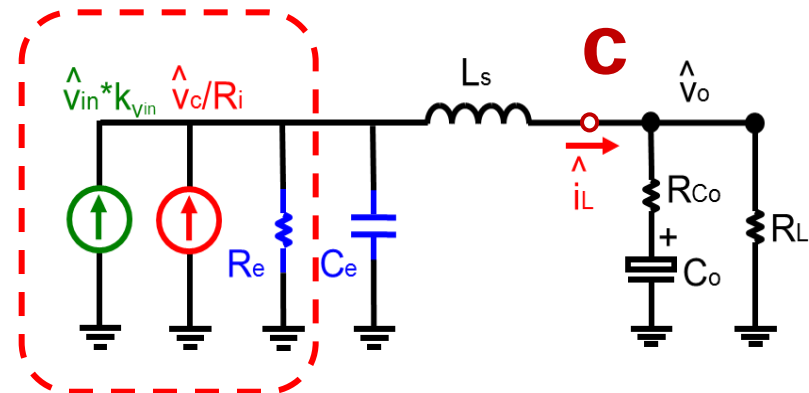
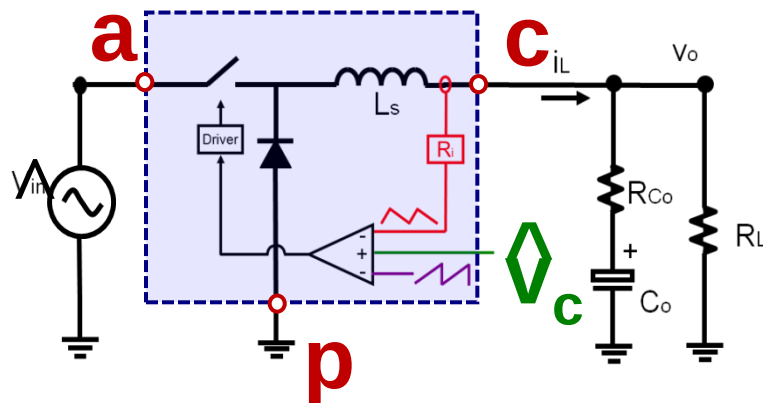


What is the equivalent circuit for current mode control PWM switch?

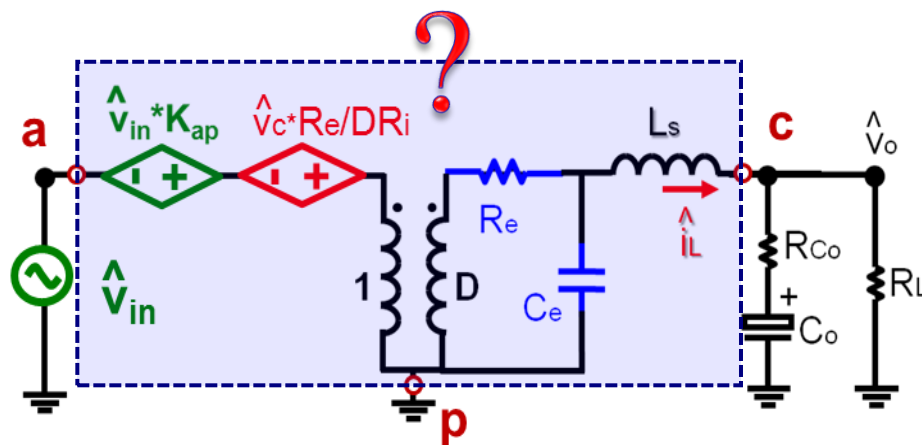
Start from Equivalent Circuit for Buck Converter



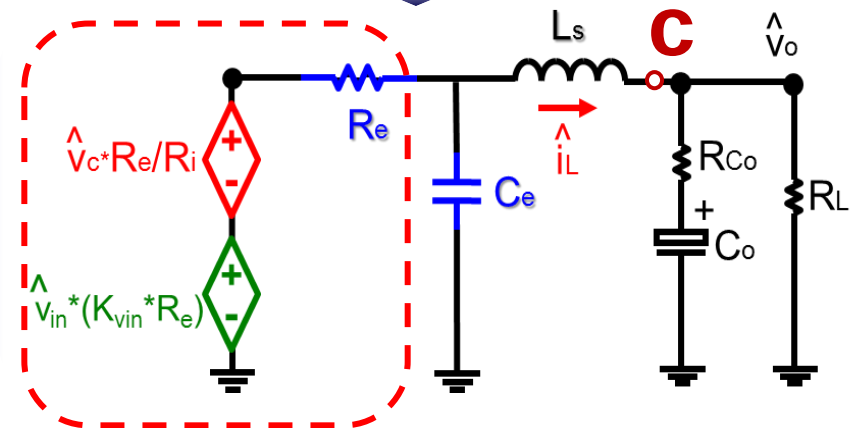
CPES



Small signal Equivalent circuit describing $\hat{V}_o / \hat{V}_c$



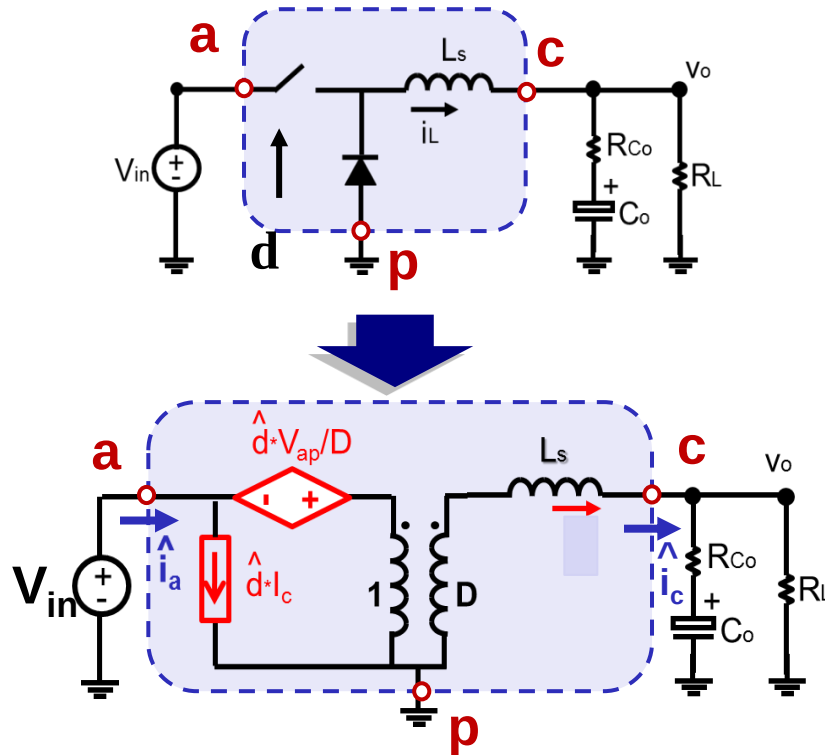
DC Transformer



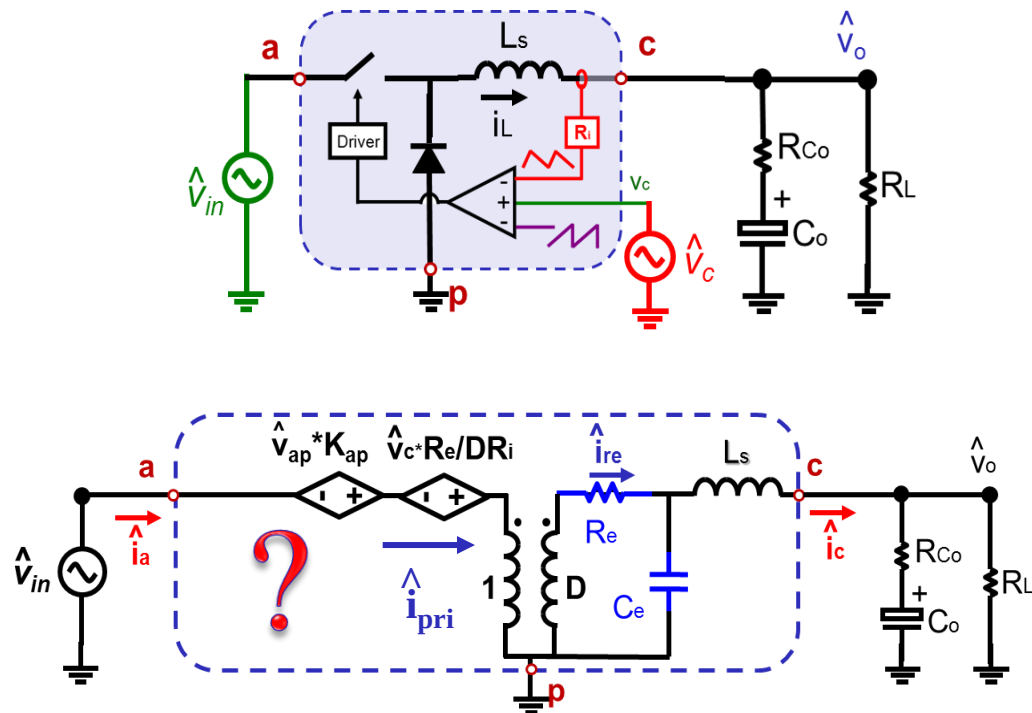
Thevenin Equivalent Circuit

Universal Property of Switch Current

➤ 3-terminal switch model



➤ 3-terminal switch model



➤ Small Signal Relationship:

$$\hat{i}_a = D \cdot \hat{i}_c + \hat{d} \cdot I_c$$

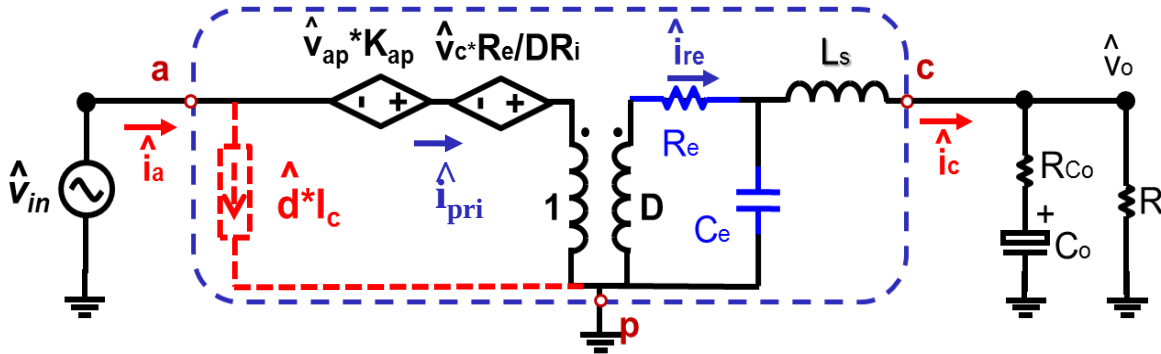
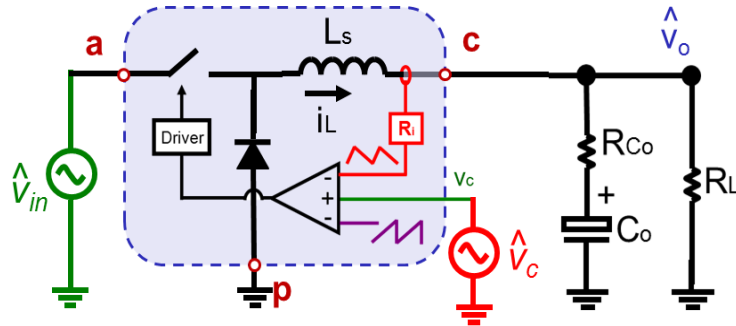
➤ Small Signal Relationship:

$$\hat{i}_a \approx D \cdot \hat{i}_c + \hat{d} \cdot I_c$$

The property of the switch current i_a should be preserved in the current mode control as well

Address the Missing Term $\hat{d} \cdot I_c$

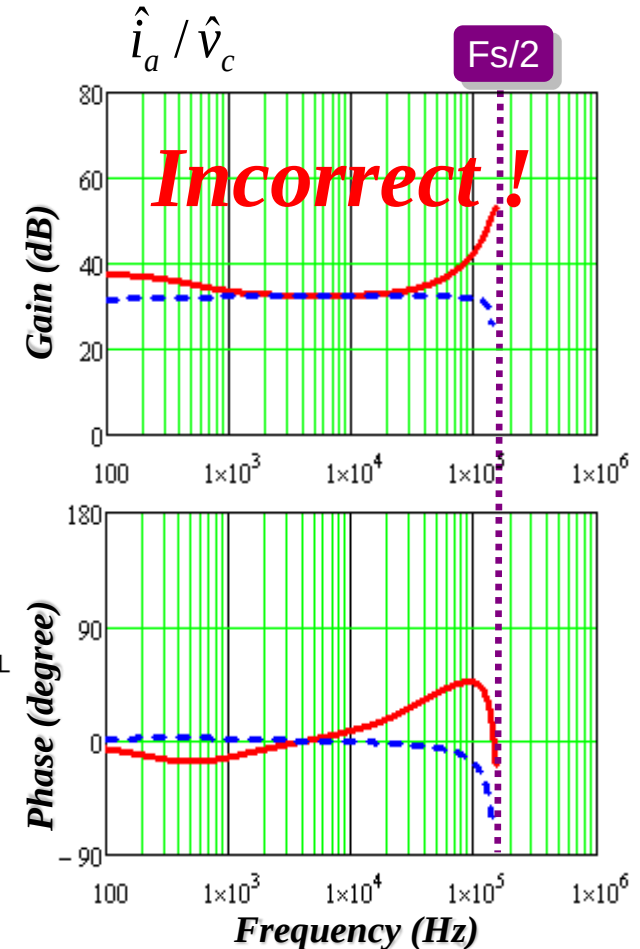
Equivalent Circuit for Buck with Current Mode Control



$$\hat{i}_{pri} = D \cdot \hat{i}_{Re} \approx D \cdot \hat{i}_c$$

$$\hat{i}_a = \hat{d} \cdot I_c + \hat{i}_{pri} \approx \hat{d} \cdot I_c + D \cdot \hat{i}_c$$

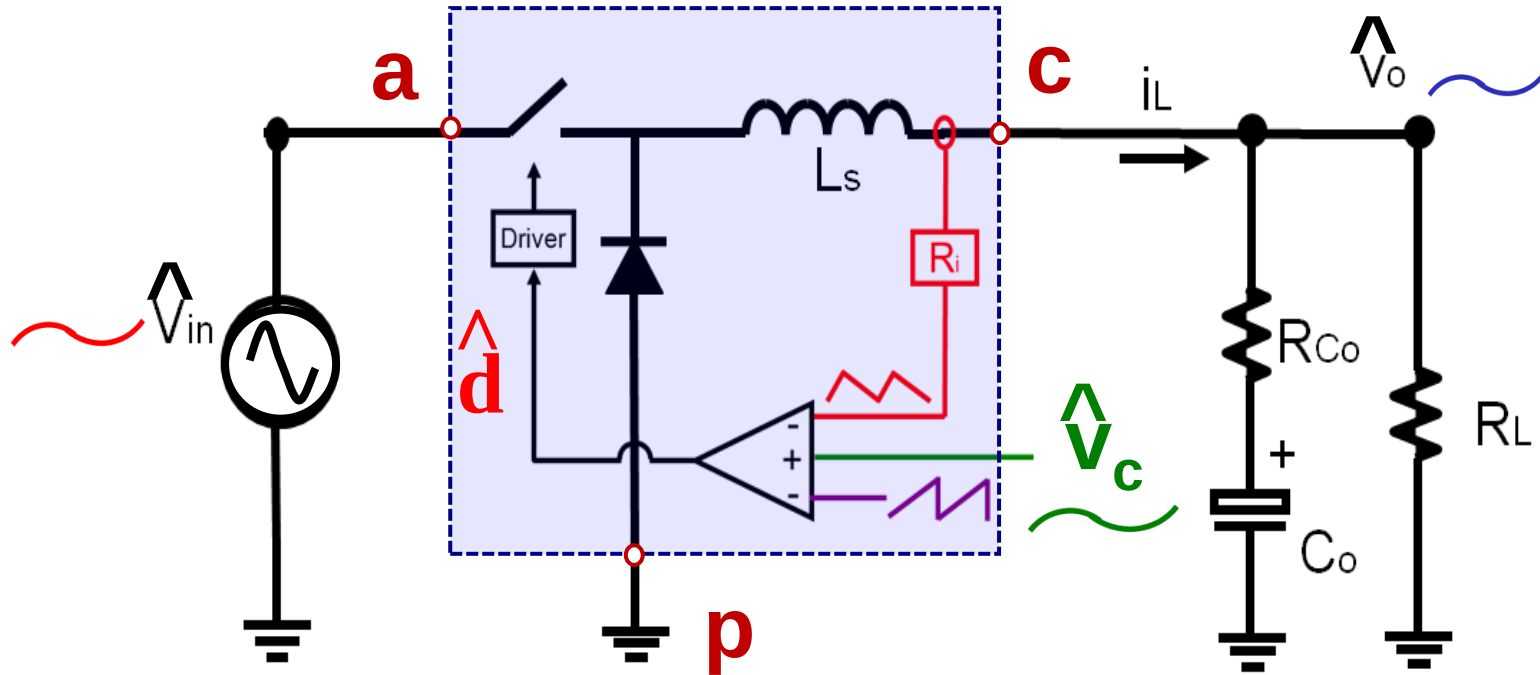
Complete the model by addressing $\hat{d} \cdot I_c$ term



◆ **Simplis Simulation**

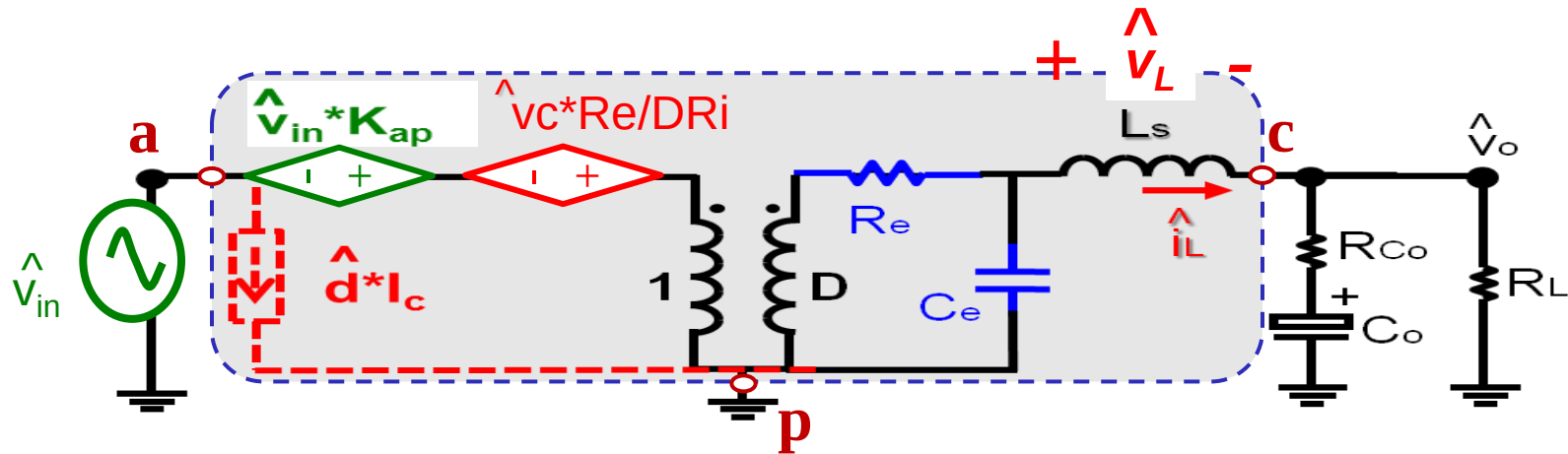
◆ **Equivalent Circuit**

Duty Cycle Modulation



To correctly represent modulation effect, input and output voltage perturbation have to be considered.

$\hat{d} \cdot I_c$ Term Representation in Equivalent Circuit



➤ From Jian Li's describing function derivation:*

$$\hat{d} \approx \hat{v}_c \cdot A(s) + \hat{v}_{ap} \cdot B(s) + \hat{v}_{cp} \cdot C(s) \quad (1)$$

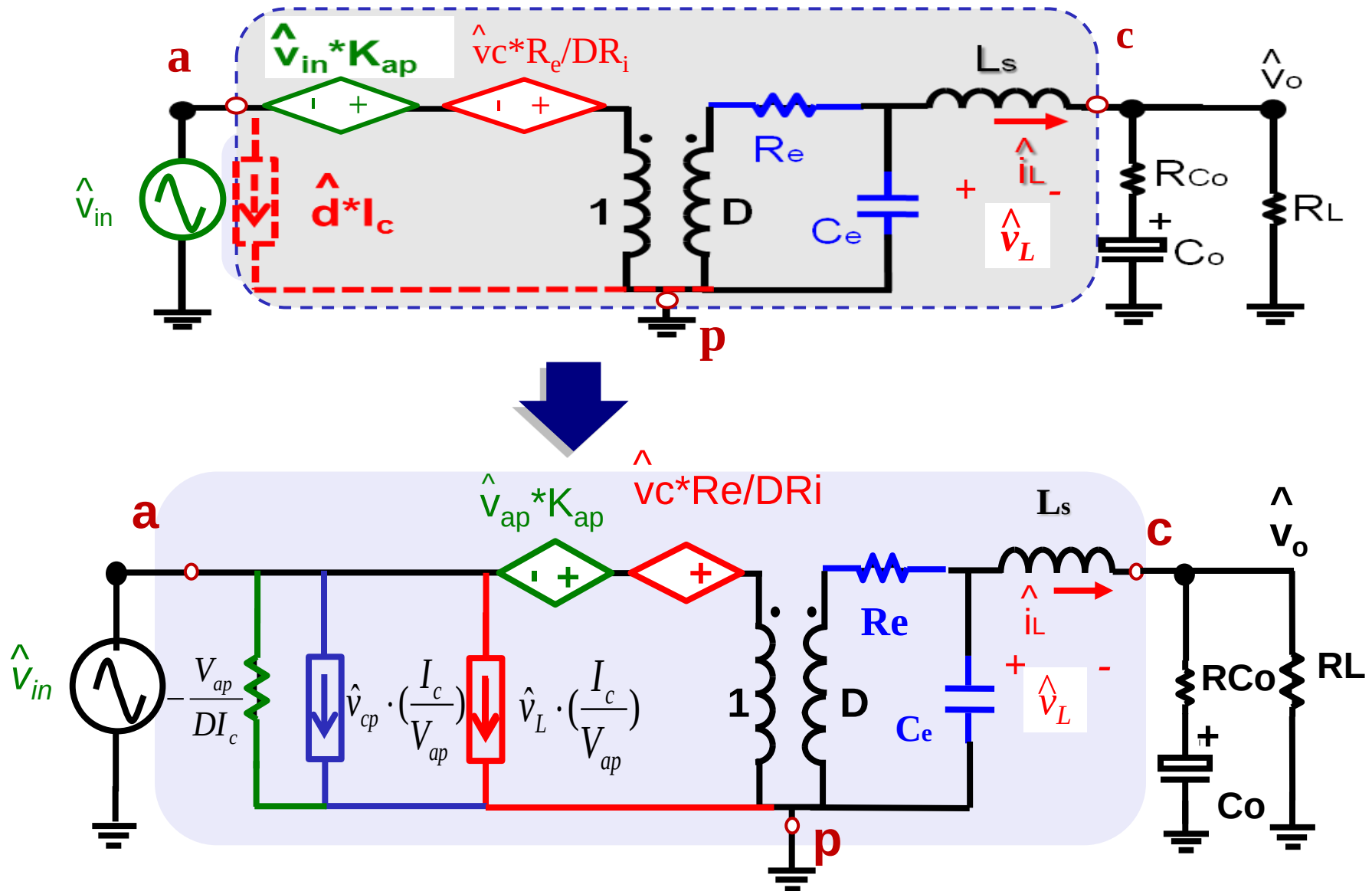
➤ From Equivalent Circuit:

$$\hat{v}_c \cdot A(s) = \hat{v}_L / V_{ap} - \hat{v}_{ap} \cdot B(s) - \hat{v}_{cp} \cdot C(s) - \hat{v}_{ap} \cdot D / V_{ap} + \hat{v}_{cp} / V_{ap} \quad (2)$$

From equation 1 and 2

$$\hat{d} \cdot I_c = \hat{v}_L \cdot \left(\frac{I_c}{V_{ap}} \right) + \hat{v}_{cp} \cdot \left(\frac{I_c}{V_{ap}} \right) + \hat{v}_{ap} / \left(-\frac{V_{ap}}{DI_c} \right)$$

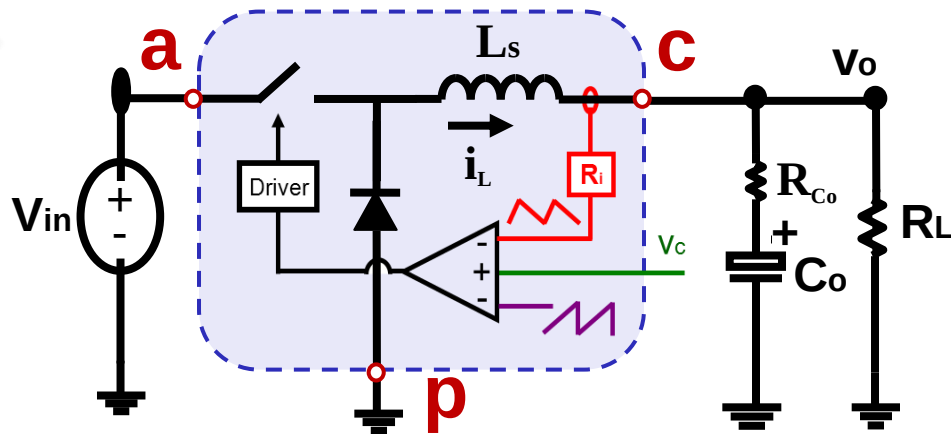
Complete Equivalent Circuit Model



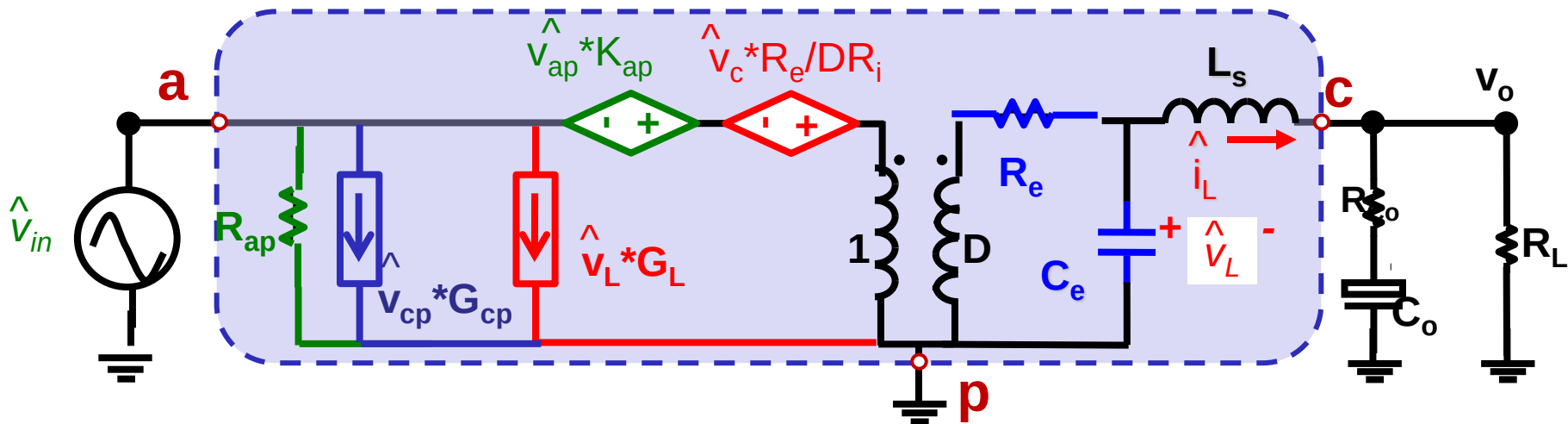
Three Terminal Switch Model

by Yingyi Yan

Buck Converter



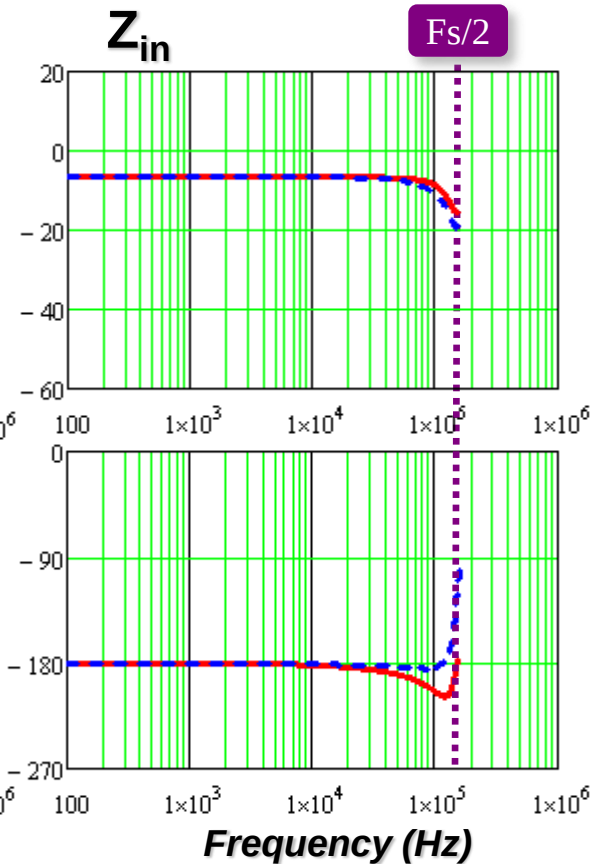
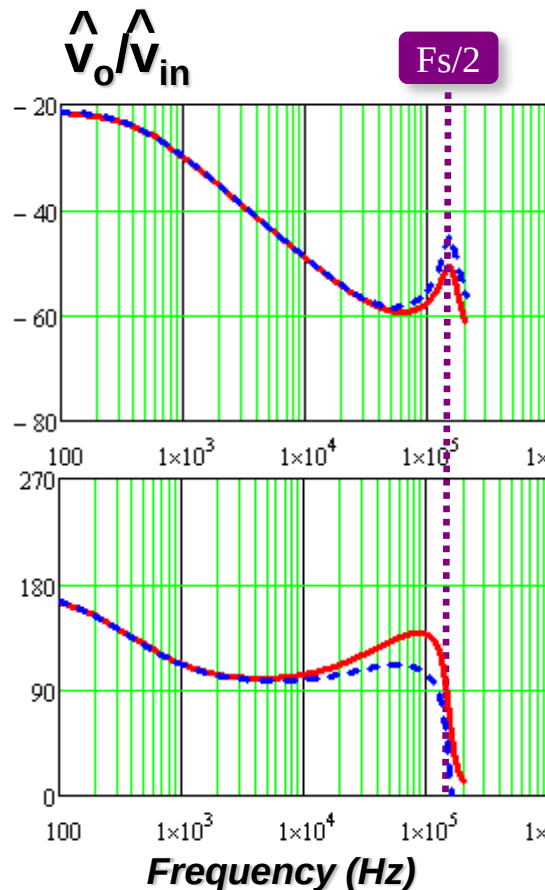
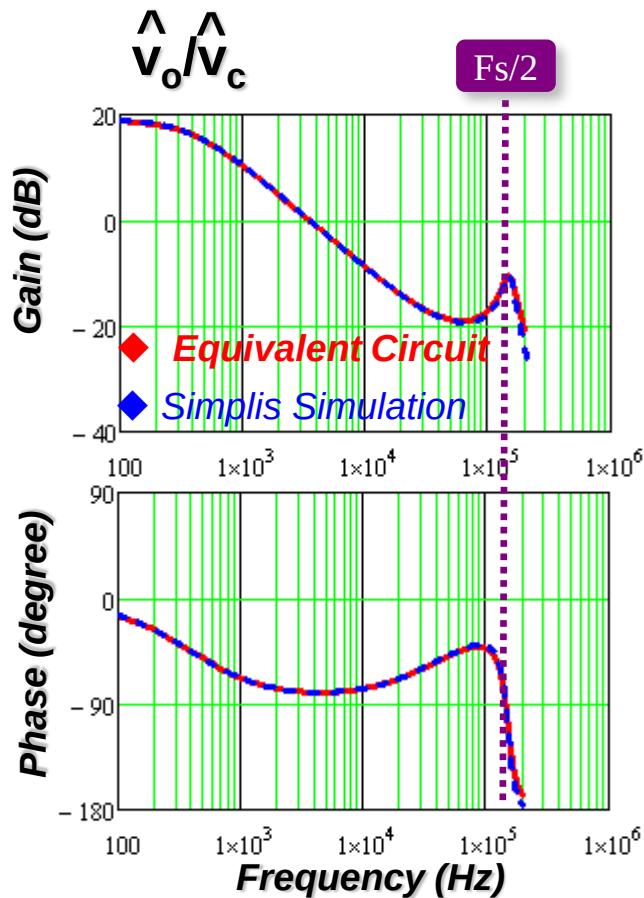
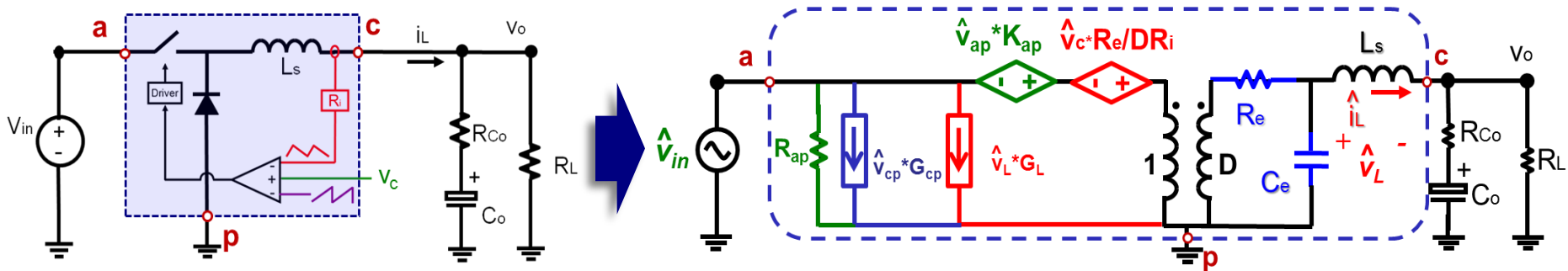
3 Terminal current-cell model



Three terminal equivalent circuit represents small signal characteristic of three terminal switch with closed current loop

Model Verification

Buck Converter with Peak current mode control



Model Extension to Other Current-Mode Controls

➤ Constant On-time Control

$$C_e = T_{on}^2 / (L_s \pi^2) \quad R_e = 2L_s / T_{on}$$

$$K_{ap} = T_{off} / T_{on}$$

Re is always positive

➤ Constant Off-time Control

$$C_e = T_{off}^2 / (L_s \pi^2) \quad R_e = 2L_s / T_{off}$$

$$K_{ap} = -1$$

Re is always positive

➤ Charge Control

$$C_e = T_{sw}^2 / (L_s \pi^2) \quad R_e = L_s / [T_{sw} (\frac{L_s I_L}{V_{cp} T_{sw}} - \frac{D}{2} + \frac{s_e}{s_f} \frac{C_T}{T_{sw}})]$$

$$K_{ap} = (T_{off} / 2L_s) \cdot R_e$$

Re can be positive or negative

➤ Peak Current-Mode Control

$$C_e = T_{sw}^2 / (L_s \pi^2) \quad R_e = L_s / [T_{sw} (\frac{s_n + s_e}{s_n + s_f} - 0.5)]$$

$$K_{ap} = -(T_{off} / 2L_s) \cdot R_e$$

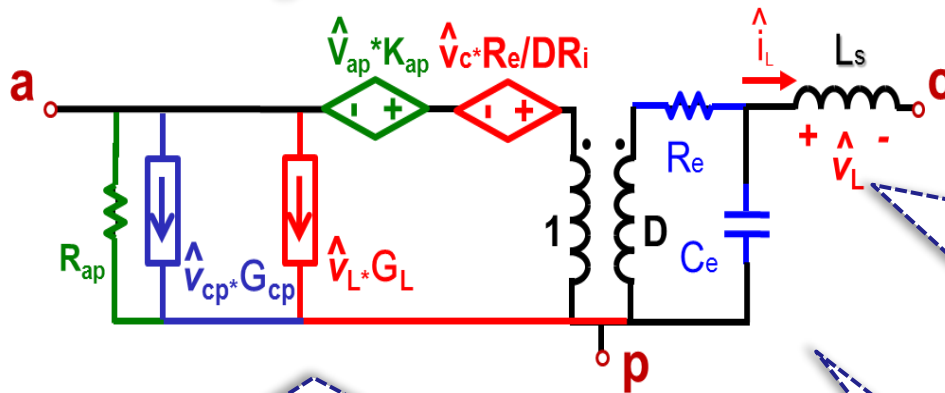
Re can be positive or negative

➤ Valley Current-Mode Control

$$C_e = T_{sw}^2 / (L_s \pi^2) \quad R_e = L_s / [T_{sw} (\frac{s_f + s_e}{s_n + s_f} - 0.5)]$$

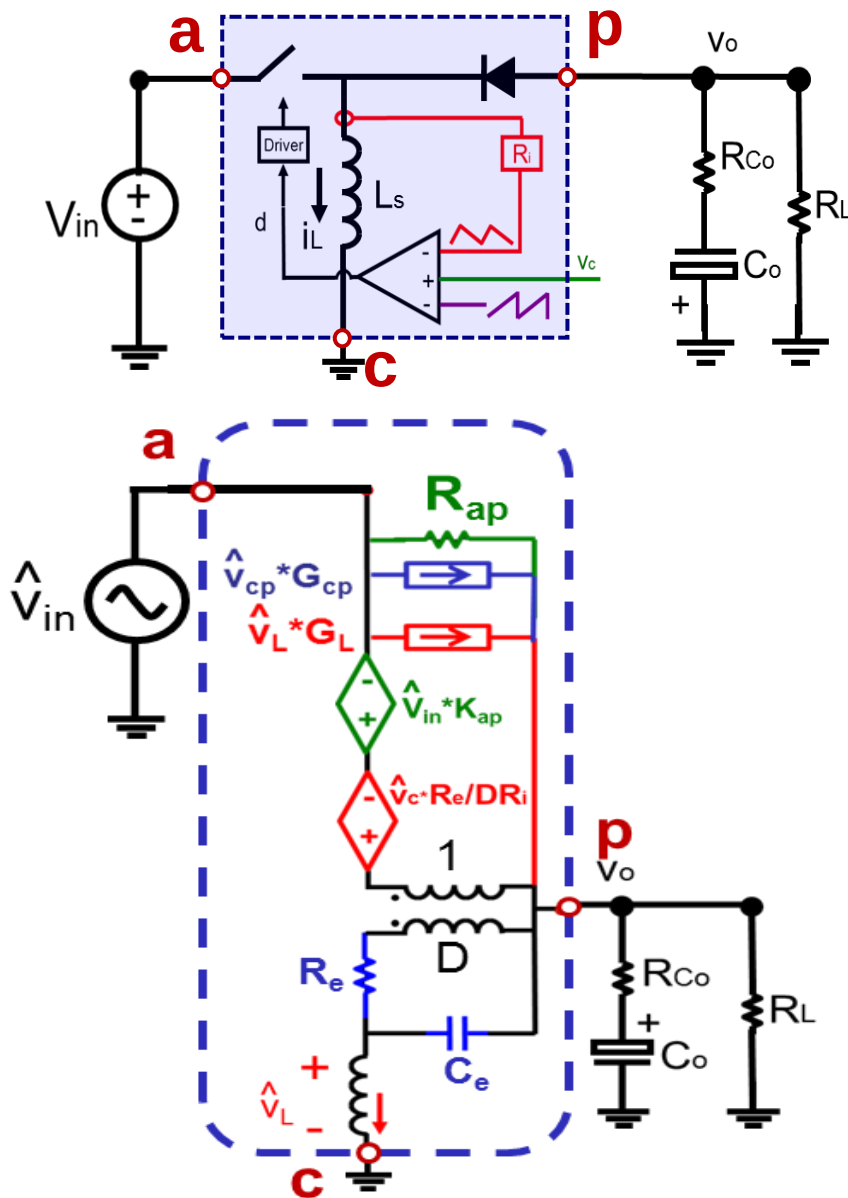
$$K_{ap} = (T_{on} / 2L_s) \cdot R_e$$

Re can be positive or negative



Control-to- v_o Transfer Function

Buck-boost with peak current mode control



$V_{in}=500V$, $V_o=400V$, $R_L=150\Omega$,
 $f_{sw}=100kHz$, $L_s=300\mu H$

